

SEMESTER 3

INDUSTRIAL ENGINEERING

SEMESTER S3
MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL
SCIENCE – 3

(Common to B & C Groups)

Course Code	GYMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	Course Type	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. (Text 1: Relevant topics from sections 11.7, 11.8, 11.9)	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w = z^2$, $w = e^z$, $w = \frac{1}{z}$, $w = \sin z$. (Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)	9
3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply	9

	connected domain (without proof), Cauchy Integral formula (without proof). (Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	
4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. (Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	K3
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	K3
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	K3
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10 th edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3 rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39 th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44 th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1 st edition, 2011

SEMESTER S3

FLUID MECHANICS AND HYDRAULIC MACHINES

Course Code	PCIET302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs 30 Mins
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise fundamental concepts on mechanics of fluids to students.
2. To acquaint the students with basic flow properties and flow measuring devices.
3. To demonstrate the working principles of hydraulic machines.
4. To enable students to apply principles in fluid statics and dynamics, working principles of hydraulic machines to solve practical problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamental concepts: Properties of fluid - density, specific weight, viscosity, surface tension, capillarity, vapour pressure, bulk modulus, compressibility, Newton's law of viscosity, Newtonian and non-Newtonian fluids, real and ideal fluids, incompressible and compressible fluids. Fluid statics: Atmospheric pressure, gauge pressure and absolute pressure. Pascal's Law, measurement of pressure - piezometer, manometers, pressure gauges, forces on planar and curved surfaces immersed in fluids, centre of pressure, buoyancy, equilibrium of floating bodies, metacentre and metacentric height.	11
2	Fluid kinematics and Dynamics : Overview on classification of flows, Continuity equation, Velocity and Acceleration of fluid flow, Euler's	11

	equation, Bernoulli's equation. Reynolds experiment, Hagen-Poiseuille equation, Major and Minor losses in pipes, Darcy-Weisbach equation, Chezy's formula. Pipes in series and parallel, siphon effect, water hammer, transmission of power through pipes and related practical problems.	
3	Flow rate measurements: venturi and orifice meters, mouth pieces, notches and weirs (derivation of discharges only for basic geometries), flow velocity measurements techniques. Time of emptying a tank, basic concepts on boundary layer theory, displacement, momentum and energy thickness. Hydraulic Turbines : Overview on Impact of jets on flat, curved symmetrical and unsymmetrical plates (stationary and moving), Impact of jet on hinged plate. impact of jet on moving series of flat and curved vanes, Hydroelectric power stations. Overview on Impulse, Reaction and Axial Flow Turbines, Working principles, constructional features, design parameters and practical problems on Pelton, Francis and Kaplan Turbines. Governing of Turbines, Draft tubes, Cavitation in Turbines	11
4	Pumps : Overview on rotary and positive displacement pumps, Working principles, applications, constructional features, design considerations, accessories, factors affecting performance characteristics and practical problems on Centrifugal pumps and Reciprocating pumps. Multi staging of pumps, Basic features and working of diaphragm pump, gear pump, peristaltic pump progressive cavity pump, drum pump, submersible pump, fire fighting pumps and Micro pumps.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain fluid properties, types of fluids, fluid statics, pressure measurements and forces, equilibrium of floating and submerged bodies.	K2
CO2	Apply concepts on fluid kinematics and dynamics and solve problems on flow through pipes, major and minor losses.	K3
CO3	Solve problems on discharge and velocity of flow, understand boundary layer theories, understand concepts and solve problems on impacts of jets and turbines.	K3
CO4	Demonstrate working of pumps, solve problems on centrifugal and reciprocating pumps and study basics on special types of pumps.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	3	-	-	1	-	-	-	-	-	2
CO2	3	-	3	-	1	3	-	-	-	-	-	2
CO3	3	-	3	-	1	3	1	-	-	-	-	3
CO4	3	-	3	-	1	3	1	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Textbook of Fluid Mechanics and Hydraulic Machines	R.K. Bansal	Laxmi Publications (P) Ltd. NewDelhi	11 Edition, 2023
2	Fluid Mechanics and Hydraulic Machines	Rajput.R.K	S Chand Publishing	6th Edition, 2017
3	Hydraulics, fluid mechanics and Hydraulic machinery	P.N. Modi & S.M. Seth	Standard Book House	22nd Edition, 2019
4	Fluid Mechanics and Hydraulic Machines	Mahesh Kumar	Pearson Education	5th Edition, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Mechanics Fundamentals and Applications	Yunus A.Cengel John M. Cimbala	McGraw Hill	4th Edition, 2019
2	Fluid Mechanics	Frank M. White, Henry Xue	McGraw Hill	8th Edition, 2015
3	Fluid Mechanics and Fluid Power Engineering	D.S. Kumar	Kataria & Sons	13th Edition, 2013
4	Instrumentation for Engineering Measurements (Chapter 12 – Fluid Flow Measurements)	James W. Dally, William E. Riley,	Wiley & Sons Inc. 2004	2nd Edition, 2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/105/112105171/
2	https://archive.nptel.ac.in/courses/112/105/112105183/
3	https://archive.nptel.ac.in/courses/103/103/103103147/ https://archive.nptel.ac.in/courses/105/103/105103192/ https://archive.nptel.ac.in/courses/112/103/112103249/
4	https://archive.nptel.ac.in/courses/112/103/112103249/

SEMESTER S3

THERMAL ENGINEERING

Course Code	PCIET303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the fundamental principles of thermodynamics and its applications.
2. To introduce principles and applications of steam engineering.
3. To acquaint the students with working principles of compressors and gas turbines.
4. To demonstrate the fundamental heat transfer principles and to describe the basic concepts of internal combustion engine operation, combustion processes, and alternative fuels.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamental of thermodynamics: Introduction and significance, basic concepts of thermodynamics, laws of thermodynamics, first and second law applications, entropy, availability and unavailability, ideal and real gases, thermodynamic relations.	10
2	Steam engineering: Fundamentals of steam (properties of steam, T-s diagram, Mollier chart), steam power cycles (Rankine cycle, Modified Rankine cycle, relative efficiency, improvement in steam cycles: reheat cycle, regenerative cycle, binary vapour cycle), steam boilers (classification, Cochran boiler, Babcock and Wilcox boiler, boiler mountings and accessories).	12
3	Compressors: Reciprocating air compressors (work, efficiency, volumetric efficiency, clearance effects), Rotary compressors, Centrifugal & Axial compressors.	10

	Gas Turbines: Open & closed cycles, Ideal cycle, Efficiencies, Regeneration, Intercooling, Reheating. (Simple problems)	
4	Heat transfer: Different modes of heat transfer - Fourier law, Newton's law of cooling, Planck's law, Kirchhoff's law, Wien's displacement law and Stefan Boltzmann's law (simple problems). IC engines: Performance testing and Combustion in IC engines - Normal and abnormal combustion in SI and CI engines, Auto ignition, Pre ignition and detonation, Factors affecting detonation, Knocking in engine. Alternate fuels in IC engines: Biodiesel, hydrogen, natural gas, LPG - IC engine pollution and control (basic concepts only).	12

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply fundamental laws of thermodynamic systems, processes, and solve related problems.	K3
CO2	Analyse steam power cycles and boiler systems by applying their knowledge of steam properties, cycle analysis, and boiler operation principles.	K4
CO3	Analyse the performance characteristics of compressors and gas turbines.	K4
CO4	Apply fundamental heat transfer principles, internal combustion engine performance and combustion phenomena in IC engines; understand the characteristics and implications of alternative fuels.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	-	-	-	-	-	-	2
CO2	3	3	-	-	1	-	-	-	-	-	-	2
CO3	3	3	2	-	-	-	-	-	-	-	-	2
CO4	3	2	2	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Thermal Engineering	Ballaney. P. L.	Khanna Publishers	25th 1st Reprint, 2017
2	Steam Table: With Mollier Diagram in S.I.Units	Khurmi. R. S. & Khurmi. N	S Chand Publishing	8th Edition, 2008
3	Engineering Thermodynamics	Nag. P. K	Tata McGraw-Hill Education	6th Edition, 2017
4	Thermal Engineering	Rajput. R. K	Laxmi publications	10th Edition, 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Heat and Mass Transfer	Bergman. T. L., Incropera F. P., DeWitt D. P., Lavine, A. S.	John Wiley & Sons, Inc.	8th Edition, 2017
2	Gas Turbine Theory	Cohen H., Saravanamuttoo H. I. H., Roger G. F. C., Straznicky P. & Nix, A. C	Pearson Education	7th Edition, 2019
3	Fundamentals of IC Engines	Ganesan, V	Tata McGraw Hill	2nd Edition, 2002
4	Thermodynamics	Holman J. P. & Bhattacharyya S	Tata McGraw-Hill	11th Edition, 2011
5	Fundamentals of Engineering Thermodynamics	Moran M. J., Shapiro H. N., Boettner D. D. & Bailey M. B.	John Wiley & Sons, Inc.	8th Edition, 2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/112105123
2	https://archive.nptel.ac.in/noc/courses/noc19/SEM2/noc19-me63/
3	https://archive.nptel.ac.in/courses/112/103/112103262/
4	http://nptel.ac.in/courses/112101097 https://archive.nptel.ac.in/courses/112/103/112103262/

SEMESTER S3

OPERATIONS MANAGEMENT

Course Code	PBIET304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To equip students with a thorough understanding of operations management, enabling them to apply demand forecasting, inventory control, aggregate planning and modern production systems effectively.
2. To familiarise mathematical models and strategic methodologies for optimizing operational processes, enhance efficiency, and make informed decisions within organizational contexts.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Operation Management: Role of Operations in an Organisation, Historical Evolution and perspectives, Operations strategies and productivity in organisations. Demand Forecasting: Forecasting methods: causal and time series models, moving average, exponential smoothing methods. Trend, cycle, and seasonality components, Winter's complete model. Analysis of forecast error, comparison of forecasting methods based on errors. Facility Location and Layout: Factors influencing location planning, Factor rating and center of gravity methods of location selection. Types facility of layouts, comparison of layouts. Systematic Layout Planning (SLP), software packages for SLP.	11

2	Inventory Models: Objectives, Types of Inventories, EOQ and EBQ models for uniform and variable demand with and without shortages. Quantity discount models, Probabilistic inventory models, Inventory systems under risk, service levels, safety stock. Selective Inventory Control: ABC Analysis, XYZ Analysis, VED Analysis.	11
3	Aggregate Planning: Definition, value of decision rules, Pure and Mixed strategies, Mathematical Models for Aggregate planning – Transportation Method, Linear programming Formulation, Linear Decision Rules. Master production schedule: Bill of Material (BOM), structuring BOM,, MRP -Lot sizing methods – Implementation issues, MRP – II, Introduction to ERP.	11
4	Job Shop Production: Activity planning, scheduling, shop loading, sequencing. Priority rules for dispatching jobs. Modern Production Systems: Introduction to Business Process Re-engineering, Lean Manufacturing, Kaizen, Toyota Production System, World Class Manufacturing Concepts. Kanban - Push vs Pull systems, Just-in-time systems.	11

Suggestion on Project Topics

- **Forecasting Model Implementation**
 - Students choose a product or service and apply different forecasting methods (e.g., causal models, time series models, moving average, exponential smoothing) to predict future demand.
- **Location Planning Analysis**
 - Students analyse factors influencing facility location decisions for a given company and apply factor rating and center of gravity methods to determine the optimal location.
- **Layout Design**
 - Students design different types of facility layouts (e.g., process layout, product layout) for a specific business scenario and compare their efficiencies.

- **EOQ/ EBQ Model Development**
 - Students develop advanced EOQ and EBQ models that account for real-world complexities such as variable demand, lead times, and shortages. They implement these models using programming languages like Python or MATLAB.
- **Optimization of Inventory Costs with Quantity Discounts**
 - Students create an optimization algorithm that determines the optimal order quantity considering quantity discounts and other cost factors. They use software like Excel Solver, Python, or MATLAB.
- **MPS Optimization for a Manufacturing Firm**
 - Students design an optimized Master Production Schedule (MPS) for a manufacturing firm using Linear Decision Rules and other relevant methodologies. They implement the optimization in software tools like Excel, Python, or MATLAB.
- **BOM Structuring and MRP System Development**
 - Students develop a comprehensive Bill of Material (BOM) for a complex product and implement a Material Requirements Planning (MRP) system using software tools like MS Project or specialized MRP software.
- **Case Study on Real-World ERP Implementation**
 - Students conduct a detailed case study on a real-world ERP implementation in a large enterprise, analysing the challenges, solutions, and outcomes.
- **Business Process Re-engineering (BPR) Case Study**
 - Students conduct a case study on Business Process Re-engineering (BPR) in a real or hypothetical company. They identify processes for re-engineering, propose improvements, and evaluate potential impacts.
- **Lean Manufacturing Implementation**
 - Students develop a Lean Manufacturing implementation plan for a manufacturing facility. They apply lean principles such as value stream mapping, 5S, and waste reduction techniques.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the role and evolution of operations management and apply its strategic impact on organizational productivity.	K3
CO2	Apply different inventory models and effectively manage inventory to optimize stock levels and reduce risks	K3
CO3	Proficient in developing aggregate plans and applying advanced planning systems for efficient operations.	K3
CO4	Effectively plan and schedule job shop production activities and apply modern production system principles to enhance operational efficiency.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	2	-	1	-	2	2	-	2
CO2	3	3	2	2	2	-	-	-	2	2	2	2
CO3	3	3	3	2	2	-	-	-	2	2	2	2
CO4	3	3	3	2	2	-	-	-	3	2	2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Operations Management: Processes and Supply Chains	Lee J. Krajewski, Manoj Malhotra	Pearson Education	14th Edition, 2024
2	Operations Management	William J. Stevenson	McGraw-Hill Education	14th Edition, 2020
3	Production and Operations Management	R. Paneerselvam	PHI Learning	3rd Edition, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Operations and Supply Chain Management	Cecil B. Bozarth, Robert B. Handfield	Pearson	5th Edition, 2019
2	Operations Management: Sustainability and Supply Chain Management	Jay Heizer, Barry Render, Chuck Munson	Pearson	14th Edition, 2022
3	Operations Management: Processes and Value Chains	Lee J. Krajewski, Larry P. Ritzman, Manoj K. Malhotra	Pearson	10th Edition, 2012
4	Production and Operations Management	S. N. Chary	McGraw-Hill	6th Edition, 2019
5	Operations Management: Theory and Practice	B. Mahadevan	Pearson India	3rd Edition, 2015

Video Links (NPTEL, SWAYAM...)	
Sl No	Link ID
1	https://archive.nptel.ac.in/courses/112/107/112107238/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to AI and Machine Learning: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11

2	Mathematical Foundations of AI and Data science: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	11
3	Applied Probability and Statistics for AI and Data Science: Basics of probability-random variables and statistical measures - rules in probability-Bayes theorem and its applications- statistical estimation-Maximum Likelihood Estimator (MLE) - statistical summaries- Correlation analysis-linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	11
4	Basics of Data Science: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	11

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	K3
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	K3
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	K3
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	-	-	-	-	-	-	-	-
CO2	3	3	3	3	-	-	-	-	-	-	-	-
CO3	3	3	3	3	-	-	-	-	-	-	-	-
CO4	3	3	3	3	-	-	-	-	-	-	-	-
CO5	3	3	3	3	-	-	-	-	-	-	-	-

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6 th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media, Inc.	2 nd edition, 2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1 st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9 th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1 st edition, 2016

Reference Books				
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2 nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1 st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1 st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1 st edition, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1 st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmn nnibpcajpcglclefindm kaj/https://www.math. arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP_{fc} from GDP_{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP_{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, $NFIA=(-200)$ and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism , Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution -Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places -accessibility and social impacts, Managing conflict , Collective bargaining, Confidentiality , Role of confidentiality in moral integrity, Codes of Ethics . Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education , employment and everyday life, History of women in Science & Technology, Gendered	6

	technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and	6

	regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

FLUID MECHANICS AND HYDRAULIC MACHINES LAB

Course Code	PCIEL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To familiarize practical fluid discharge measurements.
2. To practice performance evaluations of pumps and turbines.

Expt. No.	Experiments
1	Determination of coefficient of discharge and calibration of Notches.
2	Determination of coefficient of discharge and calibration of Orifice meter.
3	Determination of coefficient of discharge and calibration of Venturi meter.
4	Determination of hydraulic coefficients of orifices.
5	Determination of Chezy's constant and Darcy's coefficient on pipe friction apparatus.
6	Experiments on hydraulic ram.
7	Reynolds experiment.
8	Determination of metacentric height and radius of gyration of floating bodies.
9	Performance test on positive displacement pumps.
10	Performance test on centrifugal pumps, determination of operating point and efficiency.
11	Performance test on Impulse turbines.
12	Performance test on reaction turbines (Francis and Kaplan Turbines).

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Determine CD of notches/orifice/venturi meter	K3
CO2	Calibrate notches, orifice meter and Venturi meter	K3
CO3	Evaluate the losses in pipes	K3
CO4	Find the metacentric height and differentiate flows	K3
CO5	Evaluate performance of pumps and turbines	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	-	-	-	2	3	2	-	2
CO2	2	1	-	-	-	-	-	2	3	2	-	2
CO3	2	1	-	-	-	-	-	2	3	2	-	2
CO4	2	1	-	-	-	-	-	2	3	2	-	2
CO5	2	1	-	-	-	-	-	2	3	2	-	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Mechanics and Hydraulic Machines	R.K. Bansal	Laxmi Publications (P) Ltd. NewDelhi	11 Edition, 2023
2	Hydraulics, FM and HM	P.N. Modi & S.M. Seth	Standard Book House	22nd Edition 2019
3	Fluid Mechanics and Hydraulic Machines	Rajput.R.K	S. Chand Publishing	6th Edition 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Mechanics Fundamentals and Applications	Yunus A.Cengel John M. Cimbala	McGraw Hill	4th Edition, 2019
2	Fluid Mechanics	Frank M. White, Henry Xue	McGraw Hill	8th Edition, 2015
3	Fluid Mechanics and Fluid Power Engineering	D.S. Kumar	Kataria & Sons	13th Edition, 2013
4	Instrumentation for Engineering Measurements (Chapter 12 – Fluid Flow Measurements)	James W. Dally, William E. Riley,	Wiley & Sons Inc. 2004	2nd Edition, 2010

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

THERMAL ENGINEERING LAB

Course Code	PCIEL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To familiarize the performance characteristics of internal combustion engines and compressors based on experimentation.
2. To practice analysis of exhaust and flue gases, determination of thermal properties of materials and fluids, and determination of heat transfer parameters.

Expt. No.	Experiments
1	Performance test on petrol engines / MPFI engine.
2	Performance test on Diesel engines / Turbocharged engine.
3	Retardation test on IC engines.
4	Morse test on petrol engine.
5	Heat Balance test on petrol/diesel engines.
6	Determination of volumetric efficiency and air-fuel ratio of IC engines.
7	Performance test on reciprocating compressor.
8	Performance test on rotary compressor/blower.
9	Analysis of automobile exhaust gas and flue gas using exhaust gas analyser.
10	Cooling curve of IC engines.
11	Determination of flash and fire points of petroleum products -flash and fire point apparatus.
12	Determination of viscosity of lubricating oil- viscometer.
13	Determination of thermal conductivity of solids (composite wall/metal rod).
14	Determination of heat transfer coefficients in free convection.
15	Determination of emissivity of a specimen.

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse the performance of internal combustion engines and compressors	K4
CO2	Apply thermophysical properties of fluids and materials	K3
CO3	Analyse combustion emissions and engine cooling characteristics	K4
CO4	Examine heat transfer phenomena relevant to IC engine	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	2	-	-	-	2	2	-	2
CO2	3	3	-	2	2	-	-	-	2	2	-	2
CO3	3	3	-	2	2	-	2	-	2	2	-	2
CO4	3	3	2	2	2	-	2	-	2	2	-	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Thermal Engineering	Ballaney. P. L.	Khanna Publishers	25th 1st Reprint, 2017
2	Steam Table: With Mollier Diagram in S.I.Units	Khurmi. R. S. & Khurmi. N	S Chand Publishing	8th Edition, 2008
3	Engineering Thermodynamics	Nag. P. K	Tata McGraw-Hill Education	6th Edition, 2017
4	Thermal Engineering	Rajput. R. K	Laxmi publication	10th Edition, 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	IC Engine fundamentals	J.B.Heywood	McGraw-Hill	2nd Edition, 2019
2	Internal Combustion Engines	V. Ganesan	Tata McGraw-Hill	4th Edition, 2012
3	An Introduction to Combustion: Concepts and Applications	Stephen R Turns	McGraw-Hill	4th Edition, 2021
4	Thermal Engineering	Ballaney, P. L.	Khanna Publishers	1st Reprint, 2017
5	Heat and mass transfer	Rajput, R. K	S.Chand & Co	4th Edition, 2015
6	Thermal Engineering	Rajput, R. K.	Laxmi publications	10th Edition, 2018

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

INDUSTRIAL ENGINEERING

SEMESTER S4
MATHEMATICS FOR PHYSICAL SCIENCE– 4
(C Group)

Course Code	GCMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hr. 30 Min.
Prerequisites (if any)	Basic calculus.	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To provide the students with the basics of various numerical methods to develop problem solving skills used in various engineering disciplines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9

2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. [Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]	9
3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and t -distribution), Hypotheses and Test Procedures, Type I and Type II error, z Tests for Hypotheses about a Population Mean (for large sample), t Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. [Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]	9
4	Newton-Raphson Method, Gauss Elimination Method, Gauss - Jordan Method, Numerical solution of ordinary differential equations-Euler's method, Modified Euler's method, Runge - Kutta method of 2 nd Order, Numerical solution of Laplace equation –Jacobi's Method, Curve Fitting by Method of Least Squares - Straight lines, Parabola. (Text 2: Relevant topics from sections 2.5,4.2,7.5,8.4,8.5,9.4)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	K3
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	K3
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using z-tests and the one-sample <i>t</i> -test.	K3
CO4	Apply numerical methods to find solutions of linear system of equations, ordinary differential equations and Laplace equations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Introductory Methods of Numerical Analysis	S SSastry	PHI Learning Pvt Limited	5 th edition, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4 th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6 th edition, 2020
3	Numerical methods for Engineers	Steven C. Chapra, Raymond P. Canale	McGraw Hill Education	8 th edition, 2021

Video Links (NPTEL, SWAYAM...)

Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
3	https://archive.nptel.ac.in/courses/117/105/117105085/
4	https://archive.nptel.ac.in/courses/111/107/111107105/

SEMESTER S4

MACHINE TOOLS AND DIGITAL MANUFACTURING

Course Code	PCIET402	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To equip students with the foundational knowledge of metal cutting processes, including tool nomenclature, chip formation, and cutting force analysis.
2. To provide hands-on experience and theoretical understanding of operating conventional machine tools and CNC machines, emphasizing their construction, operation, and programming.
3. To familiarize students with the architecture and functions of digital manufacturing systems, Programmable Logic Controllers (PLCs), and the key principles of Industry 4.0, fostering their ability to implement advanced manufacturing strategies

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Metal Cutting: Overview of metal cutting processes, Tool Nomenclature, Attributes of each tool nomenclature, Standardised systems of nomenclature of single point cutting tool like ASA, ORS, Case studies and examples with varying tool attributes to analyse their impact on surface roughness and material removal rate (MRR), Orthogonal and Oblique Cutting. Mechanism of metal removal, Primary and secondary deformation shear zones, Mechanism of chip formation, Types of chips and types of chip breakers. Merchant's theory Friction forces in metal cutting, Analysis of cutting	11

	forces (simple problems), Work done and power required. Cutting Tool Materials, Properties and applications of various tool materials, Advanced tool materials and coatings, Thermal Aspects of Machining, Techniques for thermal management in machining. Tool Wear and Mechanisms of tool wear, Types of tool wear, Role and types of cutting fluids. Tool life, Factors affecting tool life, Taylor's tool life equation (simple problems), Economics of Machining, case studies on machining economics.	
2	Conventional Machine Tools: Lathes Machine– Types of lathes and size, specification and parts, Work holding devices. Types of operations in a lathe, plain turning, taper turning, thread cutting and facing operations. Turning Centres, C axis machining, Vertical/Inclined turrets. Milling machines - Types of milling machines, Milling operations, end milling and slot milling. Types of milling cutters, up milling and down milling. Work holding devices, Indexing and indexing heads, 5 and 6 axis milling. Reciprocating machines- shapers, planers, slotting machines, broaching machines, uses and specific applications. Grinding machines – Classification – Operations – Surface, cylindrical and centreless grinding machines.	11
3	Computer Numeric Controlled Machines: NC and CNC machines, classification and principle, Point to point, straight cut and contouring positioning - incremental and absolute systems – open loop and closed loop systems -Parts: Positional Encodes, axis and spindle drives, LM Guideways, ball screws, automatic tool changer(ATC). NC part programming: part programming fundamentals - manual	11

	programming –NC coordinate systems and axes – sequence number, preparatory functions, dimension words, speed word, feed word, tool world, miscellaneous functions, Tool nose radius compensation and constant surface speed control, Canned cycles and Multiple repetitive cycles. Programmable Logic Controllers (PLC): Need – relays - logic ladder program –timers, simple problems only.	
4	Digital Manufacturing: Definition and Illustration of Digital Manufacturing, Operation reference mode of digital manufacturing system, Architecture of digital manufacturing system, General function structure model of digital manufacturing system. Core pillars of Computer Integrated Manufacturing, Flexible manufacturing Cell (FMC), Automated Guided vehicles (AGV), Automated Storage and Retrieval Systems (ASRS), Industrial robots-major configurations and applications. PLM and its implementation, Steps in PLM, manufacturing data and its use, Overview of Industry 4.0.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Understand the different types of conventional and CNC machine tools, their parts, and their specific applications.	K2
CO2	Explain the mechanisms of metal removal, chip formation, tool wear and the role of tool angles in determining surface roughness and material removal rate (MRR).	K2
CO3	Apply knowledge to create and interpret part programs for CNC machines, including manual programming, canned cycles, and multiple repetitive cycles.	K3
CO4	Evaluate the economics of machining through case studies, considering factors such as cutting forces, tool life, and material costs to make informed decisions on machining strategies.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-		-	-	-	-	-	-	-	-	2
CO2	3	3		-	-	-	-	-	-	-	-	2
CO3	3	3	3	-	3	-	-	-	-	-	-	2
CO4	3	3	-	3	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing Process	H. N. Gupta, R.C. Gupta Arun Mittal	New Age International	2nd Edition, 2009
2	Computer control of manufacturing systems	Yoram Koren	TATA McGraw-Hill	1st Edition, 1983
3	Fundamentals of Digital Manufacturing Science	Zude Zhou, Shane (Shengquan) Xie, Dejun Chen	Springer London	1st Edition, 2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Machining and Machine Tools	Geoffrey Boothroyd, Winston A. Knight	CRC Press Inc.	3rd Edition, 2005
2	CAD/CAM Computer Aided Design and Manufacturing	P.Groover, E. M. Zimmers	Pearson Education	1st Edition, 2014
3	Programmable logic controllers	Petruszella Frank. D	McGraw-Hill	2nd Edition, 2006

Video Links (NPTEL, SWAYAM...)	
Link ID	https://onlinecourses.nptel.ac.in/noc24_me123/preview
	https://onlinecourses.nptel.ac.in/noc24_me100/preview

SEMESTER S4

WORK SYSTEM DESIGN

Course Code	PCIET403	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise concepts of work system design in organisations.
2. To enable students to utilize techniques in work system design for productivity improvement in organisations.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Productivity and Work Systems: Concepts, definition and importance of productivity, holistic approach to productivity, productivity and living standards, measuring productivity, fractional and multifactor productivity, work systems and productivity, techniques for productivity improvement, techniques in work study for productivity improvement, effect of workplace environment and workplace design in productivity improvement.	11
2	Method Study: Definition and objectives of method study, procedure of method study, selection of work, recording the work and activities, classification of activities, representation of activities, use of different charts and diagrams, examining the work, developing methods, evaluation and implementation of method study.	11

3	Work Measurement and Standard Time: Definition and objectives of work measurement, basic procedure of stop watch time study, classification of elements, breaking the job into elements, determination of sample size, performance rating systems, calculation of allowances and estimation of standard time, development of incentive schemes with examples.	11
4	Indirect methods of work measurement: Work sampling, determination of sample size, conducting work sampling study, calculation procedures, MTM and MOST in work measurement. Micro motion study: Therbligs, film analysis and SIMO chart preparation, principles of motion economy pertaining to worker, workplace and tools, worker, applications of these principles in design.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Make use of productivity concepts in an industry and identify the importance of work study and workplace environment	K3
CO2	Apply method study concepts and identify its use in productivity improvement	K3
CO3	Apply time study concepts of work measurement and identify the use of standard time in development of incentive schemes	K3
CO4	Utilize PMTS and Work sampling concepts in work measurement and identify the role of principles of motion economy in workplace design	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	-	-	-	-	-	-	2
CO2	3	2	1	-	-	-	-	-	-	-	-	2
CO3	3	2	1	-	-	-	-	-	-	-	-	2
CO4	3	2	2	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to work study	ILO	Oxford and IBH	3rd Edition, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Motion and time study design and measurement of work	Ralph M.Barnes	Wiley India Private Limited	7th Edition, 2013
2	Work systems: The Methods, Measurements and Management of works	Mikell P. Groover	Pearson International	1st Edition, 2013
3	Niebel's Methods, Standards and Work Design	Andris Freivalds, Benjamin Niebel	McGraw Hill	13th Edition, 2013

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107249/

SEMESTER S4

THEORY OF MACHINES AND DESIGN

Course Code	PBIET404	CIE Marks	60
Teaching Hours/Week (L:T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To introduce different mechanisms and enable the students to do the kinematic analysis of planar mechanisms.
2. To discuss the theory of different machine elements in order to carry out the design.
3. To discuss the theories of failure and enable the students to do the strength based design for some practical applications such as shafts, gears and structural joints viz. riveted, bolted and weld joints at the basic level.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Kinematics and Mechanisms: Mechanisms such as Quick return motion, steering mechanism, universal coupling. Kinematic analysis of planar mechanisms: Kinematic chain, kinematic diagrams, Degrees of freedom, Grashof's condition, mechanical advantage; Position vector loop, velocity and acceleration diagrams of four bar chain and slider crank mechanisms.	10

2	Cams: Radial cam- terminology; displacement, velocity, acceleration diagrams- for uniform velocity, uniform acceleration and SHM. Cam profile synthesis (knife edge follower). Friction clutch- Design of plate friction clutch; Belt drive power transmission - open and crossed belt. Brakes and dynamometers (no derivations): simple band brake, block brake, disk brake; Rope brake dynamometer.	11
3	Introduction to stress analysis: Behaviour of materials- elastic and plastic, Concept of Stress and Strain, Hooke's law (2D), principal stresses, maximum shear stress, bending and torsional stresses, combining stresses (2D), Mohr's circle, stress concentration factor; Stresses in pressure vessels. Design for strength: Theories of failure- Max. normal stress, Max. shear stress theory and von Mises theory; factor of safety. Failure criteria for fatigue, endurance limit. Design of Shafts (with single pulley/gear mounted on it), key.	12
4	Gears: Types, Gear terminology, pressure angle, law of gearing, Gear trains- simple, compound and planetary, Lewis equation. Design of spur gears. Design of structural joints: riveted joints, bolted joints and weld joints- butt and lap fillet welds.	11

Suggestion on Project Topics: Each student team can choose some simple close to real life projects from the areas discussed in the syllabus such as:

- Development and kinematic analysis of simple mechanisms.
- Cam profile synthesis.
- Design of a brake, clutch, shaft or spur gear using a design data book.
- Design of structural joints for some simple practical applications.
- Some simple case studies to understand and apply the theory/concepts discussed and learned during the lecture hours.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Analyse simple mechanisms using principles of kinematics.	K4
CO2	Execute the synthesis of radial cam profile and design of machine elements working on dry friction.	K3
CO3	Apply the principles of stress analysis to design components to resist failure by static loads.	K3
CO4	Apply principles of design for spur gear and structural joints.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	2	-	-	-	2	2	-	1
CO2	3	2	3	-	-	-	-	-	2	2	-	1
CO3	3	2	3	-	2	3	-	-	2	2	-	3
CO4	3	2	3	-	2	2	-	-	2	2	-	1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory of machines and mechanisms	Ballaney. P. L.	Khanna Publishers	21 st Edition, 2020
2	Theory of Machines	S. S. Rattan	Tata McGraw-Hill Education	5 th Edition, 2019
3	Design of Machine Elements	Bhandari. V. B.	Tata McGraw-Hill Education	5 th Edition, 2020

Design Data Books (Permitted for reference in the university examination)				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design Data Handbook.; Mechanical Engineers in SI and Metric Units.	Mahadevan, K., and K. Balaveera Reddy	CBS Publishers & Distributors	4 th Edition, 2013
2	Design Data: Data Book of Engineers	Faculty in ME, PSG College	Kalaikathir Achchagam	4 th Edition, 2020
3	Machine Design Data Handbook	NarayanaIyengar B.R & Lingaiah K	Tata McGraw Hill/Suma Publications	4 th Edition, 2006

Reference Books				
1	Theory of Mechanisms and Machines	Amithabha Ghosh and Asok Kumar Malik	East West Press	3 rd Edition, 2011
2	Theory of Machines: Kinematics and Dynamics	Sadhu Singh	Pearson Education	3 rd Edition, 2011
3	Theory of Machines	John J. Uicker Gordon R. Pennock, Joseph E. Shigley	Oxford University Press	5 th Edition, 2017
4	Machine Design	P C Sharma and D K Aggarwal	S K Kataria & Sons	1997
5	Design of Machinery	Robert L Norton	Tata McGraw-Hill	2005
6	Shigley's Mechanical Engineering Design	Richard G. Budynas, J. Keith Nisbett, Kiatfa Tangchaichit	The McGraw-Hill companies	11 th Edition, 2020

Video Links (NPTEL, SWAYAM...)

Module No.	Link ID
1	https://nptel.ac.in/courses/112104121 (Kinematics of Machines by Prof. Ashok K Mallik, IIT Kanpur)
2	https://archive.nptel.ac.in/courses/112/105/112105124/ (Design of Machine Elements-I, IIT Kharaghpur)
3	https://archive.nptel.ac.in/courses/112/105/112105124/ (Design of Machine Elements-I, IIT Kharaghpur)
4	https://archive.nptel.ac.in/courses/112/105/112105124/ (Design of Machine Elements-I, IIT Kharaghpur)

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S4

ROBOTICS AND AUTOMATION

Course Code	PEIET411	CIE Marks	40
Teaching Hours/Week (L:T: P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the concepts and applications of hydraulic and pneumatic automation.
2. To enable students to develop automation logic using PLC and other techniques.
3. To familiarise the basics of robotics and its applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to automation: classification of drives-hydraulic and pneumatic-comparison. ISO symbol of fluid power generating elements-hydraulic pumps and motor gears, vane pumps, piston pumps. Control and regulating elements: Pressure, flow and direction control valves, spool valves, electro hydraulic servo valves.	9
2	Electrical control of pneumatic and hydraulic circuits: interfacing with PLC, ladder diagrams, use of relays, timers and counters. Typical Design methods of automation-ladder diagram, circuit sequencing, combinational logic design, cascading, Karnaugh map method.	9

3	Introduction to robotics: Drive systems: AC and DC servo and stepper motors. Actuators: Pneumatic and hydraulic actuators, Sensors: Proximity and range sensors, force and torque sensors, velocity sensors, encoders, vision sensors. Classification of robots: Work envelope, Classification of work cell layout. Consideration in work cell design. Flexible automation vs robotic technology.	9
4	Industrial application of robots: Spray painting, spot-welding, arc welding, drilling, assembly operation, loading unloading, role of a robot in a manufacturing cell. Safety considerations.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Understand the basics of hydraulic and pneumatic automation.	K2
CO2	Develop automation circuits for various industrial applications using control valves.	K3
CO3	Develop automation logic using PLC and other techniques.	K3
CO4	Understand the basics of Robotics.	K2
CO5	Understand the application of robotics in various industries.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	3
CO2	3	2	-	-	-	-	-		-	-	-	3
CO3	3	2	-	-	-	-	-		-	-	-	3
CO4	3	2	-	-	-	-	-		-	-	-	3
CO5	3	2	-	-	-	-	-		-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hydraulics and Pneumatics	Jagadeesha T.	I K International Publishing House	10th Edition, 2015
2	Fluid Power with application	Antony Esposito	Prentice-Hall	7th Edition, 2008
3	Fluid power transmission a and control	Alavudeen A.	Charotar Publishing House	1st Edition, 2008
4	Introduction to Robotic Mechanic and Control	John J. Craig	Pearson Education International	4th Edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Power Logic Circuit Design	Peter Rohner	MacMillan Press	1st Edition,1994
2	Introduction to Fluid Logic	E.C.Fitch and J.B.Surjaatmadja	McGrawHill	1st Edition,1978
3	Robot Dynamics and Control	Mark.W.Spong&M.Vidyasagar	John Wiley and Sons	1st Edition, 2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108105063
2	https://nptel.ac.in/courses/108105088
3	https://nptel.ac.in/courses/112105249
4	https://onlinecourses.nptel.ac.in/noc23_de12

SEMESTER S4

INDUSTRIAL REFRIGERATION AND

AIR CONDITIONING

Course Code	PEIET412	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of the principles and applications of refrigeration and air conditioning systems.
2. To familiarise fundamental concepts of refrigeration systems, psychrometry, and air conditioning system design.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Refrigeration: History, applications, and importance of refrigeration. Thermodynamics of Refrigeration: Reversed Carnot cycle, heat pumps, refrigeration machines, limitations of the reversed Carnot cycle. Units of Refrigeration: Definition and practical examples. Air Refrigeration Systems: Reversed Joule cycle, aircraft refrigeration systems (simple, bootstrap, regenerative, and reduced ambient systems). Special Refrigeration Systems: Vortex tube refrigeration, Thermoelectric Refrigeration (Peltier Effect), Magnetic Refrigeration, Steam Jet Refrigeration.	9

2	Vapour Compression Systems: Simple cycle, representation on T-s and P-h diagrams, Coefficient of Performance (COP), and methods to improve COP (superheating, undercooling, liquid suction heat exchanger). Multi-pressure Systems: Multi-compression and multi-evaporator systems, inter-cooling, flash intercooling, flash gas removal, cascade systems. Refrigerants: Properties, eco-friendly refrigerants, mixed refrigerants, selection criteria for different applications.	9
3	Vapour Absorption Systems: Ammonia-water system, Lithium bromide-water systems, Electrolux systems, comparison with vapour compression systems and drawbacks. Applications of Refrigeration: Domestic refrigerators, water coolers, ice plants, cold storage, food preservation (plate freezing, quick-freezing). Refrigeration System Components: Compressors, condensers, expansion devices, evaporators, cooling towers (types and applications), refrigerant leakage detection, refrigerant charging, and system controls.	8
4	Psychrometry: Properties of air (saturated, unsaturated, dry, wet, dew point, humidity levels), thermodynamic equations, adiabatic saturation process, psychrometric charts, processes (sensible heating/cooling, humidifying/dehumidifying), air washers, bypass factors, sensible heat factor, RSHF, GSHF. Air Conditioning Design: Comfort and Industrial air conditioning. Comfort air conditioning factors, effective temperature, comfort charts, cooling load estimation, design conditions, supply conditions, fresh air supply. Air Conditioning Systems: Room air conditioners, split systems, packaged systems, all-air systems, chilled water systems, winter and summer air conditioning, heating systems, humidifiers, year-round AC system and controls (thermostat, humidistat), air distribution systems (duct design), air conditioning for various environments (restaurants, hospitals, retail outlets, computer centers, theaters, industrial applications).	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals and thermodynamic principles of refrigeration systems.	K3
CO2	Analyse vapour compression systems and select appropriate refrigerants for various applications.	K4
CO3	Understand the working of vapour absorption refrigeration and to explore the various applications of refrigeration, and study the components of refrigeration systems.	K3
CO4	Apply psychrometric principles to analyse and design air conditioning systems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1	-	-	-	-	-	2	3
CO2	3	2	3	2	1	-	-	-	-	-	2	3
CO3	3	2	3	2	1	-	-	-	-	-	2	3
CO4	3	3	2	2	1	2	-	-	-	2	3	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Refrigeration and Air Conditioning Technology	Bill Whitman, Bill Johnson, John Tomczyk	Cengage Learning	8th Edition, 2023
2	Refrigeration and Air Conditioning	G.F. Hundy, A.R. Trott, T.C. Welch	Butterworth-Heinemann	5th Edition, 2021
3	Refrigeration and Air Conditioning	C.P. Arora	McGraw Hill Education	5th Edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Refrigeration and Air Conditioning	Andrew D. Althouse, Carl H. Turnquist, Alfred F. Bracciano	Goodheart-Willcox	21st Edition, 2022
2	Air Conditioning Engineering	W.P. Jones	Arnold	5th Edition, 2013
3	Refrigeration and Air Conditioning Technology Handbook	J. Whitfield, P. Dincer	CRC Press	2nd Edition, 2020
4	Handbook of Air Conditioning and Refrigeration	Shan K. Wang	McGraw-Hill	3rd Edition, 2017
5	Practical Guide to Refrigeration and Air Conditioning	Minnick & Associates	Delmar Cengage Learning	2nd Edition, 2016

6	Fundamentals of HVACR	Carter Stanfield, David Skaves	Pearson	3rd Edition, 2016
7	Air Conditioning and Refrigeration Repair	Roger A. Fischer, Ken Chernoff	McGraw-Hill	2nd Edition, 2014
8	Textbook of Refrigeration and Air Conditioning	R.K. Rajput	S. Chand Publishing	6th Edition, 2016
9	Principles of Refrigeration	Roy J. Dossat	Pearson Education	5th Edition, 2019
10	Air Conditioning Principles and Systems	Edward G. Pita	Prentice Hall	4th Edition, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	http://nptel.ac.in/courses/112105128/
2	https://onlinecourses.nptel.ac.in/noc24_me77/preview
3	https://nptel.ac.in/courses/112105129
4	https://archive.nptel.ac.in/courses/112/107/112107208/

SEMESTER S4

DYNAMICS OF MACHINERY

Course Code	PEIET413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the rigid- body dynamics of kinematically driven machine components and perform the force analysis
2. To equip the students to analyze the effects of gyroscopic forces on vehicle stability, to control power and speed fluctuations in IC engines, and reduce machinery vibrations.
3. To outline the theoretical concepts of vibrations and to enable the students to perform the vibration analysis of single degree of freedom (SDOF) systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Mechanisms and machines: Mechanisms such as four bar chain, slider crank, quick return motion, steering mechanism, universal coupling. Static & Dynamic Force analysis: Static equilibrium conditions, Two & three force members, four bar and slider crank mechanisms, method of virtual work; Force & moment equilibrium, Inertial forces, D' Alembert's principle. Dynamic analysis of four-bar and slider-crank mechanisms.	9

2	Gyroscopic couple: gyroscopes, effect of gyroscopic couple on - bearings, stability of two wheel and four wheel vehicles. Balancing of rotating masses: single mass, several masses in same and in different planes; Balancing of reciprocating masses, partial balancing of single cylinder and multi-cylinder inline engines	9
3	Flywheels: Applications in punching press, IC engines; turning moment diagrams-for four stroke IC engines; Coefficient of fluctuation of energy, speed; design of flywheels. Governors: Types of governors, Watt, Porter and Proell governors-Isochronism, sensitivity and stability	9
4	Free vibrations: Undamped free vibrations of SDOF systems - natural frequency of longitudinal, transverse and torsional vibrations by Newton's method and energy method; Damped systems and critical damping, logarithmic decrement. Forced vibrations: Forced harmonic vibration, rotating unbalance, magnification factor, resonance; whirling of rotating shafts, vibration isolation and transmissibility.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Understand the functions of mechanisms and execute the force analysis of kinematic systems.	K3
CO2	Examine the balancing of rotating and reciprocating masses in machines; determine the gyroscopic effect on stability of vehicles.	K3
CO3	Understand the principle of governors/ flywheel for the control of speed/power fluctuation in engines or machines and execute its analysis and design.	K3
CO4	Determine the natural frequencies and examine free or harmonically excited vibration response of damped and undamped vibration.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	-	-	-	-	-	-	3
CO2	3	2	3	-	-	-	-	-	-	-	-	3
CO3	3	2	3	-	-	2	-	-	-	-	-	3
CO4	3	2	3	-	3	2	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory of Machines and Mechanisms	Ballaney P. L.	Khanna Publishers	21 st Edition, 2020
2	Theory of Machines	S. S. Rattan	Tata McGraw-Hill Education	4 th Edition, 2017
3	Theory of vibration with applications	W. T Thomason	CBS Publishers	1 st Indian Edition, 1990

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory of Mechanisms and Machines	Amithabha Ghosh and Asok Kumar Malik	East West Press	3rd Edition, 2011
2	Kinematics and dynamics of machinery	Robert L Norton	McGraw-Hill Education	1st Edition in SI units 2009
3	Theory of Machines	Thomas Bevan	Pearson Education Ltd.	3rd Edition, 2010

4	Theory of Machines and Mechanisms	John J. Uicker Gordon R. Pennock, Joseph E. Shigley	Oxford University Press	5th Edition, 2017
5	Mechanical vibrations and Industrial noise control	Lasithan L G	PHI Learning	1 st Edition, 2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/112104121 https://archive.nptel.ac.in/courses/112/104/112104114/
2	https://archive.nptel.ac.in/courses/112/104/112104114/
3	https://archive.nptel.ac.in/courses/112/104/112104114/
4	https://archive.nptel.ac.in/courses/112/107/112107212/

SEMESTER S4

MODERN AUTOMOTIVE TECHNOLOGIES

Course Code	PEIET414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide a comprehensive understanding of modern automotive technologies, focusing on electric and hybrid vehicles and recent advancements in automotive power plants.
2. To familiarise advanced developments in autonomous vehicle technologies and fuel cells.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Electric and Hybrid Vehicle Technology: Introduction to Vehicle Electrification: Low Emission Vehicle (LEV), Transitional Low Emission Vehicle (TLEV), Ultra Low Emission Vehicle (ULV), and Zero Emission Vehicle (ZEV). Basic components of electric vehicles: Inverters, battery packs, battery management systems, motors, and electronic power control units. Electric wiring harness: CAN Bus, Multiplex wiring. Regenerative braking. Factors to consider for converting automobiles to electric vehicles Hybrid Electric Vehicles: Types: Series and parallel hybrid layouts, Comparison of power systems and control systems, Different modes of operation for best performance, Wireless Charging, Battery Recycling and Lifecycle Management	8

2	Recent Trends in Automotive Power Plants: Stratified charge engines, Reactivity Controlled Compression Ignition engines (RCCI), Hydrogen engines, Flex fuel vehicles. Biofuels and synthetic fuels Vehicle Operation and Engine Control Systems: Application of sensors, actuators, and microprocessors for optimal fuel economy and performance, electronic engine management systems, electronic cruise control; and chassis control systems Integrated Control Systems (Basics): Advanced integration of vehicle control systems for enhanced performance and safety, AI and Machine Learning in engine control, Predictive maintenance systems.	9
3	Autonomous Vehicle Technologies: Autonomous vehicle levels (0-5), Sensor technologies (LiDAR, radar, cameras, ultrasonic sensors) Driver Assistance Systems (ADAS): Vision systems in cars, Comprehensive driver assistance: lane recognition, traffic sign recognition, road recognition, and object recognition. Connected Vehicles and IoT (Basics): Vehicle-to-Vehicle (V2V) communication, Telematics and applications, Cybersecurity in connected vehicles, Over-the-air (OTA) updates, 5G in automotive communication, Human-Machine Interfaces (HMI), Role of AI and ML in autonomous vehicles	9
4	Advanced Electronics and Microcontroller Systems in Modern Vehicles: Electronically controlled systems: Headlights, Sunroofs, Electro chromic mirrors, Head-up displays, Navigation systems, Automatic climate control, Anti-theft systems, Automatic door locks, Tyre pressure sensing, Automated wipers, Anti Lock braking system (ABS), Electronic Stability Control (ESP). Fuel Cells and Alternative Energy Systems: Introduction to fuel cells: Operational voltages, types (Proton Exchange membrane fuel	10

	cells, Alkaline fuel cells, Medium and High temperature fuel cells), Fuel choices, Fuel Processing, Fuel Stacks, Air supply, Auxiliary systems. Advanced Battery Technologies: Solid-state batteries, Fast-charging technologies, Second-life applications of EV batteries. Sustainable Automotive Materials: Use of lightweight materials, Recycling and end-of-life vehicle strategies, Vehicle-to-Grid (V2G) technology	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Understand the basic concepts and identify the components of electric and hybrid vehicles.	K3
CO2	Analyze advanced automotive power plants, control systems, and integrated vehicle control technologies to enhance performance, safety, and fuel economy.	K4
CO3	Apply the technologies used in autonomous and connected vehicles and their impact on vehicle safety and efficiency.	K3
CO4	Apply knowledge of modern automotive electronics and alternative energy systems to innovate and improve vehicle systems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	2	-	1	-	-	-	-	1
CO2	3	3	2	2	3	-	-	-	-	-	-	-
CO3	3	3	3	2	3	2	1	2	-	-	-	2
CO4	3	2	2	-	3	2	1	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electric and Hybrid Vehicles: Design Fundamentals	Iqbal Husain	CRC Press	3rd Edition, 2021
2	Advanced Engine Performance Diagnosis	James D. Halderman	Pearson	6th Edition, 2020
3	Automotive Embedded Systems Handbook	Nicolas Navet, Francoise Simonot-Lion	CRC Press	1st Edition, 2017
4	Automotive Handbook	Bosch	Robert Bosch GmbH	10th Edition, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electric Vehicle Engineering	Perry MacMullen	McGraw-Hill Education	1st Edition, 2021
2	Build Your Own Electric Vehicle	Bob Brant, Seth Leitman	McGraw-Hill	3rd Edition, 2013
3	Modern Electric, Hybrid Electric, and Fuel Cell Vehicles	Mehrdad Ehsani, Yimin Gao, Stefano Longo, Kambiz Ebrahimi	CRC Press	3rd Edition, 2018

4	Electric and Hybrid Vehicles: Technologies, Modeling, and Control - A Mechatronic Approach	Amir Khajepour, M. Saber Fallah, Avesta Goodarzi	Wiley	1st Edition, 2014
5	Automotive Control Systems: For Engine, Driveline, and Vehicle	Uwe Kiencke, Lars Nielsen	Springer	2nd Edition, 2020
6	Understanding Automotive Electronics	William B. Ribbens	Butterworth-Heinemann	8th Edition, 2017
7	Automotive Electrical and Electronic Systems	Tom Denton	Routledge	5th Edition, 2019
8	Automotive Mechatronics: Operational and Practical Issues	Bosch (Editor)	Springer	1st Edition, 2018
9	Connected and Autonomous Vehicles: Technologies, Applications, and Challenges	Rajkumar Buyya, Amir Vahid Dastjerdi	Springer	1st Edition, 2021
10	Autonomous Driving: Technical, Legal and Social Aspects	Markus Maurer, J. Christian Gerdes, Barbara Lenz, Hermann Winner	Springer	1st Edition, 2016
11	Automotive Electricity, Electronics and Computer Controls	Barry Hollembeak	Cengage Learning	6th Edition, 2018
12	Fuel Cell Fundamentals	Ryan O'Hayre, Suk-Won Cha, Whitney Colella, Fritz B. Prinz	Wiley	3rd Edition, 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	http://nptel.ac.in/courses/108103009/
	https://onlinecourses.nptel.ac.in/noc23_ee01/preview
	https://archive.nptel.ac.in/courses/108/106/108106170/
2	https://archive.nptel.ac.in/courses/106/106/106106139/
	https://onlinecourses.nptel.ac.in/noc21_ee32/preview
	https://archive.nptel.ac.in/courses/108/108/108108123/
3	https://archive.nptel.ac.in/courses/108/108/108108098/
	http://www.digimat.in/nptel/courses/video/108108123/L18.html
	https://onlinecourses.nptel.ac.in/noc23_ee61/preview
4	https://archive.nptel.ac.in/courses/108/106/108106170/
	https://archive.nptel.ac.in/courses/103/102/103102015/
	https://onlinecourses.nptel.ac.in/noc21_mm34/preview
	https://nptel.ac.in/courses/108106182

SEMESTER S4

COMPUTER AIDED DESIGN AND ANALYSIS

Course Code	PEIET415	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To initiate ideas for innovative designs and design modifications.
2. To use computer aided 3D modelling packages for demonstrating new designs.
3. To familiarize Engineering analysis of systems using software.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Design: Introduction to various stages of the design process: Formulation of problem, Generation of alternatives, Evaluation, Guided Redesign. Component design, product design and system design approaches and examples. Structural, thermal, flow kinematic and dynamic design considerations and software packages. Preliminary design and detailed design Practicing sessions for creating primary designs of components, products, and systems.	9

2	Introduction to computer Aided 3D Modelling: Historical Development, Industrial Look at CAD/CAM, Application of computers in design, Benefits of CAD. CAD Environment, Packages, and Architectures. Basics of geometric and solid modelling. Geometric Features. Two dimensional and Three-Dimensional Operations. Degree of freedom and assembly constraints. Comparison of commercial packages advantageous and limitations. (Pro-E, Fusion 360, Catia, AutoCAD inventor, Solid works). Open-source packages. Practicing sessions for creating 3D models using any 3D modelling software package.	9
3	Introduction to Structural Analysis: Engineering Analysis, History, Advantages, Classification, Basic steps, 3D modelling and mesh generation. Role of finite element analysis in computer-aided design. Material properties for structural analysis. Boundary conditions and initial conditions. Analysis methods and convergence criteria. Post processing: representation, interpretation, and evaluation methods. Practicing sessions for structural analysis using any FEM package. Comparison of analytical solutions in the text books with simulation results.	9
4	Heat Transfer and Fluid Flow Analysis: Material properties for thermal and fluid analysis, Different types of boundary conditions and initial values for thermal and fluid flow problems. Solution methods and convergence criteria. Examples for Multiphysics problems and solution strategies. Solution procedure for thermo- mechanical and fluid structure interaction problems. Practicing sessions on heat transfer and fluid flow analysis using any FEM package. Comparison of analytical solutions in the text books with simulation results.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Examination</i>	<i>Analyse</i>	<i>Evaluate</i>	<i>Total</i>
5	15	10	10	40

Criteria for Assessment (Analyse and Evaluate): 20 marks

Phases	Assessment Criteria	Marks
Analyse	<i>Problem Definition</i> <i>a. Clearly defines the real-world quality issue.</i> <i>b. Examine and identify relevant contextual factors.</i>	5
	<i>Problem Analysis</i> <i>a. Present a structured realistic solution methodology.</i> <i>b. Compare and justify the proposed solutions with evidence and logical reasoning</i>	5
Evaluate	<i>Validation of Results</i> <i>a. Thoroughly evaluate the proposed solutions.</i> <i>b. Compares trade-offs, advantages, and disadvantages.</i> <i>c. Considers feasibility, scalability, and practical implications.</i>	5
	<i>Conclusion and Report Writing</i> <i>a. Summarizes procedure, findings and insights, limitation, and scope for future work.</i> <i>b. Preparation of Report with all components of project report.</i>	5

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Develop preliminary designs with understanding of structural, thermal, kinematic and dynamics engineering concepts.	K3
CO2	Develop engineering skills in the design using model generation.	K3
CO3	Analyse engineering designs using fundamental concepts of Engineering.	K4
CO4	Evaluate engineering designs using fundamental concepts of solid mechanics, fluid mechanics and heat transfer.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	3	2	3	-	-	-	2	-	-	-
CO2	3	-	3	2	3	-	-	-	2	-	-	3
CO3	3	-	3	2	3	-	-	-	2	-	-	3
CO4	3	-	3	3	3	-	-	-	2	3	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design and Engineering	Jayasree P. K.	CBS Publishers	2 nd Edition, 2020
2	CAD/ CAM Theory and Practice	Ibrahim Zeid	McGraw Hill	2 nd Edition, 2007
3	Introduction to Finite Elements in Engineering	T. R. Chandrupatla and A. D. Belegundu,	Pearson Education	3 rd Edition, 2001
4	Computer Aided Design and Manufacturing,	M.P. Groover	Prentice Hall of India	2 nd Edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Design a project Based Approach	Dym C. L.	Wiley	4 th Edition, 2013
2	A First course in Finite Element Method	Daryl Logan	Thomson Learning	6th Edition, 2016
3	Strength of Materials	Rattan	McGraw Hills	3 rd Edition, 2017
4	Fluid Mechanics: Fundamentals and Applications	John. M. Cimbala	SIE Publishers	4th Edition, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/107/101/107101087/
2	https://archive.nptel.ac.in/courses/112/102/112102101/
3	https://archive.nptel.ac.in/courses/112/107/112107214/
4	https://archive.nptel.ac.in/courses/103/107/103107096/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Cash flow</td><td>1000</td><td>900</td><td>800</td><td>700</td><td>600</td></tr></table> (In Crores)	Year	1	2	3	4	5	Cash flow	1000	900	800	700	600	CO4	(5)
	Year	1	2	3	4	5										
Cash flow	1000	900	800	700	600											
b)	Point out any three merits of NPV method.	CO4	(3)													

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education,	6

	employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to	6

	mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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Course Assessment Method (CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks			50	

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzingler,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and

outcomes.

- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4

MACHINE TOOLS LAB

Course Code	PCIEL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To impart hands-on training on conventional and advanced machine tools.
2. To provide training in conventional and advanced welding technologies.

Expt. No.	Experiments (Minimum 10 experiments should be completed)
1	Exercises on lathe: - Plain and step turning.
2	Exercises on lathe: Ball & curve, and Taper turning.
3	Exercises on lathe: - Thread cutting.
4	Exercises on shaping machine: machining of flat surfaces.
5	Exercises on shaping machine: machining of grooves and key ways.
6	Exercises on drilling machine: drilling, boring, reaming, tapping and countersinking etc.
7	Exercises on cylindrical grinding machine: Grinding of a plain cylindrical surface.
8	Exercises on surface grinding machine: Grinding of a flat surface.
9	Exercises on tool and cutter grinding machine: Grinding of a single-point cutting tool.
10	Exercises on milling machine: Plane and pocket milling operations.
11	Exercises on milling machine: Spur gear cutting operation.
12	Study and preparation of program, simulation and exercise on CNC lathe: turning, step turning, taper turning, thread cutting, ball and cup turning etc.
13	Study and preparation of program, simulation and exercise on CNC milling machine: surface milling, pocket milling, contour milling etc.
14	Exercises on arc welding: butt welding and lap welding of M.S. plates.

15	Exercises on gas welding: butt welding and lap welding of M.S. plates.
16	Exercises on TIG welding: butt welding and lap welding of M.S. plates.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Understand and recall the fundamental operations, processes, and safety practices of various machining tools such as lathes, shaping machines, drilling machines, and grinding machines.	K2
CO2	Apply their knowledge to perform practical machining tasks, such as turning, thread cutting, grinding, milling, and welding on different machines as per manufacturing drawings.	K3
CO3	Analyze machining processes to identify and troubleshoot potential issues, ensuring precision and efficiency in operations.	K4
CO4	Apply their skills to set up and operate CNC lathe and CNC machining centre, preparing programs, and conducting simulations for precise machining operations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	1	-	-	-	-	-	-	-	2
CO2	3	-	-	3	-	-	-	-	-	-	-	2
CO3	3	-	-	2	-	-	-	-	-	-	-	2
CO4	3	-	-	1	-	-	-	-	-	-	-	2
CO5	3	-	-	1	-	-	-	-	-	-	-	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Production Technology	HMT	Tata McGraw Hill	1st Edition, 2017
2	Workshop Technology Part I	W. A. J. Chapman	ELBS & Edward Arnold Publishers	5th Edition, 1972
3	Numerical Control of Machine Tools	Yoram Koren	McGraw-Hill	1st Edition, 2014

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4

WORK SYSTEM DESIGN LAB

Course Code	PCIEL408	CIE Marks	50
Teaching Hours/Week (L:T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To equip students to carry out method study and work measurement in organisations.
2. To enable students to gain practical knowledge to improve productivity by optimising the attributes of the workplace and work environment.

Expt. No.	Experiments (Minimum 10 experiments should be completed)
1	Method analysis on assembly tasks.
2	Micromotion study on recorded tasks.
3	Determination of standard time with time study.
4	Work sampling on recorded tasks.
5	Determination of standard time with methods time measurement.
6	Application of principles of motion economy in the workplace.
7	Determination of eye fatigue after tasks.
8	Determination of visual reaction time for various tasks.
9	Determination of auditory reaction time for various tasks.
10	Determination of energy expenditure for different activities.
11	Determination of performance with variation in workplace lighting.
12	Determination of static anthropometry in standing postures.

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply method and motion study concepts in organizations	K3
CO2	Apply work measurement concepts in organizations	K3
CO3	Evaluate workers cognitive and physiological responses	K4
CO4	Analyse working environment and its effects in work places	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	-	-	-	-	-	2	2	-	2
CO2	3	2	3	-	-	-	-	-	2	2	-	2
CO3	3	2	3	-	-	-	-	-	2	2	-	2
CO4	3	2	3	-	-	-	-	-	2	2	-	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to work study	ILO	Oxford and IBH	3rd Edition 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Motion and time study design and measurement of work	Ralph M.Barnes	Wiley India Private Limited	7 th Edition 2013
2	Work systems: The Methods, Measurements and Management of works	Mikell P. Groover	Pearson International	1 st Edition 2013
3	Niebel's Methods, Standards and Work Design	Andris Freivalds, Benjamin Niebel	McGraw Hill	13 th Edition 2013

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

INDUSTRIAL ENGINEERING

SEMESTER S5

DATA ANALYTICS

Course Code	PCIET501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to apply analytical tools and deduct valid inferences from data.
2. To equip students to develop decision making and analytical skills.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to data analysis: Need for data analysis, collection of data, organization and presentation of data, primary types of measurement scales - nominal, ordinal, interval and ratio scales; Deciles, quartiles and percentiles. Descriptive statistics: Measures of central tendency, measures of variability, skewness and kurtosis, Chebyshev's theorem, the empirical rule. exploratory data analysis. Overview of discrete and continuous probability distributions.	11
2	Sampling and sampling distributions: Samples and population, sampling process, non- probability and probability sampling- different types, determination of sample size, introduction to sampling distributions, central limit theorem. Estimators: Properties, point estimates, interval estimates and confidence intervals.	11
3	Hypothesis testing: Formulation of hypothesis, type I and type II error, power of the test, One sample and two sample tests - z-test, t-test, Chi-square test, F-test. Non-parametric methods: K-S test, sign test for paired data, one sample Runs test, rank sum tests – Mann-Whitney U-test and Kruskal-Wallis test. Analysis of variance: Theory and computations of ANOVA, ANOVA table, two-way ANOVA, blocking designs, factorial design.	11

4	Correlation and regression: Correlation analysis-Pearson correlation, Spearman's rank correlation, significance of correlation, estimation using the regression line, multiple regression- k-variable multiple regression model, F-test of a multiple regression model. Introduction to index numbers: Overview of multivariate techniques - factor analysis, multidimensional scaling, cluster analysis, discriminant analysis and conjoint analysis.	11
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Interpret data based on descriptive statistics.	K2
CO2	Apply statistical techniques for analysing samples from the fields of business and industry.	K3
CO3	Infer about the population on the basis of random samples from the population.	K2
CO4	Identify the relationships, patterns, correlations and trends between variables or datasets.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	2	3	1	-	-	-	2	-	2
CO2	2	2	3	2	3	2	-	-	-	2	-	2
CO3	2	2	3	2	3	2	-	-	-	2	-	2
CO4	2	2	3	2	3	2	-	-	-	2	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Statistics for Management	Levin, I. R., Siddiqui, M. H., Rubin, D. S.	Pearson Education	8 th Edition, 2017
2	Business Statistics	Sharma, J. K.	Pearson Education	5 th Edition, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complete Business Statistics	Aczel, A. D., Sounderpandian, J., Saravanan, P. and Joshi, R.	McGraw Hill	7 th Edition, 2017
2	100 Statistical Tests	Kanji, K. G.	Sage Publication	3 rd Edition, 2007
3	John E. Freund's Mathematical Statistics with Applications	Miller, I. and Miller, M.	Prentice Hall India	8 th Edition, 2013
4	Probability and Statistics in Engineering	Hines, W. W., Montgomery, D. C., Goldsman, D. M., and Borror, C. M.	John Wiley & Sons	4 th Edition, 2004
5	Research for Marketing Decisions	Green, P. E., Tull, D. S. and Albaum, G.	Prentice Hall	5 th Edition, 2009

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/107/110107114/
	https://archive.nptel.ac.in/courses/106/106/106106179/

SEMESTER S5

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Course Code	PCIET502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable for applying tools and techniques related to supply chain management.
2. To enable for applying models and tools related to logistics management.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to supply chains: Definition, objectives, structures, decision phases, performance measures, drivers, metrics and strategic fit, design of distribution networks and models for facility location and layout decisions. Features of postponement, mass customisation and supply chain restructuring. Understanding of lean, agile, leagile, resilient, dual channel, humanitarian, and international supply chains, integration of block chain, cloud computing, internet of things (IoT), artificial intelligence (AI) and machine learning (ML) in supply chains.	11
2	Forecasting and planning in supply chains: Forecasting models for supply chains including seasonal models, de-seasonalization and forecast errors. Aggregate planning strategies, methods and models applied to example cases. Operations planning to respond predictable variability in production and manufacturing chains using sequencing, scheduling, and line balancing.	11
3	Supply chain inventory models: Application of cycle inventory models, discounting models, multi-item inventory models and production-consumption models related to supply chains including multi-echelon systems. Solutions for safety inventory models in continuous review and periodic review cases of single and multi-items with probabilistic demand included in multi-echelon chains. Apply techniques related to optimal level of product	11

	availability, bullwhip effect and risk pooling.	
4	Logistics management: Understanding transportation networks, trade-offs, risk management, 3PL, 4PL and 5PL. Models for allocation, routing, scheduling and sequencing in transportation, solutions for vehicle routing problems and multi-stage transportation problems included with node capacities. Application logistics decision models including bin packing problems, fixed charge problems and knapsack problems, models and tools related to reverse, closed loop and green logistics in uncertain situations like pandemic and rescue scenarios.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Understand general and advanced supply chain networks.	K2
CO2	Apply tools on forecasting and planning decisions in supply chains.	K3
CO3	Apply inventory models on material flow decisions in supply chains.	K3
CO4	Apply tools and models for logistics decisions.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	-	-	-	-	-	-	-	1	-	2
CO2	2	-	3	-	3	-	-	-	-	1	-	2
CO3	2	-	3	-	3	-	-	-	-	1	-	2
CO4	2	-	3	-	3	-	-	-	-	1	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Supply Chain Management: Strategy, Planning and Operation	Chopra, S., Meindl, P.	Pearson Education	6 th Edition, 2016
2	Quantitative Models in Operations and Supply Chain Management	Srinivasan, G.	Prentice Hall India	2 nd Edition, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Logistical Management: The Integrated Supply Chain Process	Bowersox, D. J., Closs, D. J.	McGraw Hill	1 st Edition, 2017
2	Logistics and supply chain management	Christopher, M.	Pearson Education	4 th Edition, 2011
3	Modelling and Supply Chain, Cengage Learning	Shapiro, J. F.	Cengage Learning	1 st Edition, 2006
4	Manufacturing Operations and Supply Chain Management.	Taylor, D., Brunt, D.	Vikas Thomson Learning	1 st Edition, 2009

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/106/110106045/
	https://archive.nptel.ac.in/courses/110/105/110105095/

SEMESTER S5

QUALITY ENGINEERING

Course Code	PCIET503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise the fundamentals of Quality Engineering and its role in quality improvement.
2. To enable students to integrate the quality engineering principles into product development and manufacturing processes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concepts related to quality engineering: Philosophies of Deming, Juran and Crosby. Basic concepts of quality- quality of design, quality of conformance, quality of performance. Quality objectives, quality policy, quality function, quality control, quality assurance, total quality management, cost of quality, loss functions, manufacturing tolerances.	9
2	Statistical process control: Process control- process variability, causes of variation, control charts for variables, control chart for attributes. State of control and process out of control identification in control charts, patterns in control charts. Other control charts - cumulative sum control chart for process mean and exponentially weighted moving average control chart. Specifications and process capability - process capability indices, process capability analysis, tolerances for assemblies and subassemblies.	9
3	Acceptance sampling plans: Single, double, multiple and sequential sampling plans. Operating characteristic curves for single and double sampling plans. Evaluating sampling plans. Standard sampling plans - MIL-STD 105E sampling method and its equivalents, Dodge - Romig tables, ABC standards, AOQL and LTPD plans.	9

4	Quality improvement tools and techniques: Fault tree analysis, event tree analysis, failure mode and effect analysis, quality function deployment, Six Sigma- introduction, definition, DMAIC method, roles and responsibilities, Taguchi's methods - robust design, control and noise factors, S/N ratios, Quality circles - concepts, objectives, organisational structure, KAIZEN - meaning - management practices, quality standards- ISO 9000 Series, ISO 14000 standards.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the philosophies and concepts related to quality engineering.	K2
CO2	Apply statistical quality control tools for process control.	K3
CO3	Apply the knowledge on acceptance sampling for sampling inspection.	K3
CO4	Explain the quality improvement tools and techniques and apply the same for process improvement.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	-	-	-	-	2	2	-	-	2
CO2	3	2	3	2	-	-	-	2	2	-	-	2
CO3	3	2	3	2	-	-	-	2	2	-	-	2
CO4	3	2	3	2	-	-	-	2	2	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Quality Control and Improvement	Mithra A.	Wiley	5 th Edition, 2021
2	Statistical Quality Control	Mahajan M.	Dhanpat Rai & Co	4 th Edition, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Statistical Quality Control	Grant, E., Leavenworth R.	McGraw Hill	7 th Edition, 2017
2	Introduction to Statistical Quality Control	Montgomery D. C.	Wiley	8 th Edition, 2020
3	Total Quality Management	Besterfield, D.H., Besterfield, C	Pearson Education	5 th Edition, 2019
4	The Handbook for Quality Management	Pyzdek, T., Keller, P.A	McGraw Hill	2 nd Edition, 2012

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/104/110104080/

SEMESTER S5
OPTIMISATION TECHNIQUES

Course Code	PBIET504	CIE Marks	60
Teaching Hours/Week (L:T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to solve large scale optimization problems with single as well as multiple objectives.
2. To enable students to apply advanced tools and techniques to handle linear, non - linear problems and network models.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Linear Programming: Advanced solution techniques - Revised simplex method, dual simplex method, two phase method, interior point method, sensitivity analysis. Integer programming: Formulation and solution techniques - Introduction to dynamic programming and software tools for linear programming.	11
2	Queuing theory: Markovian queue models, single server & multiple server models. Game theory: Practical application of game theory, two-person, zero -sum games, solution with and without saddle point, rules of dominance, other solution methods.	11
3	Nonlinear programming: Definition, formulation of non-linear programming problems (NLPs), Convex and non-convex functions, solving NLPs with one variable (unconstrained maximization and minimization), unconstrained maximization and minimization with several variables, the method of steepest descent, Newton's method and quasi - Newton method, conjugate gradient and conjugate direction methods. Constrained optimization methods: Lagrange multipliers, Kuhn-Tucker condition, method of feasible directions, pareto optimality and trade-off curves.	11

4	Network problems: Introduction to graph theory and basic definitions. Minimum spanning tree problem - Prim's algorithm, Kruskal's algorithm, Shortest path problems - Dijkstra's algorithm, Floyd's algorithm, successive shortest path algorithm. Maximum flow problems, flow augmenting path, labelling algorithm, maximum flow and minimum cut, shortest augmenting path algorithm, minimum cost flow problem, network simplex method. Travelling salesman problem (TSP): Branch and bound and heuristic algorithms for the TSP, Chinese postman problem. Vehicle Routing Problems: Optimal solutions, Little's algorithm and heuristic solutions, savings based algorithm, Holmes and Parker refinement.	11
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Suggestion on Project Topics: Each student team can choose some simple close to real life projects from the areas discussed in the syllabus, such as

- Production Planning problem
- Cutting Stock Problem
- Fixed Charge Problem
- Resource allocation
- Scheduling
- Parameter optimization
- Analysis of practical problems using network models
- Portfolio optimization

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 subdivisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply advanced tools to solve linear and linear integer programming problems and analyse the sensitivity of solutions to dynamic situations.	K4
CO2	Apply queuing and game theory to develop solutions for industrial and allied areas.	K3
CO3	Solve the constrained and unrestricted nonlinear optimization problems using gradient non-gradient based techniques.	K3
CO4	Construct appropriate network models and build optimal solutions to industrial problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	3	3	-	-	2	2	3	-	3
CO2	3	3	2	3	3	-	-	2	2	3	-	3
CO3	3	3	2	3	3	-	-	2	2	3	-	3
CO4	3	3	2	3	3	-	-	2	2	3	-	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Operations Research: Theory And Application	Sharma J. K	Laxmi Publications	6 th Edition, 2017
2	Engineering Optimization: Theory and Practice	Singaresu S. Rao	New Age International	3 rd Edition, 2004
3	Operations Research	Srinivasan G.	Prentice Hall India	2 nd Edition, 2010
4	Quantitative Methods in Management	Vohra N. D.	Tata McGrew Hill	5 th Edition, 2017
5	Introduction to Management Science	Bernard W. Taylor	Pearson Education	11 th Edition, 2013
6	Principles of Operations Research with Applications to Managerial Decisions	Harvey M. Wagner	Prentice Hall India	2 nd Edition, 1975

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Operations Research	Frederick S. Hillier, Gerald J. Lieberman	Tata McGrew Hill	11 th Edition, 2020
2	Optimization in Operations Research	Ronald L. Rardin	Pearson Education	2 nd Edition, 2016
3	Introduction to Operations Research	Hamdy A. Taha	Pearson Education	10 th Edition, 2019
4	Operations Research: Applications and Algorithms	Wayne L. Winston	Duxbury Press	4 th Edition, 2003
5	Convex Optimization	Stephen Boyd, Lieven Vandenberghe	Cambridge University Press	1 st Edition, 2004
6	Convex Optimization Algorithms	Dimitri P Bertsekas	Athena Scientific	1 st Edition, 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/noc/courses/noc21/SEM1/noc21-mg43/
2	https://archive.nptel.ac.in/noc/courses/noc17/SEM2/noc17-ma18/
3	https://archive.nptel.ac.in/courses/112/106/112106131/
4	https://archive.nptel.ac.in/courses/112/106/112106131/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S5

ADVANCED MANUFACTURING PROCESSES

Course Code	PEIET521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of advanced manufacturing methods and its processes parameters.
2. To enable students to apply the process capabilities of the modern manufacturing methods in industries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Mechanical processes: Ultrasonic machining- elements of process, effect of parameters, applications, limitations of the process, advantages and disadvantages. Abrasive jet machining (AJM) - variables in AJM, metal removal rate in AJM. Water jet machining- jet cutting equipment, process details, advantages and applications.	9
2	Electrochemical processes: Electrochemical machining (ECM) - elements of ECM process, tool work gap, chemistry of the process, metal removal rate, accuracy and surface finish, advantages, applications, limitations. Electrochemical grinding - material removal, surface finish, accuracy, advantages, applications.	9
3	Thermal processes: Electric discharge machining (EDM) or spark erosion machining processes, mechanism of metal removal, electrode feed control, dielectric fluids, selection of electrode material, surface finish, machining accuracy, applications, wire cut EDM. Laser beam machining (LBM)- apparatus, material removal, cutting speed and accuracy of cut, advantages and limitations.	9

4	Plasma arc and Electron beam processes: Plasma arc machining (PAM) - plasma, non-thermal generation of plasma, mechanism of metal removal, PAM parameters, safety precautions, other applications of plasma jets. Electron beam machining (EBM) - generation and control of electron beam, theory of electron beam machining, process capabilities and limitations. High energy rate forming processes: Explosive forming processes- principles, effect of process in material properties, types of explosives forming. Magnetic pulse forming processes - general principles, applications.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply the concepts of non conventional mechanical material removal processes.	K3
CO2	Identify the various electrochemical machining processes.	K3
CO3	Apply the principles and concepts of thermal material removal processes.	K3
CO4	Make use of the concepts of high energy rate forming, plasma arc and electron beam machining processes.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	-	3	-	-	-	-	-	-	2
CO2	3	1	2	-	3	-	-	-	-	-	-	2
CO3	3	1	2	-	3	-	-	-	-	-	-	2
CO4	3	1	2	-	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing Science	Amitabha Ghosh A. K., Mallik	East West Press	2 nd Edition, 2010
2	Production Technology	HMT	McGraw Hill Education	1 st Edition, 2017
3	Modern Machining Processes	P. C. Pandey, H. S. Shan	McGraw Hill Education	1 st Edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Unconventional Machining Processes	M. Adithan	Atlantic	1 st Edition, 2018
2	Advanced Methods of Machining	Mc Geongh J. A.	Chapman and Hall	1 st Edition, 1988
3	Non-Traditional Machining Processes	Jagadeesha T.	I K International Publishing House	0 th Edition, 2016

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/103/112103202/

SEMESTER S5

MECHANICS OF METAL CUTTING

Course Code	PEIET522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with a thorough understanding of the principles and mechanics underlying metal cutting processes.
2. To equip students with the skills to analyze and apply advanced machining techniques.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of metal cutting: Definition and importance of metal cutting, types of metal cutting processes- turning, milling, drilling and grinding. Overview of cutting tools and their materials. Mechanics of cutting - chip formation and types of chips, orthogonal and oblique cutting. Forces in cutting - cutting force, thrust force, radial force. Cutting tool geometry - single-point and multi-point cutting tools, tool angles and their significance. Tool life and wear mechanisms. Cutting fluids- types and properties of cutting fluids, functions and applications of cutting fluids.	9
2	Cutting tool materials and coatings: High-speed steels, carbides, ceramics, and diamond tools, properties and applications of different tool materials. Tool coatings - types of coating - TiN, TiC and Al ₂ O ₃ , benefits of coatings in cutting performance. Tool material selection - criteria for selecting tool materials. Cost benefit analysis of different tool materials and coatings.	9
3	Cutting parameters and surface integrity: Cutting parameters - speed, feed, and depth of cut, influence of cutting parameters on cutting forces and tool life, optimization of cutting parameters for different materials. Surface integrity- surface roughness and its measurement, factors affecting surface	9

	finish in machining, impact of cutting conditions on surface integrity. Inspection methods for machined components, statistical process control in machining, tolerances and fits in metal cutting.	
4	Advanced Machining Techniques and Future Trends: Introduction to non-traditional machining processes - EDM, ECM, LBM, applications and advantages of non-traditional machining, high-speed and ultra-precision machining, principles of high-speed machining, equipment and techniques for ultra-precision machining, applications in aerospace, automotive, and electronics industries. Emerging technologies in machining- additive manufacturing, hybrid machining.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of metal cutting processes and cutting tool geometry.	K2
CO2	Identify cutting tool materials and coatings for their suitability in various machining applications.	K3
CO3	Apply cutting parameters to improve surface integrity and enhance the overall quality of machined components.	K3
CO4	Make use of advanced machining techniques and automation technologies to address the challenges in the industry.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	2	-	1	-	-	-	-	-	-	2
CO2	3	-	2	-	2	-	-	-	-	-	-	2
CO3	3	-	2	-	2	-	-	-	-	-	-	2
CO4	3	-	2	-	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Metal Cutting Principles	Milton C. Shaw	Oxford University Press	2 nd Edition, 2005
2	Machining and Machine Tools	A. B. Chattopadhyay, P. C. Mishra	Wiley India	2 nd Edition, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Elements of Workshop Technology	S. K. Hajra Choudhury	Media Promoters and Publishers	15 th Edition, 2010
2	Fundamentals of Metal Cutting and Machine Tools	B. L. Juneja	New Age Publishers	2 nd Edition 2017

Video Links (NPTEL, SWAYAM...)	
LINK ID	https://archive.nptel.ac.in/courses/112/105/112105233/

SEMESTER S5

ADVANCED METAL JOINING TECHNIQUES

Course Code	PEIET523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To equip students with fundamental knowledge of advanced welding processes.
2. To enable students to apply advanced welding processes for industrial applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Radiant energy welding: Electron beam welding - guns, weld environment, welding in different degrees of vacuum, equipment and safety, joint design, applications. Laser beam welding - principle of operation, process parameters, applications and limitations. Explosive welding- process parameters, applications, advantages and limitations, joint design, materials and applications.	9
2	Plasma arc welding: Plasma arc welding- theory and principles, transferred arc and non-transferred arc techniques, equipment and tooling, advantages, disadvantages and applications. Laser TIG hybrid welding, double side arc welding, magnetic arc welding.	9
3	Friction welding: Basic principles, process variants, different stages of friction welding, process parameters, advantages, limitations and applications. Friction stir welding- process variables and applications. Diffusion welding- theory and principle of process, key variables	9
4	Modeling of welding processes: Wire arc additive manufacturing (WAAM), (concept only), modelling of welding processes, welding modelling software's and its applications, major welding defects and its detection. Heat Affected zone (HAZ) formation in different welding methods, welding distortions and residual stresses in different welding methods.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply radiant energy welding processes in industries.	K3
CO2	Make use of plasma arc welding and other modern welding techniques in manufacturing.	K3
CO3	Identify the applications of friction and diffusion welding for various fabrication requirements.	K3
CO4	Analyse the performance of a welding process using modern tools.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	3	-	-	-	-	-	-	2
CO2	3	-	-	-	3	-	-	-	-	-	-	2
CO3	3	-	-	-	3	-	-	-	-	-	-	2
CO4	3	3	-	-	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Welding Processes and Technology	R.S. Parmar	Khanna Publishers	3 rd Edition, 1996
2	Advanced Welding Technology	B. R. Senthil Kumar	Notion Press	1 st Edition, 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Textbook of Welding Technology	G.D. Garg	S.K. Kataria & Sons	4 th Edition, 2012
2	Welding and Welding Technology	Richard Little	McGraw Hill Education	2nd Edition, 2017
3	Advanced Welding Technology	S.P. Tewari	S.K. Kataria & Sons	1 st Edition, 2011

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/113/106/113106087/

SEMESTER S5

METAL FORMING AND TOOL DESIGN

Course Code	PEIET524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of the principles, processes, and techniques involved in metal forming.
2. To equip students with the skills to design effective tools for various metal forming processes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of metal forming: Definition and classification of metal forming processes, advantages and limitations of metal forming, applications in various industries, stress and strain in metal forming. Concepts of stress and strain in deformable bodies, flow stress, true stress, and true strain, yield criteria for ductile materials (Von Mises and Tresca). Plastic deformation and work hardening, forming limit diagrams, influence of temperature on forming processes. Hot, warm, and cold working - strain rate sensitivity and its impact on material behavior.	9
2	Bulk deformation processes: Forging - types (open-die, closed-die, impression-die), equipment, and applications. Rolling - types (hot and cold rolling), rolling mills, and products. Extrusion - direct and indirect extrusion, equipment, and applications. Drawing - wire drawing, tube drawing, and deep drawing. Sheet metal forming processes: Bending- types of bending operations, bending allowances, and springback. Shearing - types of shearing operations, shearing force calculation. Deep drawing - mechanics of deep drawing. Stamping and punching - operations, tools and applications, defects of various metal forming processes and its remedies.	9

3	Introduction to tool design: Role of tool design in manufacturing, types of tools used in metal forming, material selection for tools and dies, design of forging dies, types of forging dies, design considerations and calculations for forging dies, lubrication and cooling of forging dies, die design for extrusion processes, tooling for tube and wire drawing, design of dies and punches for blanking and piercing, design of deep drawing dies and tooling for stamping operations.	9
4	Advanced topics in metal forming and tool design: Finite element analysis in metal forming, basics of finite element method (FEM) in metal forming, application of FEM in analyzing forming processes, role of automation in metal forming, CNC and robotic applications in forming processes, maintenance strategies for forming tools, additive manufacturing for tool design and repair.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts and principles of metal forming processes, including stress, strain, and material behavior	K2
CO2	Apply various metal forming processes such as forging, rolling, extrusion, and sheet metal forming in industry.	K3
CO3	Identify tools for different metal forming processes, considering factors such as material selection, tool wear and maintenance.	K3
CO4	Apply finite element analysis (FEA) and other advanced techniques to optimize metal forming processes and tool design.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	2	-	2	-	-	-	-	-	-	2
CO2	3	-	2	-	2	-	-	-	-	-	-	2
CO3	3	-	2	-	2	-	-	-	-	-	-	2
CO4	3	-	2	-	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Metal Forming: Mechanics and Metallurgy	William F. Hosford, Robert M. Caddell	Cambridge University Press	4 th Edition, 2011
2	Fundamentals of Tool Design	Society of Manufacturing Engineers (SME)	SME	6 th Edition, 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Metal Forming Analysis	R. H. Wagoner, J. L. Chenot	Cambridge University Press	1 st Edition, 2001
2	Tool Design	Cyril Donaldson, George H. LeCain, V. C. Goold	Tata McGraw-Hill Education	4 th Edition, 2012
3	Sheet Metal Forming: Processes and Applications	Taylan Altan, A. Erman Tekkaya	ASM International	1 st Edition, 2012

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107250/

SEMESTER S5

ADDITIVE MANUFACTURING AND 3D PRINTING

Course Code	PEIET525	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to familiarise the concepts and various software tools in Additive Manufacturing (AM).
2. To equip students with knowledge on applications of additive manufacturing in real world scenarios.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to additive manufacturing (AM): Importance of additive manufacturing, basic principle of additive manufacturing, procedure of product development in additive manufacturing, classification of additive manufacturing processes, materials used in additive manufacturing, benefits and challenges in additive manufacturing.	9
2	Design for additive manufacturing (DFAM): Concepts and objectives of AM, unique capabilities - part consolidation topology. Optimization, generative design, CAD model preparation, AM file formats - STL, problems with STL, AMF Design for part quality improvement, part orientation, support structure, slicing, tool path generation, design rules for extrusion based AM. Introduction to slicing software - cura, creality print, modelling and converting CAD models into STL file, design and fabrication of parts by varying part orientation and support structures.	9
3	Additive manufacturing processes: Fused deposition modelling (FDM), Selective laser sintering (SLS), Selective laser melting (SLM), Electron beam melting (EBM), Stereolithography (SLA), Laser engineering net	9

	shaping (LENS), hands on sessions using FDM and resin 3D printers, design and fabrication of topology optimized parts.	
4	Other additive manufacturing processes: Laminated object manufacturing (LOM), binder Jetting, three-dimensional printing - materials, process, benefits, limitations, applications. Material jetting, multijet modeling - materials, process, benefits, limitations, applications. Applications and case studies of additive manufacturing: Biomedical, manufacturing, aerospace, automotive, food and electronics domains.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Examination</i>	<i>Analyse</i>	<i>Evaluate</i>	<i>Total</i>
5	15	10	10	40

Criteria for Assessment (Analyse and Evaluate): 20 marks

Phases	Assessment Criteria	Marks
Analyse	Problem Definition a. Clearly defines the real-world quality issue. b. Examine and identify relevant contextual factors.	5
	Problem Analysis a. Present a structured realistic solution methodology. b. Compare and justify the proposed solutions with evidence and logical reasoning	5
Evaluate	Validation of Results a. Thoroughly evaluate the proposed solutions. b. Compares trade-offs, advantages, and disadvantages. c. Considers feasibility, scalability, and practical implications.	5
	Conclusion and Report Writing a. Summarizes procedure, findings and insights, limitation, and scope for future work. b. Preparation of Report with all components of project report.	5

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain the principles and product development in Additive manufacturing technology.	K2
CO2	Apply the process of transforming a concept into the final product in AM technology.	K3
CO3	Identify the processes of additive manufacturing for various fabrication requirements.	K3
CO4	Apply the additive manufacturing techniques in real world scenarios.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	3	-	2	-	-	-	-	2
CO2	3	3	2	-	3	-	2	-	-	-	-	2
CO3	3	3	3	-	3	-	2	-	-	-	-	2
CO4	3	3	3	-	3	2	2	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Additive manufacturing technologies	Ian Gibson, David Rosen, Brent Stucker, Mahyar Khorasani	Springer	3 rd Edition, 2021
2	Rapid prototyping: Principles and applications	Chua C. K., Leong K. F., and Lim C. S.,	World Scientific Publishers	3 rd Edition, 2010
3	3D Printing and Additive Manufacturing: Principles and Applications: Fourth Edition of Rapid Prototyping	Chee Kai Chua, Kah Fai Leong,	World Scientific Publishers	4 th Edition, 2014
4	Additive Manufacturing	Amit Bandyopadhyay, Susmita Bose	CRC Press	1 st Edition 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Understanding Additive Manufacturing: Rapid Prototyping, Rapid Manufacturing	Andreas Gebhardt	Hanser Gardner Publication, Cincinnati, Ohio	3 rd Edition, 2011
2	Laser Additive Manufacturing: Materials, Design, Technologies, and Applications	Milan Brandt	Woodhead Publishing., United Kingdom	2 nd Edition, 2016
3	Rapid Prototyping: Theory and practice	Kamrani A. K., Nasr E. A.	Springer., United States	1 st Edition, 2006
4	Rapid Prototyping and Engineering applications: A tool box for prototype development	Liou L. A. W., Liou F. W.	CRC Press., United States	3 rd Edition, 2011

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/103/112103306/

SEMESTER S5

QUALITY AND MATERIAL TESTING LAB

Course Code	PCIEL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To enable the students gain practical knowledge and skills related with the theoretical studies of statistical quality control.
2. To develop skills to inspect and test materials and components for mechanical properties, defects and flaws.

Expt. No.	Experiments ((Minimum 8 experiments from Part A and 4 experiments from Part B should be completed)
PART A: Quality Control Lab	
1	Verification of central limit theorem using rectangular distribution.
2	Verification of central limit theorem using triangular distribution.
3	Verification of central limit theorem using Henry line.
4	Construction of variable control charts (X bar and R charts) using actual vernier measurements.
5	Construction of variable control charts (X bar and R charts) using actual micrometer measurements.
6	Construction of p - chart.
7	Construction of c- chart.
8	Construction of np - chart.
9	Construction of u - chart.
10	Construction of Demerit chart (D-chart).
11	Construction of Quality score chart (Q-chart).
12	Construction of CUSUM chart.
13	Construction of OC-curve.
14	Experiment on double sampling plan.

15	Experiment on multiple sampling plan.
16	Study and construction OC curve for p-chart.
17	Study on finished product inspection (visual) and certification procedures.
18	Performance test on finished products.
PART B: Material Testing Lab	
1	Study and experiment on Universal Testing Machine (UTM) to determine tensile characteristics of ferrous materials.
2	Study and experiment on Universal Testing Machine (UTM) to determine tensile characteristics of non-ferrous materials.
3	Study and experiment on Vickers Hardness Testing Machine to determine hardness of the given specimen.
4	Study and experiment on non-destructive examination using dye penetrant test.
5	Study and experiment on non-destructive evaluation (NDE) using yoke type magnetic crack detector.
6	Study and experiment on portable magnetic inspection unit (Model Radent - 2000).
7	Study and experiment on bench type wet horizontal magnetic particle machine (model Radent – B 2000) and black light.
8	Study and experiment on ultrasonic flaw detector (Model Einstein - II TFT).
9	Study and experiment on non-destructive evaluation (NDE) using radiographic film viewer.
10	Study and experiment on eddy current sorter.
11	Study and experiment on non-destructive evaluation using acoustic emission.
12	Study and experiment on non-destructive evaluation using thermal camera.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply SQC techniques / principles like central limit theorem, control charts, sampling plans and OC curve.	K3
CO2	Construct charts / graphs related to SQC techniques / principles and infer on the same.	K3
CO3	Experiment with various destructive testing techniques to determine mechanical properties of materials.	K3
CO4	Experiment with non-destructive evaluation on different test specimens using various NDT equipment.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	-	-	-	-	-	2	2	-	2
CO2	3	2	3	-	-	-	-	-	2	2	-	2
CO3	3	2	3	-	-	-	-	-	2	2	-	2
CO4	3	2	3	-	-	-	-	-	2	2	-	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Statistical Quality Control	Grant E., Leavenworth R.	McGraw Hill	7 th Edition, 2017
2	Mechanical Metallurgy	G. E. Dieter	McGraw Hill	3 rd Edition, 2017
3	Practical Non-Destructive Testing	Balde, R., Jayakumar T., Thavasimuthu T.	Alpha Science	3 rd Edition, 2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Statistical Quality Control	Gupta R. C.	Khanna Publishers	10 th Edition, 2003
2	Fundamentals of Quality Control and Improvement	Mithra A.	Wiley	4 th Edition, 2016
3	Introduction to Statistical Quality Control	Montgomery	John Wiley & Sons	8 th Edition, 2019
4	Experimental Stress analysis	Dally J. W., Railey W. P.	Mcgraw Hill	3 rd Edition, 1991
5	Non-Destructive Testing	Hull B., John V	Macmillan	1 st Edition, 1998
6	Ultrasonic Testing of Materials	Krautkramer J., Krautkramer, H.	Springer-Verlag	4 th Edition, 1990

Video Links (NPTEL, SWAYAM...)	
Link ID	https://nptel.ac.in/courses/112/107/112107259/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S5

DATA ANALYTICS LAB

Course Code	PCIEL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To enable students to apply data analysis for appropriate decision making in industrial and allied areas.
2. To equip students to develop skills in data management and data visualization.

Expt. No.	Experiments (Minimum 10 experiments should be completed)
1	Implement and Infer descriptive statistics in Excel.
2	Implement Parametric Tests in Excel.
3	Implement and Infer one-way and two-way ANOVA in Excel.
4	Conduct correlation and regression analysis in Excel.
5	Fitting Probability Distribution using EasyFit / StatFit.
6	Exploring data using SPSS / Minitab / Systat.
7	Hypothesis Testing using SPSS / Minitab / Systat.
8	Conduct Normality Test using SPSS / Minitab / Systat.
9	Conduct Parametric Tests in SPSS / Minitab / Systat.
10	Conduct Non-Parametric tests in SPSS / Minitab / Systat.
11	Conduct Correlation and Regression Analysis in SPSS / Minitab / Systat.
12	Multivariate Analysis in SPSS / Minitab / Systat.
13	Descriptive Analysis in R programming.
14	ANOVA Test in R programming.
15	Time Series Analysis in R programming.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Classify, organize and summarize data sets.	K2
CO2	Apply concepts of data science to solve real world context problems.	K3
CO3	Apply inferential statistical analysis techniques to describe data sets.	K3
CO4	Examine data using exploratory statistical techniques to derive insightful analysis.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	1	3	2	-	1	-	-	-	2
CO2	2	2	2	2	3	2	-	1	-	-	-	2
CO3	2	2	2	2	3	2	-	1	-	-	-	2
CO4	2	2	2	2	3	2	-	1	-	-	-	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complete Business Statistics	Aczel A. D., Sounderpandian, J., Saravanan, P., Joshi, R.	McGraw Hill	7 th Edition, 2017
2	Discovering Statistics using IBM SPSS Statistics	Andy Field	SAGE South Asia	4 th Edition, 2019
3	The Book of R – A First Course in Programming and Statistics	Tilman M. Davies	No Starch Press	1 st Edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	SPSS for Dummies	Keith McCormick, Jesus Salcedo, Aaron Poh	Wiley	3 rd Edition, 2015
2	An Introduction to Secondary Data Analysis with IBM SPSS Statistics.	John MacInnes	SAGE	1 st Edition, 2017
3	SPSS Survival Manual: A Step by Step Guide to Data Analysis using SPSS for Windows	Julie Pallant	Open University Press	4 th Edition, 2010
4	Microsoft Excel 2019: Data Analysis and Business Modeling	Winston, Wayne L.	PHI Learning	6 th Edition, 2019
5	R for Data Science: Import, Tidy, Transform, Visualize, and Model Data	Garrett Golemund, Hadley Wickham	O'Reilly Media	1 st Edition, 2017
6	R Programming – An Approach to Data Analytics	G. Sudhamathy, C. Jothi Venkateswaran	MJP Publishers	1 st Edition, 2018

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/106/110106072/
	https://archive.nptel.ac.in/courses/110/104/110104094/
	https://archive.nptel.ac.in/courses/110/107/110107092/
	https://archive.nptel.ac.in/courses/111/104/111104146/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

INDUSTRIAL ENGINEERING

SEMESTER S6

SYSTEM MODELLING AND SIMULATION

Course Code	PCIET601	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to construct a theoretical base to model discrete and continuous systems.
2. To equip students to model and simulate discrete and continuous systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to system modelling: Concepts of a system, system environment, continuous and discrete systems, system models, comparison of analytical methods and simulation, areas of application of simulation, steps in discrete event simulation study. Random Numbers: Properties of random numbers, generation of pseudo-random numbers, random number generators. Tests for random numbers-frequency, gap, run, and Poker tests, test for autocorrelation, Monte Carlo method.	11
2	Generation of random variates: Inverse transformation method for exponential, uniform, Weibull, triangular, and discrete distributions, Direct transformation method for normal and lognormal distributions, Acceptance-rejection technique for Poisson and Gamma distributions. Input modelling: Data collection, identifying the distribution with the collected data, goodness of fit tests, selecting input models without data. Verification and validation of simulation models, Output data analysis, Variance reduction techniques. Time Advance Mechanisms for discrete event simulation: Next-Event time advance and fixed increment time advance methods.	11

3	System dynamic modelling: Evolution of system dynamics as a system enquiry methodology, elements of system dynamics modelling- physical flows, information flows, level & rate variables, delays, information smoothing, table functions and table function multipliers, causal loop diagramming, flow diagrams. Steps for Modelling in system dynamics.	11
4	Behaviour of linear lower order systems: Analytical approach - first order positive and negative feedback systems, pure second order positive and negative feedback systems. Modes of dynamic behaviour: Exponential growth, goal seeking, oscillation, interactions of fundamental modes. Principles of simulation modelling, developing model equations, algorithms for Euler integration, hand simulation of system dynamics models. S-Shaped growth- epidemics, innovation, diffusion and growth of new products. Overview of Software packages for System modelling and simulation.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of the system and its simulation, Demonstrate the generation of random numbers and its testing.	K2
CO2	Perform input modeling for discrete simulation, verify and validate simulation models and perform output data analysis	K3
CO3	Illustrate various elements of system dynamics modelling, Construct causal loop diagrams and Flow diagrams.	K3
CO4	Develop the fundamental modes of dynamic behaviour and their interactions; Model and Analyze systems using the principles of system dynamics.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	3	1	-	-	1	1	1	2
CO2	3	3	3	2	3	2	-	-	1	1	1	2
CO3	3	3	2	2	3	3	-	-	1	1	1	2
CO4	3	3	3	3	3	3	-	-	1	1	1	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Discrete Event System Simulation	Jerry Banks	Pearson Education	5 th Edition, 2013
2	Introduction to System Dynamic Modeling	M C Bora, Pratap K J Mohapatra and P Mandal	Universities Press	1 st Edition, 1994

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Simulation Modeling and Analysis	A M Law	McGraw Hill	6 th Edition, 2024
2	System Simulation	Geoffrey Gordon	Prentice Hall India	2 nd Edition, 1979
3	Discrete Event Simulation: Modeling, Programming and Analysis	George Fishman	Springer series in Operations Research and Financial Engg.	1 st Edition, 2001
4	System Simulation with Digital Computer	Deo Narsingh	Prentice Hall India	24 th Edition, 2011
5	Simulation	Sheldon M. Ross	Elsevier	6 th Edition, 2023
6	Business Dynamics – Systems Thinking and Modeling for a Complex World	John D. Sterman	McGraw Hill	1 st Edition, 2000

Video Links (NPTEL, SWAYAM...)	
Link ID	https://nptel.ac.in/courses/112107220
	https://onlinecourses.nptel.ac.in/noc22_me18

SEMESTER S6

FINANCIAL ENGINEERING

Course Code	PCIET602	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable to understand financial engineering, investment and apply mathematics in financial engineering
2. To enable tools related to capital budgeting, cost of capital, working capital and capital structure.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to finance and financial engineering: Definitions, financial literacy - needs, financial decisions - goals, financial systems - functions, financial markets - types, financial investment - investment techniques, financial intermediaries, asset types - real estate, securities, commodities, derivatives, and futures.	9
2	Financial mathematics: Principal and interest - simple interest and compound interest - continuous compounding. Interest rate - risk free interest rates, term structure of interest rates and short rate models. Time value of money - investment and maturity period, doubling period, present value of annuity and future value of annuity - calculations. Equated monthly instalment (EMI) - EMI calculation. Stocks - valuation of stocks and intrinsic value of stock.	9
3	Capital budgeting and capital cost: Capital budgeting process - investment criteria, conventional and discounted cash flow methods - net present value (NPV) and internal rate of return (IRR) - Computation of NPV, IRR, benefit cost ratio, equivalent maintenance cost and payback period.	9

	Cost of capital: Weighted average cost and weighted marginal cost - factors affecting. Company's cost of capital and project cost of capital - cost of debt - cost of preference - cost of equity - capital asset pricing model (CAPM).	
4	Working capital and capital structure theories: Working capital - factors affecting capital requirements. Cash management - models of cash management, operating cycle and cash cycle calculations. Capital structure - net income approach, net operating income approach, traditional position, and Modigliani-Miller (MM) hypothesis. Trade off and signalling theory.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Understand finance, financial engineering, and related areas.	K2
CO2	Apply financial mathematics including time value of money.	K3
CO3	Apply tools and techniques for capital budgeting and cost of capital.	K3
CO4	Understand capital structure theories and apply techniques of working capital and cash.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	-	-	-	1	-	-	1	-	-	2
CO2	2	1	3	-	-	1	-	-	-	-	-	2
CO3	2	1	3	-	-	1	-	-	-	-	3	2
CO4	2	1	3	-	-	1	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation²

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Financial Management	Prasanna Chandra	McGraw Hill	10 th Edition, 2019
2	Financial Accounting	R Narayanaswamy	PHI Learning	4 th Edition, 2017
3	Understanding the Mathematics of Personal Finance	Lawrence N. Dworsky	Wiley Publication	1 st Edition, 2009
4	Fundamentals of Financial Management	Eugene F. Brigham, Joel F. Houston	Cengage Publication	15 th Edition, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Applied corporate finance	Aswath Damodaran	Wiley publication	4 th Edition, 2015
2	Financial Management	I M Pandey	Vikas publishing house	11 th Edition, 2016

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/107/110107144/

SEMESTER S6

TOTAL QUALITY MANAGEMENT AND SIX SIGMA

Course Code	PEIET631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable the students to understand the principles and philosophy of Total Quality Management and Six Sigma.
2. To equip students to apply TQM and Six Sigma principles to real world scenarios.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to quality and TQM: Quality - definition, quality quantified, dimensions of quality. TQM – definition, new and old cultures. Gurus of TQM – Deming, Juran, Shewhart, Crosby, Fiegenbaum, Ishikawa, Taguchi – their main contributions, obstacles to TQM, benefits of TQM. quality circles, cost of quality, Taguchi quality loss function.	9
2	Axioms of TQM, Customer satisfaction and Employee involvement: TQM axioms - commitment, scientific knowledge and involvement, Deming's fourteen points on quality management, quality council, quality statements, customer satisfaction, customer perception of quality, customer retention, employee involvement, motivation, empowerment, team and teamwork, suggestion system.	9
3	Quality Management Systems (QMS): Introduction, benefits of ISO registration, ISO 9000 series of standards, sector-specific standards - AS 9100, ISO/TS 16949 and TL 9000, ISO 9001 requirements - implementation, documentation, internal audits, registration. Environmental Management System. Continuous process improvement and QMS: Continuous process	9

	improvement - Juran's quality trilogy, PDCA cycle, Kaizen, seven basic tools of quality, seven new management tools, Benchmarking.	
4	Quality improvement through Six Sigma: Six sigma –meaning, history of six sigma, working of six sigma. Process capability – meaning, significance and measurement. Six sigma roles and responsibilities, DMAIC process- various phases. Case examples of DMAIC, six sigma in manufacturing and service industries.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of quality highlighting the contributions given by quality gurus.	K2
CO2	Demonstrate knowledge on TQM axioms and focus on customer-centric approach with teamwork.	K2
CO3	Apply various tools and techniques for continuous process improvement.	K3
CO4	Make use of six sigma and its methodology in industries.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	-	-	-	-	-	-	2
CO2	3	2	2	-	-	-	-	-	-	-	-	2
CO3	3	2	2	-	-	-	-	-	-	-	-	2
CO4	3	2	2	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Total Quality Management	Besterfield Dale H. et al.	Pearson Education	5 th Edition, 2018
2	Transactional Six Sigma and Lean Servicing	Betsi Harris Ehrlich	St. Lucie Press	1 st Edition, 2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Total Quality Management	Subburaj Ramasamy	McGraw Hill Education	4 th Edition, 2017
2	Total Quality Management	Arora K. C.	S K Kataria and Sons	4 th Edition, 2016
3	The Management and Control of Quality	James R. E., William M. L.	Cengage Learning	8 th Edition, 2012
4	Managing Quality – Concepts and Tasks	Narayana V., Sreenivasan, N.S	New Age International	1 th Edition, 2005
5	The Six Sigma Handbook	Pyzdek T., Keller, P	McGraw-Hill	5 th Edition 2018

Video Links (NPTEL, SWAYAM...)	
Link ID	https://nptel.ac.in/courses/110104080
	https://nptel.ac.in/courses/110105123

SEMESTER S6

RELIABILITY ENGINEERING

Course Code	PEIET632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to understand important concepts of reliability engineering
2. To equip students to apply various tools, techniques and procedures of reliability engineering for practical scenarios.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Reliability concepts: Definition of reliability, reliability vs. quality, reliability function, MTTF, hazard rate function, bathtub curve, derivation of the reliability function, failure and failure modes, causes of failures and unreliability. Reliability models: Constant failure rate model, time dependent failure models. Weibull distribution, normal distribution, lognormal distribution. Serial configuration, parallel configuration, combined series parallel systems, K-out-of-m systems.	10
2	Redundancy techniques in system design: Component vs unit redundancy, weakest-link technique, mixed redundancy, standby redundancy, redundancy optimization, double failures and redundancy, Markov analysis, load sharing systems, standby systems, degraded systems, three state devices, covariate models.	8
3	Reliability in design: Reliability in design process, system effectiveness, economic analysis and life cycle cost. Reliability allocation- optimal allocations, ARINC and AGREE methods. System safety and fault tree analysis- Tieset and Cut-set methods. Use of boolean algebra in reliability analysis.	8

4	Maintainability and availability: Definitions and basic concepts, relationship between reliability, availability and maintainability, inherent availability, achieved availability, operational availability. Economics of reliability: Economic issues, manufacturers cost, customers cost, reliability achievement cost models, reliability utility cost models, depreciation cost models, availability cost model for parallel systems. Reliability management and reliability management by objectives.	10
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of reliability, various models of reliability and failure concepts.	K2
CO2	Apply mathematical models of reliability and failure modes.	K3
CO3	Make use of the design process of reliability.	K3
CO4	Demonstrate the relation between reliability, availability, maintainability and economic aspects of reliability.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	1	1	-	-	-	-	-	-	2
CO2	3	3	-	3	3	-	-	-	-	-	-	2
CO3	3	2	3	1	1	-	-	-	-	-	-	2
CO4	3	2	2	1	1	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliability Engineering	Balagurusamy E.	Tata McGraw Hill	1 st Edition, 2017
2	Reliability Engineering	Srinath L. S.	East West Press	4 th Edition, 2005
3	Reliability and Maintainability Engineering	Charles E. Ebeling	Tata McGraw Hill	1 st Edition, 2017
4	Practical Reliability Engineering	Patrick D. T., O'Connor, Andre Kleyner	John Wiley & Sons Ltd	5 th Edition, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Reliability Engineering	E. E. Lewis	Wiley	2 nd Edition, 1994
2	Quality Planning and Analysis	J.M. Juran, Frank M. Gryna	Tata McGraw Hill	5 th Edition, 2007

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/127/105/127105234/

SEMESTER S6

INDUSTRIAL SCHEDULING

Course Code	PEIET633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to solve scheduling problems.
2. To equip students to apply mathematical and heuristic methods in industries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to scheduling: Definition of scheduling, role of scheduling, terminologies involved in scheduling, classification of scheduling models - Graham et al. $\alpha/\beta/\gamma$ notations to classify scheduling problems, performance measures, relationship between performance measures, theorems related to performance measures (proof and problems). Classes of schedules - non-delay schedules, active schedules and semi-active schedules. Single machine models without due dates (proof of theorems and problems) - permutation schedules theorems, shortest processing time (SPT) rule, weighted shortest processing time (WSPT) rule.	9
2	Single machine models with due dates: Problems with due dates, minimizing total lateness, minimizing maximum lateness and maximum tardiness, minimizing the number of tardy jobs - Moore and Hodgson algorithm, minimizing mean tardiness - Wilkerson-Irwin algorithm, minimizing regular performance measures with precedence constraints - Lawler's algorithm, application of Smith's algorithm. General purpose methodologies for single machine problem - neighborhood search techniques, dynamic programming approach, branch & bound approach.	9
3	Parallel machine problems and flow shop problems: Parallel identical processors and independent jobs with preemptions and without preemptions,	9

	parallel identical processors and dependent jobs, permutation schedule- Johnson's algorithm, two jobs and m machine problems - Aker's graphical method. Special cases of three-machine problem - extension of Johnson's algorithm, branch & bound algorithms for makespan problems, heuristic approaches - Palmer's algorithm, CDS algorithm and insertion heuristics.	
4	Job shop scheduling and stochastic single machine models: Types of schedules, schedule generation, heuristic procedure - shifting bottleneck procedure, integer programming formulation, simulation studies of the dynamic job shop (overview only). Stochastic models - preliminaries, stochastic based on expectation, stochastic dominance based on variance, impact of randomness on fixed schedules, classes of policies - non-preemptive static list policy, preemptive static list policy, non-preemptive dynamic policy and preemptive dynamic policy, stochastic single machine models - arbitrary distributions without preemptions, arbitrary distributions with preemptions - the Gittins Index, likelihood ratio ordered distributions and exponential distributions.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply a variety of problem solving approaches for single machine problems in scheduling.	K3
CO2	Develop computational and algorithmic skills to apply in solving single machine problems.	K3
CO3	Apply mathematical as well as applied methods in parallel and flow shop scheduling problems.	K3
CO4	Identify job shop and stochastic scheduling problems and to apply various heuristics for the given job shop scheduling.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	1	1	2	-	-	-	-	-	2
CO2	3	1	2	1	1	2	-	-	-	-	-	2
CO3	3	2	2	2	3	3	-	-	-	-	-	2
CO4	3	2	2	3	3	3	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Sequencing and Scheduling – An Introduction to the mathematics of the job-shop	Simon French	Ellis Horwood Series	1 st Edition, 1981
2	An Introduction to the Mathematics of Planning and Scheduling	Geza Paul Bottlik	Routledge	1 st Edition, 2017
3	Scheduling: Theory, Algorithms, and Systems	Michael L. Pinedo	Springer	5 th Edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Sequencing and Scheduling	Kenneth R. Baker, Dan Trietsch	John Wiley & Sons, Inc.	2 nd Edition 2019
2	Handbook of Scheduling: Algorithms, Models, and Performance Analysis	Joseph Y-T. Leung	Chapman & Hall/CRC	1 st Edition 2004

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/106/110106045/

SEMESTER S6

HEURISTICS FOR DECISION MAKING

Course Code	PEIET634	CIE Marks	40
Teaching Hours/Week (L: T: P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide basic understanding of decision theory and various heuristic solution methods.
2. To enable students to apply these heuristic algorithms for various decision situations.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Decision theory: Types of decision making environment, decision making under uncertainty and under risk, decision tree analysis. Basic concepts & evolutionary algorithms: Genetic Algorithm (GA) - types of encoding, selection, crossover and mutation, binary, continuous hybrid and parallel GA, applications of GA in solving constrained and combinatorial optimization problems.	9
2	Ant colony algorithms: Overview, basic algorithm, variants, Formalization and properties of ant colony optimization, applications in scheduling and vehicle routing problems. Particle swarm optimization (PSO): Basic concepts, swarm intelligence principles, computational characteristics, PSO in real number space, velocity updating, topology of the particle swarm, parameter selection; discrete PSO, PSO variants and PSO applications.	9
3	Search based algorithms: Scatter search - components, algorithm, applications, greedy randomized adaptive search procedure, Tabu Search (TS) - principles, neighbourhood, candidate list, short term and long term memory, threshold accepting, application of TS in decision making. Simulated Annealing: Main components of simulated annealing, homogenous vs. in-homogenous simulated annealing, annealing schedules,	9

	applications in sequencing and scheduling, travelling salesman problems. variants of simulated annealing.	
4	Artificial Neural Networks: Biological and artificial neural networks, basic concepts, generic algorithm, application areas, application of ANN to solve transportation problems, Knapsack Problems. Constraint Programming: Problem Formulation in constraint programming, basic search and constraint propagation, constraint programming vs mathematical programming, applications.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply decision theory concepts and evolutionary algorithms for decision making.	K3
CO2	Solve combinatorial optimization problems using nature inspired algorithms.	K3
CO3	Make use of effective searches to arrive at smart solutions to complex decision problems in engineering and management.	K3
CO4	Apply artificial neural network and constraint programming to solve complex problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	3	-	-	-	-	-	-	3
CO2	3	3	3	-	3	-	-	-	-	-	-	3
CO3	3	3	3	-	3	-	-	-	-	-	-	3
CO4	3	3	3	-	3	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Heuristic Techniques for Combinatorial Problems	Colin R. Reeves	Orient Black Swan	1 st Edition, 1993
2	Metaheuristics for Hard Optimization	Johann Dréo, Alain Pétrowski, Patrick Siarry, Eric Taillard	Springer-Verlag Berlin and Heidelberg Gmbh	1 st Edition, 2010
3	First Course in Artificial Intelligence	Deepak Khemani	McGraw Hill Education	5 th Edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Metaheuristic Search Concepts-A Tutorial with Applications to Production and Logistics	Günther Zäpfel, Roland Braune, Michael Bögl	Springer-Verlag Berlin and Heidelberg Gmbh	20 th Edition, 2014
2	Meta-Heuristics: Theory and Applications	Ibrahim H. Osman, James P. Kelly	Springer	1 st Edition, 1996
3	Meta-Heuristics: Advances and Trends in Local Search Paradigms for optimization	Stefan Voß, Silvano Martello, Ibrahim H. Osman, Cathérine Roucairol	Springer	1 st Edition, 1998
4	Genetic Algorithms + Data Structures = Evolution Programs	Z. Michalewicz	Springer-Verlag Berlin and Heidelberg Gmbh	2 nd Edition, 1994

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/106/106/106106226/
	https://archive.nptel.ac.in/courses/110/106/110106134/

SEMESTER S6

ENTERPRISE RESOURCE PLANNING

Course Code	PEIET635	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart knowledge in architecture, different packages and modern trends in ERP .
2. To enable students to identify the important issues pertaining to implementation of ERP software in industries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Enterprise Resource Planning (ERP): Overview, need, history, risks and benefits, enterprise applications, ERP and related technologies, ERP in manufacturing perspective. Business process reengineering: Business process modelling, business modeling, procedure for implementation, applications, case studies.	9
2	ERP software packages: Selection of ERP package, various approaches to ERP selection, procurement process for ERP package, features of various modules of ERP. ERP implementation: Life cycle, methodologies, issues, hidden costs, vendors, consultants and users, ERP project management, ERP security, ERP training, change management, application support.	9
3	ERP Functional Modules: Human capital management, financial management, procurement and inventory management, supplier relationship management, production planning and execution, supply chain planning, sales and service, warehouse and transport management, customer relationship management, quality management, maintenance management and enterprise asset management, product lifecycle management.	9

4	ERP Market: SAP AG, Baan Company, PeopleSoft, Oracle Corporation, Microsoft Dynamics, JD Edwards world solution company, QUAD system software associates, Epicor ERP and Lawson ERP, open source ERP packages. ERP and eBusiness: ERP II, future directions and trends in ERP, ERP resources on the web. Case studies in ERP: HRM, finance, production, materials, sales and distribution.	9
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**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Examination</i>	<i>Analyse</i>	<i>Evaluate</i>	<i>Total</i>
5	15	10	10	40

Criteria for Assessment (Analyse and Evaluate): 20 marks

Phases	Assessment Criteria	Marks
Analyse	<i>Problem Definition</i> <i>a. Clearly defines the real-world quality issue.</i> <i>b. Examine and identify relevant contextual factors.</i>	5
	<i>Problem Analysis</i> <i>a. Present a structured realistic solution methodology.</i> <i>b. Compare and justify the proposed solutions with evidence and logical reasoning</i>	5
Evaluate	<i>Validation of Results</i> <i>a. Thoroughly evaluate the proposed solutions.</i> <i>b. Compares trade-offs, advantages, and disadvantages.</i> <i>c. Considers feasibility, scalability, and practical implications.</i>	5
	<i>Conclusion and Report Writing</i> <i>a. Summarizes procedure, findings and insights, limitation, and scope for future work.</i> <i>b. Preparation of Report with all components of project report.</i>	5

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain basics of ERP system and apply business process reengineering in organisations.	K3
CO2	Demonstrate software packages and its implementation in industries.	K2
CO3	Apply functional modules of ERP in organisations.	K3
CO4	Identify ERP solution providers and to apply ERP in various sections in an organisation.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	-	2	2	-	-	-	3	3	-	3
CO2	2	2	-	2	2	-	-	-	3	3	-	3
CO3	3	3	-	3	3	-	-	2	3	3	-	3
CO4	3	3	-	3	3	-	-	-	3	3	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	ERP Demystified	Alexis Leon	McGraw-Hill Education India Pvt. Ltd.	3 rd Edition, 2015
2	Enterprise Resource Planning	Rajesh Ray	McGraw-Hill Education India Pvt. Ltd.	1 st Edition, 2011
3	Enterprise Resource Planning	Mary Sumner	Pearson	1 st Edition, 2004
4	Enterprise Resource Planning: A Managerial Perspective	Veena Bansal	Pearson	1 st Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Enterprise Resource Planning: Fundamentals of Design and Implementation	K. Ganesh, Sanjay Mohapatra, S. P. Anbu Udayasankar, P. Sivakumar	Springer	1 st Edition, 2014
2	Enterprise Resource Planning	Daniel E. O’Leary	Cambridge University Press	1 st Edition, 2000
3	Concepts in Enterprise Resource Planning	Ellan F. Monk, Bret J. Wagner	Thomson Course Technology	2 nd Edition, 2006

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107238/

SEMESTER S6
APPLIED ERGONOMICS

Course Code	PBIET604	CIE Marks	60
Teaching Hours/Week (L:T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide a detailed understanding of ergonomics and its application in the real working environments.
2. To provide an adequate insight about the functioning and capability of humans in work settings.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to ergonomics: Definition, multidisciplinary engineering approach in ergonomics, various disciplines of ergonomics- industrial ergonomics, visual ergonomics and organizational ergonomics. Human Technological System (HTS): Concept and classifications, design of HTS, conceptual design, advanced development, detailed design and development, system reliability and modelling.	11
2	Design of displays and controls: Definition and measurement of information, information processing in human, information processing model, design of visual, auditory, tactual and olfactory displays, input and output in displays, usage of text, graphics, symbols and codes in design of visual displays, motor skills in human, human control of systems, design of controls, design of hand tools and devices.	11
3	Biomechanics and occupational stress: Biomechanics, mechanics of upper extremity of hand, lower extremity, and foot, bending, lifting and carrying,	11

	human physiology, oxygen consumption, energy expenditure, cardiovascular and respiratory effects, work capacity and fatigue, occupational stress and musculoskeletal disorders.	
4	Design of workplace and work environment: Anthropometry and workplace, workspace and work surface design, principles of seat design, environmental factors influencing human performance, effects of illumination, noise and vibration in design of workplaces, design of virtual environments and applications, safety and security aspects in workplaces, assessment of risks in workplaces.	11

Suggestion on Project Topics: Each student team can choose some simple close to real life projects from the areas discussed in the syllabus, such as

- Workplace design with digital human modelling software
- Design of thermal environment with software support
- Physiological measurement with heart rate monitor
- Anthropometric data collection of different demographic segments

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 2 sub divisions. • Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply ergonomic concepts in design and modelling of human technological systems.	K3
CO2	Make use of display and control concepts in design of systems.	K3
CO3	Apply biomechanics and human physiology in reduction of workplace risks.	K3
CO4	Develop workplaces that fits to the worker and to identify safe work environments.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	-	3	-	-	-	2	-	-	3
CO2	3	3	2	-	3	-	-	-	2	-	-	3
CO3	3	3	2	-	3	-	-	-	2	-	-	3
CO4	3	3	2	-	3	-	-	-	2	-	-	3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Human Factors and Ergonomics	Bridger, R. S.	CRC	4 th Edition, 2017
2	Human Factors in Engineering and Design	Sanders, M. S.	McGraw Hill	7 th Edition, 1992
3	Human Factors Engineering.	Phillips, C. A.	John Wiley & Sons.	1 th Edition, 2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Guide to Human Factors and Ergonomics	Helander M.	CRC Press	2 nd Edition, 2002
2	Designing for People: An Introduction to Human Factors Engineering	John D. Lee, Christopher D. Wickens, Liu Y., Boyle L. N.	CreateSpace	3 rd Edition, 2017
3	Fitting the Human - Introduction to Ergonomics / Human Factors Engineering	Kroemer K. H. E.	CRC Press	7 th Edition, 2017
4	Handbook of Standards and Guidelines in Human Factors and Ergonomics	Karwowski W., Szopa A., Soares M. M.	CRC Press	2 nd Edition, 2021

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/105/110105162/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study / Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S6

STATISTICAL QUALITY CONTROL

Course Code	OEIET611	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable the students to understand the principles of statistical quality control and its importance in industry.
2. To apply statistical quality control techniques to achieve continuous quality improvement in products and services.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concepts related to statistical quality control (SQC): Introduction, definitions of quality, basic concept of quality, quality control, definition of SQC, benefits and limitations of SQC, quality objectives, quality policy, quality function, quality assurance, quality costs. Total quality management - -philosophies, models, implementation.	9
2	Statistical process control: Process variability, process control, control chart for variables - $\bar{x}$ and R charts, $\bar{x}$ and s charts, control chart for attributes - p chart, np chart, c chart, u chart and D chart. State of control and process out of control identification in control charts. Process capability - process capability indices and process capability analysis.	9

3	Acceptance Sampling Plans: The concept of acceptance sampling, economics of inspections, types of acceptance sampling plans, probability of acceptance in single, double and multiple sampling techniques. Operating characteristic curve, AQL, LTPD, producer's risk, consumer's risk, AOQL and ATI. Standard sampling plans - MIL-STD 105E sampling method and its equivalents, Dodge - Romig sampling plans.	9
4	Six Sigma: Meaning, history of six sigma, working of six sigma. six sigma roles and responsibilities, DMAIC process- various phases, six sigma in manufacturing and service industries. Quality Management Systems: Benefits of ISO registration, ISO 9000 series of standards, ISO 9001 requirements- implementation, documentation, internal audits, registration, sector-specific standards, ISO 14000 standards.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the concepts related to statistical quality control for quality improvement.	K2
CO2	Apply statistical tools to monitor and improve process performance.	K3
CO3	Apply the concepts and methodology of accepting sampling in industries.	K3
CO4	Demonstrate expertise in six sigma methodology and quality improvement tools.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	-	-	-	-	-	-	2
CO2	3	2	2	2	-	-	-	-	-	-	-	2
CO3	3	2	2	-	-	-	-	-	-	-	-	2
CO4	3	2	2	2	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Statistical Quality Control	Grant E. L	McGraw Hill	7 th Edition, 2000
2	Fundamentals of Quality Control and Improvement	Amitava Mitra	Pearson Education	7 th Edition, 2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Statistical Quality Control	Gupta R. C.	Khanna Publishers	10 th Edition, 2003
2	Introduction to Statistical Quality Control	Montgomery	John Wiley & Sons	8 th Edition, 2019
3	Statistical Quality Control	Mahajan M.	Dhanpat Rai & Sons	3 rd Edition, 2001
4	Transactional Six Sigma and Lean Servicing	Betsi Harris Ehrlich	St. Lucie Press	1 st Edition, 2002

Video Links (NPTEL, SWAYAM...)	
Link ID	https://nptel.ac.in/courses/110104080

SEMESTER S6

QUALITY MANAGEMENT

Course Code	OEIET612	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable the students to understand the fundamentals of quality management and its importance in organisations.
2. To apply quality management principles to real world scenarios.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Quality Management: definitions of the terms - quality, quality planning, quality control, quality assurance, quality management. Total Quality Management (TQM) - overview on TQM, TQM axioms, consequences of total quality, barriers to TQM, Deming approach to TQM, Juran's quality trilogy, Crosby's fourteen steps for quality improvement.	9
2	Strategic quality management: cost of quality, customer satisfaction, quality function deployment (QFD), integrating quality into strategic management, quality and the management cycle, obstacles to achieving successful strategic quality management, supplier selection, concepts of 5S, six sigma and Kaizen. Lean principles.	9
3	Human dimensions of TQM: top management commitment, leadership for TQM, change management, resources for quality activities, training for quality, employee involvement, motivation, empowerment, team work, self managing teams, role of the quality director.	9

4	Supporting tools, activities and techniques in TQM projects: affinity diagram, brainstorming, cause and effect analysis, process flow chart, check sheets, scatter diagram, Pareto chart, histogram, control charts, process capability, Taguchi's robust design, total productive maintenance, failure mode and effect analysis. Quality systems: ISO 9000 family of standards, ISO 14000 standards, quality auditing - types and benefits.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of quality management and its importance in organizations.	K2
CO2	Explain the strategic management principles and its implementation.	K2
CO3	Demonstrate the various human dimensions of TQM.	K2
CO4	Apply quality improvement tools and techniques in industries.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	-	-	-	-	-	-	-	-	2
CO2	2	2	2	-	-	-	-	-	-	-	-	2
CO3	2	2	2	-	-	-	-	-	-	-	-	2
CO4	3	3	3	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Total Quality Management	Besterfield Dale H, et al.	Pearson Education	5 th Edition, 2018
2	Total Quality Management	Subburaj Ramasamy	McGraw Hill Education	4 th Edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Total Quality Management	K.C. Arora	S K Kataria and Sons	4 th Edition, 2016
2	Juran on "Leadership for Quality" An Executive Handbook	Juran J. M.	The Free Press	1 st Edition, 1989
3	Managing for Total Quality - From Deming to Taguchi and SPC	Logothetics N.	Prentice Hall Ltd	1 st Edition, 1993
4	Quality Planning and Analysis - From Product Development through Use	Juran J. M., Gryna F. M.	Tata McGraw Hill	3 rd Edition, 2004

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/104/110104080/

SEMESTER S6

MAINTENANCE ENGINEERING AND MANAGEMENT

Course Code	OEIET613	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to acquire knowledge in theory and practical application of Maintenance Engineering
2. To equip students in various tools and strategies for effective maintenance of a system.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Maintenance and reliability concepts: Maintenance - need/purpose, basic and composite functions of maintenance, principles, objectives and benefits of maintenance. Defects and failures - definitions, basics of failures, sources of defects, failure rate, causes and types of failures. Reliability - basic concepts, bathtub curve. MTTF, MTBF, MTTR, MTBM, maintainability, availability, inter-relationship between productivity, quality, reliability, availability and maintainability in maintenance.	9
2	Maintenance strategies and classifications: Maintenance strategies/ systems/ types – classifications, basis/factors considered for selection. Breakdown maintenance, corrective maintenance - concept, features. Routine maintenance and opportunistic maintenance - concept, features. Proactive maintenance - concept, features. Preventive maintenance- basics. Predictive maintenance - basics, advantages and disadvantages. Condition based maintenance - features. Merits and demerits of maintenance strategies. Condition monitoring concepts, monitoring systems (offline and online) and techniques/methods.	9

3	Tools, strategies and methods for effective maintenance: Reliability centred maintenance (RCM) and its advantages. Fault tree analysis (FTA) - steps, features and merits. Event tree analysis (ETA) - features, merits and procedure. Root cause analysis (RCA), cause and effect analysis. Failure modes and effects analysis (FMEA). Failure mode effect and criticality analysis (FMECA). Six sigma maintenance, lean maintenance concept – features and benefits. zero maintenance concept, 5-S maintenance concept, maintenance effectiveness, overall equipment effectiveness (OEE).	9
4	Maintenance planning, scheduling, cost and budgeting: Maintenance planning, procedure/steps in maintenance planning. Maintenance scheduling - basic techniques. Maintenance organization - factors affecting size and type of maintenance organization, objectives and characteristics, types of maintenance organisation - formal and informal organisation, classifications of maintenance organisation - line & staff, functional, centralized and decentralized maintenance organisation, merits and demerits. Maintenance costs - classification of maintenance costs, maintenance cost analysis. Maintenance Budget - types of maintenance budget, preparation of maintenance budget (basics).	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the concept of maintenance and reliability and their inter-relationship.	K2
CO2	Demonstrate the different types and strategies of maintenance and different techniques used in condition monitoring.	K2
CO3	Explain the tools, strategies and methods for assessing and ensuring effective maintenance .	K2
CO4	Apply planning, scheduling, costing & budgeting in maintenance and preparation of maintenance budget.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	-	-	-	-	-	-	-	-	2
CO2	3	3	1	-	-	-	-	-	-	-	-	2
CO3	3	3	1	1	-	-	-	-	-	-	-	2
CO4	3	3	1	2	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliability, Maintenance and Safety Engineering, 1st edition,	Gupta A. K.	Laxmi Publications	1 st Edition, 2009
2	Reliability-Based Design	Rao S. S.	McGraw-Hill	1 st Edition, 1992
3	Maintenance Engineering and Management	Srivastava, S.K.	S. Chand	Reprint Edition, 2020
4	Maintenance Engineering and Management	Venkataraman.	Prentice-Hall	4 th Edition, 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Maintenance	Brumbach M., Clade J.	Delmar Cengage Learning	2 nd Edition, 2013
2	Handbook of Condition Monitoring: Techniques and Methodology	Davies A.	Springer	8 th Edition, 1997
3	Maintenance Engineering and Management	Mishra R. C., Pathak K.	PHI Learning.	2 nd Edition, 2012
4	Maintenance Engineering Handbook	Mobley K.	McGraw-Hill	8 th Edition, 2008

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/105/112105232/
	http://vlabs.iitkgp.ac.in/mssp/exp9/index.html

SEMESTER S6

HEURISTICS SOLUTION TECHNIQUES

Course Code	OEIET614	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide basic understanding of various meta-heuristic solution algorithms.
2. To enable students to apply heuristic algorithms for solving optimization problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic concepts & Evolutionary algorithms: Genetic algorithm (GA), basic concepts, types of encoding, selection, crossover and mutation, binary, continuous, hybrid and parallel GA, application of GA in solving constrained and combinatorial optimization problems, sequencing and scheduling problems and transportation problems.	8
2	Ant Colony Algorithms: Overview, basic algorithm, variants, formalization and properties of ant colony optimization, applications in scheduling and vehicle routing problem. Particle Swarm Optimization (PSO): Basic concepts, swarm intelligence principles, computational characteristics, parameter selection, discrete PSO, variants and applications.	10
3	Search based algorithms: Scatter search - Components, algorithm, applications, greedy randomized adaptive search procedure, Tabu Search (TS) - principles, neighbourhood, candidate list, short - term and long - term memory, threshold accepting, application of TS in decision making. Simulated Annealing: Main components of simulated annealing, homogeneous vs. in-homogenous simulated annealing, annealing schedules, applications in sequencing and scheduling, travelling salesman problem, variants of simulated annealing.	10

4	Artificial Neural Networks (ANN): Basic concepts, biological and artificial neurons, learning rules and activation functions, ANN architecture, single layer and multi layer feed forward networks, back propagation networks and applications. Fuzzy Systems: Crisp logic, predicate logic, fuzzy logic, fuzzy rule based system, defuzzification methods, applications.	8
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply evolutionary algorithms for decision making.	K3
CO2	Solve combinatorial optimization problems using nature inspired algorithms.	K3
CO3	Make use of effective searches to arrive at smart solutions to complex decision problems in engineering and management.	K3
CO4	Apply artificial neural network and fuzzy logic concepts to solve complex problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	3	3	-	-	2	3	3	-	3
CO2	3	3	-	3	3	-	-	2	3	3	-	3
CO3	3	3	-	3	3	-	-	2	3	3	-	3
CO4	3	3	-	3	3	-	-	2	3	3	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Heuristic Techniques for Combinatorial Problems	Colin R. Reeves	Orient Black Swan	1 st Edition, 1993
2	Metaheuristics for Hard Optimization	Johann Dréo, Alain Pétrowski, Patrick Siarry, Eric Taillard	Springer-Verlag Berlin and Heidelberg Gmbh	1 st Edition, 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Metaheuristic Search Concepts-A Tutorial with Applications to Production and Logistics	Günther Zäpfel , Roland Braune, Michael Bögl	Springer-Verlag Berlin and Heidelberg GmbhGmbh	20 th Edition, 2014
2	Meta-Heuristics: Theory and Applications	Ibrahim H. Osman, James P. Kelly	Springer	1 st Edition, 1996
3	Meta-Heuristics: Advances and Trends in Local Search Paradigms for optimization	Stefan Voß, Silvano Martello, Ibrahim H. Osman, Cathérine Roucairol	Springer	1 st Edition, 1998
4	Genetic Algorithms + Data Structures = Evolution Programs	Z. Michalewicz	Springer-Verlag Berlin and Heidelberg Gmbh	2 nd Edition, 1994

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/106/106/106106226/
	https://archive.nptel.ac.in/courses/110/106/110106134/

SEMESTER S6

MODELLING AND SIMULATION LAB

Course Code	PCIEL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To make the students gain practical skills in applying system simulation techniques in various industrial and interdisciplinary areas.
2. To provide hands-on training on system modelling and behaviour.

Expt. No.	Experiments
1	Exercise on random number generation and test of fitness using Excel and Statistical softwares
2	Exercise on statistical distribution fitting using SPSS/ Minitab / Systat etc.
3	Exercise on Monte-Carlo simulation using Excel.
4	Simulation of queuing systems using Simio / Arena / Promodel / Witness etc.
5	Simulation of manufacturing systems using Simio / Arena / Promodel / Witness etc.
6	Simulation of healthcare systems using Simio / Arena / Promodel / Witness etc.
7	Simulation of biological systems using Vensim / Powersim / Stella etc.
8	Simulation of social systems using Vensim / Powersim / Stella etc.
9	Simulation of industrial systems using ensim / Powersim / Stella etc.
10	Apply system dynamics models for forecasting, financial analysis and decision making.

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the system concepts to understand a real world system.	K3
CO2	Distinguish a discrete and continuous system with respect to modelling.	K4
CO3	Build various scenarios for a discrete system and simulate the same.	K3
CO4	Develop cause and effect relationships for continuous systems and simulate the same.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	2	2	3	-	-	1	1	-	2
CO2	2	1	2	1	2	1	-	-	1	1	-	2
CO3	2	2	3	2	3	2	-	-	1	1	-	2
CO4	2	2	3	2	3	2	-	-	1	1	-	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Excel Data Analysis : Modeling and Simulation	Hector Guerrero	Springer	2nd Edition, 2019
2	Simio and Simulation - Modeling, Analysis, Applications	Jeffrey S. Smith, David T. Sturrock	Simio LLC	7 th Edition, 2024
3	System Dynamics Modelling with Vensim	Juan Martin Garcia	Independently published	1 st Edition, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Simulation Modeling and Analysis	A. M. Law	McGraw Hill	6 th Edition, 2024
2	System Simulation	Geoffrey Gordon	Prentice Hall India	2 nd Edition, 1979
3	Discrete Event Simulation: Modeling, Programming and Analysis	George Fishman	Springer series in Operations Research and Financial Engg.	1 st Edition, 2001
4	System Simulation with Digital Computer	Deo Narsingh	Prentice Hall India	24 th Edition, 2011
5	Simulation	Sheldon M Ross	Elsevier	6 th Edition, 2023
6	Business Dynamics – Systems Thinking and Modeling for a Complex World	John D Sterman	McGraw Hill	1 st Edition, 2000

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107220/
	https://archive.nptel.ac.in/courses/112/107/112107214/
	https://archive.nptel.ac.in/courses/110/101/110101142/
	https://archive.nptel.ac.in/courses/110/107/110107157/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

INDUSTRIAL ENGINEERING

SEMESTER S7

MANAGEMENT ACCOUNTING FOR ENGINEERS

Course Code	PEIET741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable the students to understand the concepts of management accounting and its importance in organisations.
2. To apply various tools and techniques to ascertain the financial condition of organizations.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of accounting: Management Accounting, Financial accounting, Cost accounting, comparison of financial accounting, cost accounting and management accounting, generally accepted accounting principles, accounting standards, accounting cycle, construction of journals and ledgers.	9
2	Financial accounting: Salient features of balance sheet and profit and loss statement, cash flow and fund flow analysis, limitations of cash flow and fund flow statements, working capital management, ratio analysis - types, computation and interpretation of financial ratios. Depreciation-definition and methods.	9
3	Cost accounting systems: Cost concepts relating to income measurement and profit planning, cost concepts for control and decision making, job costing, process costing, allocation of overheads, activity-based costing, standard costing and variance analysis, marginal costing, cost - volume-profit analysis.	9
4	Budgeting systems: Introduction, planning process, budget - definition, meaning and purpose, requirements for a sound budget, operating budgets, financial budgets, flexible budgets, zero based budgets and budgetary control, construction of various budgets.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain fundamental concepts of accounting principles and construct preliminary accounts.	K3
CO2	Apply techniques of accounting principles to prepare financial statements.	K3
CO3	Apply cost accounting concepts for managerial decision making.	K3
CO4	Construct various types of budgets for planning and decision making in organisations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	-	3	-	-	-	-	-	3	2
CO2	2	3	2	-	3	-	-	-	-	-	3	2
CO3	2	3	2	-	3	-	-	-	-	-	3	2
CO4	2	3	2	-	3	-	-	-	-	-	3	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Management Accounting	M. Y. Khan, P. K. Jain	McGraw Hill	8 th Edition, 2021
2	Management Accounting	Gupta S.K., Sharma R. K.	Kalyani Publishers	14 th Edition, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Management Accounting	Hornngren, Sundem, Stratton, Burgstahler, Schatzberg	PHI Learning	16 th Edition, 2013
2	Financial accounting - A Managerial Perspective	Narayanaswamy R.	PHI Learning	6 th Edition, 2017
3	Financial and Managerial accounting - The basis for business decisions	Jan Williams	Tata McGraw Hill	16 th Edition, 2017
4	Fundamental of Financial Management	James C., Van Horne	Pearson Education	13 th Edition, 2015

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/101/110101003/

SEMESTER S7

FINANCIAL TIME SERIES ANALYSIS

Course Code	PEIET742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to develop linear models for time series analysis.
2. To equip students to model and predict financial market behaviors using time series analysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Linear time series models: Financial data and their properties, financial time series and their characteristics, means, variances, covariances, correlation and autocorrelation, correlogram, stationarity, trends, seasonal variation, univariate and multivariate data, decomposition of series, the spectral density, smoothing and filtering, white noise, autoregressive models, moving average models, ARMA and ARIMA models.	9
2	Conditional heteroscedastic models: Characteristics of volatility, ARCH model, GARCH model, GARCH-M model, EGARCH model, threshold GARCH model, CHARMA model, random coefficient autoregressive model, asymmetric power ARCH model, nonsymmetric GARCH model, stochastic volatility model, long-memory stochastic volatility model, applications.	9
3	Nonlinear time series and nonparametric models: Nonlinear time series, univariate parametric nonlinear models - general formulation, threshold autoregressive models, Markov switching models, smooth transition autoregressive models, time-varying coefficient models. Univariate nonparametric models - kernel smoothing, local conditional mean, local polynomial fitting, splines, wavelet smoothing, nonlinear additive models, index model and sliced inverse regression.	9

4	High frequency financial data: Nonsynchronous trading, bid-ask spread of trading prices, empirical characteristics of trading data, models for price changes, duration models, value at risk (VaR), risk metrics, extreme value theory. Multivariate linear time series: Stationary vector autoregressive time series, vector autoregressive moving average time series. Multivariate volatility models: Multivariate GARCH models, GARCH models for bivariate returns, higher dimensional volatility models, factor-volatility models, multivariate t distribution.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Develop, estimate, and critically evaluate suitable linear time series models to forecast time series data.	K3
CO2	Develop conditional heteroscedastic models for various applications.	K3
CO3	Apply nonlinear time series models and nonparametric estimation techniques.	K3
CO4	Develop multivariate time series and volatility models.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	2	2	-	-	-	-	-	-	2
CO2	2	3	2	2	2	-	-	-	-	-	-	2
CO3	2	3	2	2	2	-	-	-	-	-	-	2
CO4	2	3	2	2	2	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	An Introduction to Analysis of Financial Data with R	Ruey S. Tsay	John Wiley & Sons, Inc.	1 st Edition, 2014
2	Nonlinear Time Series Analysis	Ruey S. Tsay, Rong Chen	John Wiley & Sons, Inc.	1 st Edition, 2018
3	Multivariate time series analysis: with R and financial applications	Ruey S. Tsay	John Wiley & Sons, Inc.	1 st Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Statistics and Data Analysis for Financial Engineering with R examples	David Ruppert, David S. Matteson	Springer	2 nd Edition, 2015
2	Time Series Analysis: Univariate and Multivariate Methods	William W. S. Wei	Pearson	2 nd Edition, 2005
3	Time Series Analysis and its Applications with R examples	Robert H. Shumway, David S. Stoffer	Springer	2 rd Edition, 2011
4	Time Series for Data Science: Analysis and Forecasting	Wayne A. Woodward, Bivin P. Sadler, Stephen D. Robertson	CRC Press	1 st Edition, 2024

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/103/106/103106123/

SEMESTER S7

FINANCIAL MANAGEMENT

Course Code	PEIET743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable for applying techniques of financial decisions, sources, and analysis.
2. To enable for applying tools of capital structure and working capital management.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Financial management: Introduction to finance and financial management. Scope of financial management, objectives of financial management, nature of financial management, goal of financial management, and financial management decisions. Time value of money - basic computations using compounding and discounting techniques. Major financial decision areas - factors influencing financial decisions, internal and external factors. Financial risk and return concepts. Methods for profit and wealth maximizations.	9
2	Financial sources: Identification of major financial sources - long term sources of finance - equity shares, debentures, preferred stock. Short term sources - bank sources, trade credit, overdrafts, commercial papers, money markets and mutual funds. Financial statement and analysis: Introduction concepts of financial accounting, financial statements, and financial analysis. Major financial statements - income statement, balance sheet, fund flow statement and cash flow statement. Ratio analysis - balance sheet ratios, income statement ratios, cash flow statement ratios, leverage ratios and composite ratios.	9
3	Capital structure decisions and theories: Capital structure - meaning, scope and patterns of capital structure, capital structure decisions, cost of	9

	capital - cost of specific sources of capital. Measurement of cost of capital - weighted average cost of capital. Capital structure theories / approaches - net income approach, net operating income approach, traditional approach, and Modigliani-Miller (MM) approach. Planning of domestic and international sources of finance.	
4	Working capital decisions and management: Working capital - working capital planning, working capital decisions, working capital management. theoretical models of cash management, inventories (including risk analysis), trade receivables, trade payables and short-term financing. Operating cycles. Capital budgeting - conventional and discounted cash flow (DCF) methods. Decision criteria in capital budgeting, net present value (NPV) - internal rate of return (IRR) comparisons, capital rationing, risk analysis and international capital budgeting.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply tools for financial management decisions.	K3
CO2	Make use of techniques for identifying financial sources and analysis.	K3
CO3	Identify the tools for capital structure decisions and management.	K3
CO4	Apply techniques for working capital decisions and management.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	3	-	3	-	-	-	-	-	-	2
CO2	2	-	3	-	3	-	-	-	-	-	-	2
CO3	2	-	3	-	3	-	-	-	-	-	-	2
CO4	2	-	3	-	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Financial Management: Theory and Practice	Chandra, P	Tata McGraw Hill	9 th Edition, 2017
2	Management Accounting	Gupta, S. K., Sharma, R. K.	Kalyani Publishers	1 st Edition, 2017
3	Financial Management, Text, Problems & Cases	Khan, M. Y., Jain, P. K.	Tata McGraw-Hill	8 th Edition, 2018
4	Financial Management	Pandey, I. M.	Vikas Publishing House	11 th Edition, 2016.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Financial Management and Policy: Text and Cases	Bhalla V. K.	Anmol Publications	9 th Edition, 2009
2	Fundamentals of Financial Management	Brigham, E. F., Houston, J. F.	Cengage Learning	10 th Edition, 2015
3	Principles of Managerial Finance	Gitman, L. J., Zutter, C. J	Pearson Education	13 th Edition, 2017
4	Principles of Financial Management	Horne, J. C. V., Wachowicz, J	Pearson Education Limited	13 th Edition, 2010
5	Financial Management	V. R. Palanivelu	S. Chand and Company Limited	2 nd Edition, 2011

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/107/110107144/
	https://archive.nptel.ac.in/noc/courses/noc21/SEM1/noc21-mg06/

SEMESTER S7

CORPORATE FINANCE AND PORTFOLIO MANAGEMENT

Course Code	PEIET744	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to understand the basic concepts of corporate finance.
2. To provide students an insight into different aspects of portfolio management.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Capital budgeting: Basic principles of capital budgeting, investment decision criteria - NPV, IRR, payback period, average accounting rate of return, profitability index, NPV profile, multiple IRR problem and no IRR, cash flow projections, project analysis and evaluation. Cost of capital: Costs of the different sources of capital, cost of debt, preferred stock, common equity, cost of capital estimation, beta estimation and project beta, country risk, marginal cost of capital schedule, flotation costs.	9
2	Financial leverage: Business risk and financial risk, capital structure - capital structure decision. Dividends and share repurchase: Dividends- forms, payment chronology, dividend policy and company value, factors affecting dividend policy, pay-out policies, analysis of dividend safety; Share repurchases- methods, effects of repurchases, valuation equivalence of cash dividends and share repurchases. Working capital management: Managing and measuring liquidity, managing the cash position, investing short-term funds, managing accounts receivable, managing inventory, managing accounts payable, managing	9

	short-term financing.	
3	Corporate governance: Importance, definitions, corporate governance considerations, board, management, shareowner rights, forms of business and conflicts of interest, sources of conflict, corporate governance evaluation. Mergers and acquisitions: Definitions and classifications, motives for merger, transaction characteristics, takeovers, regulation, merger analysis, corporate restructuring. Portfolio management: Overview, portfolio perspective on investing, investment clients, steps in the portfolio management process, pooled investments, portfolio concepts, mean-variance analysis, multifactor models, active portfolio management.	9
4	Portfolio risk and return: Investment characteristics of assets, risk aversion and portfolio selection, portfolio risk, efficient frontier and investor's optimal portfolio, capital market theory, pricing of risk and computation of expected return, capital asset pricing model (CAPM). Portfolio planning and construction: Portfolio Planning- investment policy statement (IPS), major components of an IPS, gathering client information, portfolio construction, capital market expectations, strategic asset allocation, steps toward an actual portfolio, additional portfolio organizing principles.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Make use of the basic principles of capital budgeting and different costs of capital.	K3
CO2	Apply difference measures of leverage, dividends and management of working capital	K3
CO3	Identify the overall corporate governance process and basics of portfolios.	K3
CO4	Apply the basic principles of portfolios to assess portfolio risk and returns.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	3	-	-	-	-	3	2
CO2	2	2	2	-	-	3	-	-	-	-	3	2
CO3	2	2	2	-	-	3	-	-	-	-	3	2
CO4	3	3	3	-	-	3	-	-	-	-	3	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Capital Market Expectations Managing Investment Portfolios; A Dynamic Process	Calverly, John P.	Wiley	3 rd Edition, 2007
2	Asset Allocation	Sharpe William F., Peng Chen, Jerald E. Pinto, Dennis W. McLeavy	Wiley	2 nd Edition, 2007
3	Credit Suisse Global Investment Returns Sourcebook	Dimson, Elroy, Paul Marsh , Mike Staunton	Credit Suisse Research Institute	3 rd Edition, 2009
4	The Black Swan: The impact of highly improbable	Taleb, Nassim N.	New York: Random House Inc,	2 nd Edition, 2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Corporate Finance and Portfolio Management	CFA Institute	CFA Institute	1 st Edition, 2011
2	Mergers, Acquisitions and Corporate Restructuring: Text and Cases	Chandrasekhar Krishnamurti, Viswanath S. R.	Atlantic	2 nd Edition, 2023

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/105/110105156/

SEMESTER S7

INTRODUCTION TO STOCK MARKETS AND TRADING

Course Code	PEIET745	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to apply techniques on stock markets and trading.
2. To equip students to identify tools for fundamental and technical analysis, and select derivatives and portfolios.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to stock markets: Economic markets, types, stock markets- history and evolution of modern stock market, financial securities, types of securities, primary markets and secondary markets, trading instruments in the market, international stock exchanges, Indian stock exchanges -NSE-BSE, stock market indices, SEBI and its functions, financial Intermediaries, initial public offering (IPO), investment methods in stock markets.	9
2	Trading and trading psychology: Trading process, types of trading strategies- scalping intraday method, momentum trading, swing positional trading. Risk in trading, types of risks, risk- reward ratio, risk management, target and stop loss, hedging in trading. Trading psychology- emotions of traders, methods to overcome the emotions, biases of traders, traits of successful traders.	9
3	Fundamental and technical analysis: Fundamental analysis- pillars of fundamental analysis, top down and bottom up approach, Porter's five forces. Financial statements- balance sheet, profit & loss statement, cash flow statement. Financial ratios- leverage, liquidity, profitability and solvency banking ratios. Valuation of stocks- intrinsic value of stocks. Technical	9

	analysis- RWT theory, Dows theory, candlestick patterns, support and resistance, bullish, bearish and neutral candlestick patterns, types of technical indicators, break out theory.	
4	Derivatives and portfolios: Options trading - fundamentals, terminologies, option Greeks, strategies, option premium calculation problems. Forex trading - forex market, currency pairs, transactions. Commodity trading - benefits, type of commodities. Portfolio management: Ways of portfolio management, steps in portfolio management and strategies, portfolio diversification.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Examination</i>	<i>Analyse</i>	<i>Evaluate</i>	<i>Total</i>
5	15	10	10	40

Criteria for Assessment (Analyse and Evaluate): 20 marks

Phases	Assessment Criteria	Marks
Analyse	<i>Problem Definition</i> - Clearly defines the real-world quality issue. - Examine and identify relevant contextual factors.	5
	<i>Problem Analysis</i> - Present a structured realistic solution methodology. - Compare and justify the proposed solutions with evidence and logical reasoning	5
Evaluate	<i>Validation of Results</i> - Thoroughly evaluate the proposed solutions. - Compares trade-offs, advantages, and disadvantages. - Considers feasibility, scalability, and practical implications.	5
	<i>Conclusion and Report Writing</i> - Summarizes procedure, findings and insights, limitation, and scope for future work. - Preparation of Report with all components of project report.	5

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply the techniques of stock markets.	K3
CO2	Identify trading and trading psychology.	K3
CO3	Apply tools on fundamental and technical analysis	K3
CO4	Make use of techniques to select derivatives and portfolios.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	-	-	-	-	--	-	-	-	2
CO2	2	2	3	-	-	-	-	-	-	-	-	2
CO3	2	2	3	-	-	-	-	-	-	-	-	2
CO4	2	2	3	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Beginners Guide to Stock Market	Matthew R Kratter	Trader University	1 st Edition, 2023
2	Fundamental Analysis for Investors	Raghu Palat	Vision Books	4 th Edition, 2016
3	Technical Analysis for Dummies	Barbara Rockefeller	Wiley Publication	3 rd Edition, 2014
4	Option traders guide to Probability, Volatility and timing	Jay Kaeppel	Wiley Publications	Illustrated Edition, 2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Equity Investing Strategies	Anatoly B Schmidt	World Scientific	1 st Edition, 2021
2	Technical Analysis of Financial Markets	John J Murphy	New York Institute of Finance	Special Indian Edition, 2020

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/105/110105121/

SEMESTER S7

PERSONNEL MANAGEMENT AND INDUSTRIAL RELATIONS

Course Code	PEIET751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To develop basic skills in students for maintaining good industrial relations and personnel functions in organisations.
2. To enable the student to apply the interaction pattern among labour, management and organization.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to personnel management: Meaning, concept, functions and evolution of personnel management, importance of personnel management, evolution of human relation department from personnel management, relationship between labour, management and state. Human resource planning: Meaning and concept, need for human resource planning, types of human resource planning, concept of job analysis, uses of job analysis information.	9
2	Performance appraisal and compensation: Defining performance appraisal, importance of performance appraisal, performance appraisal policy, compensation policies and methods, wage determination process and factors influencing wage and salary administration, types of wage, minimum wage, fair wage, living wage, money and real wage.	9
3	Industrial relations (IR): Meaning, definition, concept and scope, objectives, and elements of IR, factors affecting industrial relations, defining industrial conflict and disputes, causes and effect of industrial dispute, source of grievances, grievances handling procedures, concept of social justice,	9

	origin and growth of concept of labour welfare, role of government in labour welfare, role of trade union in industrial relation.	
4	Industrial dispute settlement mechanisms: Various dispute settlement machinery, mediation conciliation process, role, concept of arbitration and methods of arbitration, compulsory and voluntary arbitration, court of enquiry, labour courts, industrial tribunals, functions of industrial tribunal in state level and national level, distinction between conciliation, arbitration and adjudication, worker's participation in management, collective bargaining.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the concepts of personnel management and human resource planning.	K2
CO2	Apply the knowledge of performance appraisal and compensation of employees.	K3
CO3	Demonstrate the concepts of industrial relations in organisations.	K2
CO4	Apply the concepts of dispute settlement mechanism and workers participation in management	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	2	2	-	-	-	-	-	-	-	2
CO2	2	-	3	3	-	-	-	-	-	-	-	2
CO3	2	-	3	3	-	-	-	-	-	-	-	2
CO4	2	-	2	2	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Personal Management: Text and Cases	Mamoria C. B., Gankar S. V.	Himalaya Publishing House	1 st Edition, 2011
2	Essentials of HRM and Industrial Relation	Subba Rao P.	Himalaya Publishing House	5 th Edition, 2014
3	Personal Management & Industrial Relations	Tripathi P. C.	S. Chand & Sons	22 nd Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Relations	Venkata Ratnam C. S., Manoranjan Dhal	Oxford University Press, New Delhi	2 nd Edition, 2017
2	Industrial relation, Trade Union & Labour Relation	P. R N. Sinha, Indu Bala Sinha	Pearson	3 rd Edition, 2017

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/105/110105069/
	https://archive.nptel.ac.in/courses/122/105/122105020/

SEMESTER S7

MARKETING MANAGEMENT

Course Code	PEIET752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To equip the student to understand the concepts of marketing and its role in the business environment.
2. To enable the student to apply the marketing mix elements, the buyer behaviour and the contemporary marketing domains.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to marketing: Defining marketing, core concepts in marketing, evolution of marketing, marketing planning process. Scanning business environment: Internal and external, value chain, competitor analysis, strategies and core competencies, PESTEL framework, SWOT analysis, marketing interface with other functional areas, Marketing Research and Marketing Information System, prospects and challenges in marketing.	9
2	Marketing strategy: Strategic market planning, corporate strategy, business strategy, product strategy, services marketing, product and brand management, consumer markets, business/industrial markets, analysis of consumer and industrial markets, defining Market Segmentation, Targeting and Positioning (STP) and strategies in STP.	9
3	Marketing mix decisions: Marketing mix elements, product planning and development, Product Life Cycle (PLC), new product	9

	development, pricing objectives, methods and strategies, factors affecting price decisions, managing integrated marketing channels, managing retailing, wholesaling and logistics, Integrated Marketing Communications (IMC), advertising and sales promotion, public relations and publicity.	
4	Buyer behaviour: Understanding industrial and consumer buyer, factors influencing buyer behaviour, buyer behaviour models, building and measuring customer satisfaction. Trends in marketing management: Relationship marketing, relationship management, customer relationship management (CRM), CRM practices, online marketing, big data and artificial intelligence (AI) in marketing.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the concepts in marketing and marketing environment	K2
CO2	Apply the appropriate strategic knowledge in marketing	K3
CO3	Demonstrate the marketing mix elements and decisions	K2
CO4	Make use of the knowledge of buyer behaviour and trends in marketing	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	2	2	-	-	-	-	-	-	-	2
CO2	2	-	3	3	-	-	-	-	-	-	-	2
CO3	2	-	3	3	-	-	-	-	-	-	-	2
CO4	2	-	2	2	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marketing Management	Ramaswamy, V.S., Namakumari S	Sage Publication	6 th Edition, 2018
2	Marketing Management	Chandrasekar K S	Vijay Nicole	2 nd Edition, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marketing Management	Kotler Philip, Keller	Pearson India Education	15 th Edition, 2016
2	Fundamentals of Marketing	Stanton, W.J, Michael Etezel, Bruce J.Walker	McGraw Hill	9 th Edition, 1991
3	Marketing Management	Saxena Rajan	Tata McGraw Hill.	6 th Edition, 2019

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/104/110104068/
	https://archive.nptel.ac.in/courses/110/104/110104070/

SEMESTER S7

GREEN MANUFACTURING

Course Code	PEIET753	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to familiarise the concepts of environmental sustainability and green energy systems.
2. To equip students with knowledge on energy saving concepts in the manufacturing sector and the GreenCo rating system.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Environmental sustainability and impact assessment: Environmental impact assessment objectives, legislative development, European community directive, Hungarian directive. Strategic environmental assessment and sustainability appraisal, regional spatial planning and environmental policy.	9
2	Lean manufacturing and green energy system: Conventional manufacturing versus lean manufacturing, principles of lean manufacturing. World energy consumption - greenhouse effect, global warming. Energy conservation and measurement principles with their applicability in engineering and process industries.	9
3	Energy saving machinery: Electricity billing, components and costs, need and control, determination of kVA demand and consumption. Selection of fans, pumps and compressors, performance evaluation, cause for inefficient operation, scope for energy conservation.	10

	Solid waste management: Definition, terminology, classification and sources, need for hazardous waste management - need, handling, methods of collection, storage and transport with suitable examples. Solid waste management- need, waste prevention and life cycle assessment, collection, storage, reuse and recycling of solid waste with suitable examples.	
4	GreenCo rating: Ecological footprint - need for GreenCo rating, GreenCo rating system, parameters, intent, system approach, weightage, assessment process, types of rating, GreenCo benefits, case studies of GreenCo rating - manufacturing sector, automobile, FMCG and fertilisers, service sector, logistics, airports and hotels.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply the concepts of environmental sustainability and environmental impact assessment.	K3
CO2	Apply suitable schemes towards design of green manufacturing requirements.	K3
CO3	Analyze manufacturing processes towards conservation of energy and prevention of hazardous and solid wastes.	K4
CO4	Apply the concepts of green co-rating in organisations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	-	-	-	3	-	-	-	-	2
CO2	2	2	2	-	-	-	3	-	-	-	-	2
CO3	2	2	2	-	-	-	3	-	-	-	-	2
CO4	2	2	2	-	-	-	3	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Green Manufacturing Fundamentals and Applications	David A. Dornfeld	Springer Science & Business Media	1 st Edition, 2012
2	Design and Analysis of Lean Production Systems	Ronald G. Askin and Jeffrey B. Goldberg	John Wiley and Sons	1 st Edition, 2001
3	Environment and Sustainability Policy: Creation, implementation, Evaluation	Stephen Doven	The Federation Press	1 st Edition, 2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Impact Assessment and Sustainable Development: European Practice and Experience (Evaluating Sustainable Development series)	Clive George, Collin C., Kirkpolarice. H,	Edward Elgar Publishing Ltd.	1 st Edition, 2007
2	Green Manufacturing: Case Studies in Lean and Sustainability	Association for Manufacturing Excellence	Productivity Press	1 st Edition, 2017
3	Energy Auditing and Conservation, Methods Measurements, management and Case Study	Yacov Y. Haimes Marguerite A., H. Ruffner	Taylor & Francis Inc	1 st Edition, 1980
4	Advances in sustainable manufacturing	Seliger G., Khraisheh, M. M. and Jawahir I. S.	Springer Science & Business Media.	1 st Edition, 2011
5	Green Design and Manufacturing for Sustainability	Nand K. Jha	CRC Press	1 st Edition, 2015

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/104/112104225/
	https://archive.nptel.ac.in/courses/110/104/110104119/

SEMESTER S7

INDUSTRIAL SAFETY AND HEALTH

Course Code	PEIET754	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide a detailed understanding of occupational safety and its impacts in worker's health.
2. To provide an adequate background for applying the concept of safety in working environments.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to industrial safety: Definitions, safety systems, safety information system, basic concepts, definition of accidents, analysis of causes of accident. Fire protection systems: Automated fire fighting system, chemistry of fire, water sprinkler, fire hydrant, alarm and detection system. Suppression systems - CO2 system, foam system, DCP system, halon system, portable extinguisher.	9
2	Managing safety: Safety inspection, procedure, periodicity, checklist, report forms. Planning for safety and productivity, safety sampling, safety audit, safety survey, JSA, accident prevention. Work permit and lock out system, accident analysis, safety education, communication and safety performance analysis. Personal protective equipment testing, usage.	9
3	Safety in material handling: Selection of material handling equipment, equipment used, ropes, chains, slings, hooks, clamps,	9

	procedure for testing and checking as per standard. Design conservation, conveyor systems, belt, roller chain and elevator and lifts, industrial hoists, mobile crane, forklift, operation maintenance and checking procedure.	
4	Hazard analysis: Electrical, physical and chemical hazards, detailed hazard analysis, cost effectiveness in hazard eliminations, fault tree analysis and HAZOP, safety legislation- provisions in factory act for safety, explosive act, workmen compensation act, compensation calculation, boiler act, pollution control act.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of industrial safety and to identify the use of fire protection systems in organisations.	K3
CO2	Make use of safety systems and procedures in organisations.	K3
CO3	Identify the safety of material handling systems.	K3
CO4	Apply hazard analysis and to make use of provisions of safety legislations in organisations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	3	-	-	2	-	-	-	-	-	2
CO2	2	-	3	-	-	2	-	-	-	-	-	2
CO3	2	-	3	-	-	2	-	-	-	-	-	2
CO4	2	-	3	-	-	2	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Safety Management: 21st Century Perspectives of Asia	J Maiti, Pradip Kumar Ray	Springer	Online Edition, 2017
2	Industrial Safety Management: Hazard Identification and Risk Control	L. M. Deshmukh	McGraw-Hill Education	1 st Edition, 2017
3	Safety at Work	John Ridley, John Channing	Routledge	7 th Edition, 2007
4	Industrial Safety, Health and Environment Management Systems	R.K. Jain, Prof. Sunil S. Rao	Khanna Publishers	4 th edition, 2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Accident Prevention Manual for Business & Industry: Engineering & Technology	National Safety Council	National safety council, Chicago.	14 th Edition, 2015
2	Fault Tree Handbook	W. E. Vesely, F. F. Goldberg	Create space Independent Pub	1 st Edition, 2014
3	Fire Prevention Handbook	Derek James	Buttes Worths and Co.	1 st Edition, 1986
4	Industrial Accident Prevention: A safety management approach	Herbert William Heinrich, Dan Petersen, Nestor R. Roos	McGraw Hill Inc.	5 th Edition, 1980

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/105/110105094/

SEMESTER S7

MARKETING RESEARCH

Course Code	PEIET 755	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable the student to understand the importance of marketing research and research interventions.
2. To equip the student to apply knowledge to analyse the market with appropriate research methods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to marketing research: Define marketing research, scope, importance, classification, functions, role of marketing research in decision making, marketing research process. Defining marketing research problem: Identifying the problem, discussions and interviews with experts, insights from secondary data, marketing environment analysis, problem definition, decisions towards approaching the problem.	9
2	Research design: Defining the research design, classification of research design, exploratory research, descriptive research, causal research design, qualitative research techniques. Sources of data: Secondary data - internal Sources and external sources, international secondary data, role of secondary data in marketing research, primary data - survey and observation, experimentation.	9

3	Sampling and scaling: Sampling design process, probability sampling methods, non-probability sampling methods, online sampling techniques, determination of sample size. Scale characteristics, levels of measurement, primary scales, comparative scaling techniques, non-comparative scales, scale evaluation, relation between reliability and validity. Data collection: Data instruments, questionnaires, observation forms, Designing Questionnaire, determinants in question wording and order, form and layout.	9
4	Data analysis and research report writing: Data preparation process for analysis. procedures for hypothesis testing, parametric tests and non-parametric tests, basic concepts of analysis of variance and Correlation and regression, statistical software, importance of report preparation and presentation, elements of quality report, report format and layout. Applications of marketing research: Product research, advertising research, market and sales analysis research, social media marketing, branding, case studies in marketing research.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Examination</i>	<i>Analyse</i>	<i>Evaluate</i>	<i>Total</i>
5	15	10	10	40

Criteria for Assessment (Analyse and Evaluate): 20 marks

Phases	Assessment Criteria	Marks
Analyse	Problem Definition	5
	a. Clearly defines the real-world quality issue.	
	b. Examine and identify relevant contextual factors.	
	Problem Analysis	5
	a. Present a structured realistic solution	

	<i>methodology.</i> <i>b. Compare and justify the proposed solutions with evidence and logical reasoning</i>	
Evaluate	<i>Validation of Results</i> <i>a. Thoroughly evaluate the proposed solutions.</i> <i>b. Compares trade-offs, advantages, and disadvantages.</i> <i>c. Considers feasibility, scalability, and practical implications.</i>	5
	<i>Conclusion and Report Writing</i> <i>a. Summarizes procedure, findings and insights, limitation, and scope for future work.</i> <i>b. Preparation of Report with all components of project report.</i>	5

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain the market research process and outline the market research problem.	K2
CO2	Apply appropriate research designs in real world scenario.	K3
CO3	Identify suitable methods of sampling and data collection.	K3
CO4	Analyse the data for better decision making with the support of software.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	2	2	-	-	-	-	-	-	-	3
CO2	2	3	3	3	-	-	-	-	-	-	-	3
CO3	2	3	3	3	-	-	-	-	-	-	-	3
CO4	2	3	3	3	-	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marketing Research: An Applied Orientation	Naresh K. Malhotra	Tata McGraw Hill	6 th Edition, 2009
2	Marketing Research: Text and Cases	Nargundkar R.	Tata McGraw Hill	4 th Edition, 2020
3	Market Research	Beri G. C.	McGraw Hill	6 th Edition, 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marketing Research	Boyd, Westfall, Statsch	Irwin Inc	7 th Edition, 1989
2	Business Research Methods	William G. Zikmund	Cengage Learning	9 th Edition, 2012
3	Marketing Research, Concept & Cases	Cooper, Schindler	Tata McGraw Hill	1 st Edition, (Indian Edition) 2005

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/107/110107080/

SEMESTER S7

HUMAN RESOURCE MANAGEMENT

Course Code	OEIET721	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable the student to understand the strategies and processes of human resource management.
2. To equip the student to analyse the implications of human resource practices at individual, and organisational level.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to human resources management (HRM): Significance of HRM, definition, scope, functions, and objectives of HRM, roles of HR managers, introduction to HR analytics, HR accounting, HR audit, human resource information system, human resource planning (HRP), factors affecting HRP, process and methods of forecasting the demand and supply of human resources.	9
2	Job analysis and job design: Defining job analysis and its process, job description, job specification, methods of job design. Recruitment and selection: Defining recruitment, methods, and process in recruitment, Defining selection, process, and methods of selection, selection tests and its importance, types of interviews, induction/orientation, definition and types of transfer, promotion and demotion.	9
3	Performance assessment and training: Defining performance appraisal, its process and methods, training need	9

	assessment/identification (TNA/TNI), importance of training, on the job vs. off the job training, evaluation of training, past oriented and future oriented employee training. Wages and salary administration: Defining job evaluation, process and methods of job evaluation, basics of compensation, definition and types of compensation, differentiate wages and salary, methods of wage determination, wage theories and concepts, minimum wage, fair wage, living wage.	
4	Employee relations: Defining industrial relations (IR), objectives of IR, parties to IR, Trade Unions in India, defining participative management, methods of workers participatory management, Collective Bargaining. Employee discipline and grievance: Definition and settlement of grievance, open door vs. step ladder technique, defining industrial disputes, dispute settlement machinery, bipartite and tripartite bodies in India, industrial disputes act, factories act.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Make use of HRM strategies and processes of modern organizations.	K3
CO2	Apply the concepts of job analysis, recruitment and selection processes.	K3
CO3	Make use of performance appraisals and appropriate wage administration.	K3
CO4	Identify the implications of employee relations and industrial legislations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	2	2	-	-	-	-	-	-	-	2
CO2	2	-	3	3	-	-	-	-	-	-	-	2
CO3	2	-	3	3	-	-	-	-	-	-	-	2
CO4	2	-	2	2	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Resource and Personnel Management - Text and Cases	Aswathappa K.	Tata McGraw-Hill Publishing Co. Ltd	4 th Edition, 2005
2	Essentials of Human Resource Management and Industrial Relations	Subba Rao P.	Himalaya Publishing House	5 th Edition, 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human resource management: Strategic and international perspectives.	Crawshaw, J., Budhwar, P., Davis, A.	Sage Publications	1 st Edition, 2014
2	A handbook of human resource management practice	Armstrong, M.	Kogan Page Publishers	1 st Edition, 2006
3	Industrial Relations	Venkata Ratnam S.	Oxford University Press	5 th Edition, 2006

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/105/110105069/
	https://archive.nptel.ac.in/courses/122/105/122105020/

SEMESTER S7

INDUSTRIAL ENGINEERING AND MANAGEMENT

Course Code	OEIET722	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to familiarise the concepts of industrial engineering and management.
2. To equip the students to apply the concepts of industrial engineering and management in real world scenarios.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Evolution of Scientific Management and Industrial Engineering: Activities of industrial engineering, techniques of industrial engineering, functions of Management -brief description of each function. Types of Organization structures, types of companies and their formation. Productivity and work study: Definitions, methods study and work measurement, use of charts and diagrams. Performance rating and methods, types of allowances, computation of basic time and standard time.	9
2	Facility location: Factors influencing plant location, plant layout, different types of layouts, material flow pattern, layout planning, computerized layout planning techniques. Production planning and control: Functions and objectives, job, batch, mass and continuous production. Materials Management - inventory, determination of economic order quantity, inventory control techniques.	9

3	Quality engineering: Quality control, control chart for variables and attributes, introduction to ISO-9000 series, ISO 14000 series, total quality management, six sigma concepts, quality information systems, benchmarking and documentation. Reliability concepts: Definition of reliability, reliability vs. quality, reliability function, MTTF, hazard rate function, bathtub curve, derivation of the reliability function, failure and failure modes, causes of failures and unreliability.	9
4	Financial management: Scope of financial management, functions and objectives of financial management, working capital, factors affecting working capital, working capital cycle, depreciation and methods of calculating depreciation. Personnel Management: Objectives and functions, recruitment, selection, training, induction, concepts and techniques of wages and incentive systems. Industrial relations: Fatigue and methods of eliminating fatigue, industrial disputes, collective bargaining, trade unions.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain the concepts of industrial engineering and management and to apply work study techniques for improvement of productivity.	K3
CO2	Make use of concepts of facility location and production planning control in industries.	K3
CO3	Apply the concepts of quality and reliability in industries.	K3
CO4	Make use of the concepts of financial management, personnel management and industrial relations in organisations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	-	-	2	-	-	-	-	-	2
CO2	2	3	2	-	-	2	-	-	-	-	-	2
CO3	2	3	2	-	-	2	-	-	-	-	-	2
CO4	2	3	2	-	-	2	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Essentials of Management: An International Perspective	H. Koontz, H. Weihrich	McGraw-Hill	8 th Edition, 2009
2	Principles of management	P C Tripathi, P N Reddy	TMH	4 th Edition, 2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Work Study	ILO	Oxford and IBH	3 rd Edition, 2015
2	Operations Management	William J. Stevenson	McGraw-Hill Education	14 th Edition, 2020
3	Production and Operations Management	R. Paneerselvam	PHI Learning	3 rd Edition, 2012
4	Total Quality Management	Besterfield Dale H, et al.	Pearson Education	5 th Edition, 2018
5	Reliability Engineering	Balagurusamy E.	Tata McGraw Hill	1 st Edition, 2017
6	Reliability Engineering	Srinath L. S.	East West Press	4 th Edition, 2005
7	Financial Management	M. Y. Khan, P. K. Jain	Tata-McGraw Hill	8 th Edition, 2018.
8	Human Resources and Personnel Management: Text and Cases	K. Ashwathappa	Tata-McGraw Hill	4 th Edition, 2005
9	Industrial Relations	Venkata Ratnam C. S., Manoranjan Dhal	Oxford University Press, New Delhi	2 nd Edition, 2017

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107292/
	https://archive.nptel.ac.in/courses/112/107/112107249/
	https://archive.nptel.ac.in/courses/110/107/110107141/
	https://archive.nptel.ac.in/courses/112/107/112107143/
	https://archive.nptel.ac.in/courses/110/105/110105088/
	https://archive.nptel.ac.in/courses/110/107/110107144/
	https://archive.nptel.ac.in/courses/110/105/110105069/
	https://archive.nptel.ac.in/courses/122/105/122105020/

SEMESTER S7

PRODUCTION AND OPERATIONS MANAGEMENT

Course Code	OEIET723	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to apply production and operations management concepts including product design, process selection, modern production systems and sustainability.
2. To equip students to demonstrate proficiency in forecasting, aggregate planning, material and capacity management and production activity control.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to production and operations management (POM): Objectives and scope of POM, historical evolution and perspectives, operations strategy and productivity, relationship with other functional areas, manufacturing vs service sectors, operations decision making, phases in product design and development, product life cycle management, process selection, sustainability in operations.	9
2	Forecasting for operations management: Need for forecasting and determinants of demand, forecasting methods - qualitative methods - delphi technique, market research, nominal group technique, quantitative methods - moving average, exponential smoothing, regression analysis, Winter's model, analysis of forecast errors. Aggregate planning: Managerial inputs, pure and mixed strategies, mathematical models - transportation method, linear programming, linear decision rules.	9

3	Material requirement planning: Master production schedule (MPS) - development and implementation, material requirement planning (MRP) and MRP-II, enterprise resource planning (ERP). Capacity management: Capacity measures and influencing factors, long-term vs short-term capacity planning, systematic capacity planning approach, tools and techniques for capacity management, capacity requirement planning (CRP), business process outsourcing (BPO).	9
4	Production activity control (PAC): Objectives of PAC, scheduling techniques, job shop production, activity planning, shop loading, sequencing and dispatching rules. Modern production systems: Just-in-time (JIT) and kanban systems, lean manufacturing, kaizen and continuous improvement, Toyota production system (TPS), business process re-engineering, world class manufacturing concepts.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Explain the concepts of production operations management, and to apply operations strategies, product life cycle management, and process selection with a focus on sustainability.	K3
CO2	Demonstrate the ability to forecast demand using various techniques and develop aggregate production plans using both qualitative and quantitative methods.	K3
CO3	Develop Material Requirement Planning (MRP), and capacity planning techniques.	K3
CO4	Apply production activities and modern production systems in industries.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	-	-	-	3	-	-	-	-	2
CO2	3	2	3	-	-	-	2	-	-	-	-	2
CO3	3	2	3	-	-	-	2	-	-	-	-	2
CO4	3	2	3	-	-	-	2	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Operations Management: Processes and Supply Chains	Lee J. Krajewski, Manoj Malhotra	Pearson Education	14 th Edition, 2024
2	Operations Management	William J. Stevenson	McGraw-Hill Education	14 th Edition, 2020
3	Production and Operations Management	R. Paneerselvam	PHI Learning	3 rd Edition, 2012

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Operations and Supply Chain Management	Cecil B. Bozarth, Robert B. Handfield	Pearson	5 th Edition, 2019
2	Operations Management: Sustainability and Supply Chain Management	Jay Heizer, Barry Render, Chuck Munson	Pearson	14 th Edition, 2022
3	Operations Management: Processes and Value Chains	Lee J. Krajewski, Larry P. Ritzman, Manoj K. Malhotra	Pearson	10 th Edition, 2012
4	Production and Operations Management	S. N. Chary	McGraw-Hill	6 th Edition, 2019
5	Operations Management: Theory and Practice	B. Mahadevan	Pearson India	3 rd Edition, 2015

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107238/
	https://archive.nptel.ac.in/courses/110/107/110107141/

SEMESTER S7

SYSTEM SIMULATION

Course Code	OEIET724	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to construct a theoretical base for applying simulation in various interdisciplinary areas.
2. To enable students to model and simulate discrete and continuous systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	System concepts: Concepts of a system, system environment, continuous and discrete systems, system models, areas of application of simulation, steps in discrete event simulation study. Random numbers: Properties of random numbers, generation of pseudo-random numbers, random number generators. Tests for random numbers: Frequency, gap, run, and Poker tests, test for autocorrelation, Monte Carlo method.	9
2	Random variates: Generation of random variates, inverse transformation method for exponential, uniform, Weibull, triangular, and discrete distributions. Direct transformation method for normal and lognormal distributions. Acceptance-rejection technique for Poisson and Gamma distributions. Input modelling: Data collection, identifying the distribution with the collected data, goodness of fit tests, selecting input models without data. Verification and validation of simulation models, output data analysis, variance reduction techniques. Time advance mechanisms for discrete event simulation: Next-event time	9

	advance and fixed increment time advance methods.	
3	Introduction to system dynamics: Evolution of system dynamics as a system enquiry methodology – general systems theory, cybernetics, the systems approach, system dynamics approach. Elements of system dynamics: Modelling, physical flows, information flows, level & rate variables, delays, information smoothing, table functions and table function multipliers, causal loop diagramming, flow diagrams. steps for modelling in system dynamics.	9
4	Behaviour of systems: Analytical approach to behaviour of linear low-order systems - first order positive and negative feedback systems, pure second order positive and negative feedback systems. Fundamental modes of dynamic behaviour - exponential growth, goal seeking, oscillation, Interactions of fundamental modes. System dynamics modelling: Principles of simulation modelling, developing model equations, algorithm for Euler integration, hand simulation of system dynamics models, S-shaped growth, epidemics, innovation diffusion and growth of new products. Overview of software packages for system modelling and simulation. Case studies.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of the system and its simulation in interdisciplinary areas; demonstrate the generation of random numbers and its testing.	K2
CO2	Perform input modeling for discrete simulation, verify and validate simulation models and perform output data analysis.	K3
CO3	Illustrate various elements of system dynamics modelling; construct causal loop diagrams and Flow diagrams.	K3
CO4	Develop the fundamental modes of dynamic behaviour and their interactions; Model and Analyze systems using the principles of system dynamics.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	3	2	-	-	-	-	-	2
CO2	3	3	3	2	3	2	-	-	-	-	-	2
CO3	3	3	2	2	3	3	-	-	-	-	-	2
CO4	3	3	3	3	3	3	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Discrete Event System Simulation	Jerry Banks	Pearson Education	5 th Edition, 2013
2	Introduction to System Dynamic Modeling	M. C. Bora, Pratap K. J. Mohapatra, P. Mandal	Universities Press	1 st Edition, 1994

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Simulation Modeling and Analysis	A. M. Law	McGraw Hill	6 th Edition, 2024
2	System Simulation	Geoffrey Gordon	Prentice Hall India	2 nd Edition 2020
3	Discrete Event Simulation: Modeling, Programming and Analysis	George Fishman	Springer series in Operations Research and Financial Engg.	1 st Edition, 2001
4	System Simulation with Digital Computer	Deo Narsingh	Prentice Hall India	24 th Edition, 2011
5	Simulation	Sheldon M. Ross	Elsevier	6 th Edition 2023
6	Business Dynamics – Systems Thinking and Modeling for a Complex World	John D. Sterman	McGraw Hill	1 st Edition, 2000

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/112/107/112107220/
	https://archive.nptel.ac.in/courses/110/101/110101142/

SEMESTER 8

INDUSTRIAL ENGINEERING

SEMESTER S8

DATA SCIENCE WITH JULIA

Course Code	PEIET861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart a thorough understanding of the basics of Julia software and develop skills in using Julia's package ecosystem for various applications.
2. To perform advanced analytics using Julia's powerful libraries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic programming in Julia: Overview of Julia software package, installation and set up, variables, data types and operations, data structures - arrays, tuples, dictionaries, sets; conditionals and loops, functions, methods and constructors, modules and packages, input and output.	9
2	Advanced programming in Julia: Object oriented programming and functional programming, parallel programming, metaprogramming, working with dates and time, working with data frames.	9
3	Statistical analysis with Julia: Exploratory data analysis, hypothesis testing, time series analysis. Machine learning with Julia: Regression analysis, classification, clustering and dimensionality reduction.	9
4	Optimization with Julia: Solving linear programming problems, solving mixed integer programming and vehicle routing problems, solving non-linear and mixed integer non-linear programming problems.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Develop various programs to implement projects using best practices in coding.	K3
CO2	Develop parallel computing strategies, and efficiently handle concurrent tasks to solve complex, large-scale computational problems.	K3
CO3	Make use of machine learning models using Julia, and apply these techniques to derive actionable insights from real-world data sets.	K3
CO4	Apply Julia's optimization libraries to design and solve various optimization problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	2	2	-	-	-	-	-	-	2
CO2	2	3	2	2	2	-	-	-	-	-	-	2
CO3	2	3	2	2	2	-	-	-	-	-	-	2
CO4	2	3	2	2	2	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data Science with Julia	Paul D., McNicholas, Peter Tait	CRC Press	1 st Edition, 2021
2	Julia for Data Science	Zacharias Voulgaris	Technics Publications	1 st Edition, 2016
3	Statistics with Julia: Fundamentals for Data Science, Machine Learning and Artificial Intelligence	Yoni Nazarathy, Hayden Klok	Springer	1 st Edition, 2021
4	Think Julia: How to Think Like a Computer Scientist	Ben Lauwens, Allen Downey	O'Reilly Media	1 st Edition, 2019
5	Julia for Machine Learning	Anshul Joshi, Rahul Lakhanpal	Packt Publishing	1 st Edition, 2021

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Julia Programming Projects	Adrian Salceanu	Packt Publishing	1 st Edition, 2018
2	Julia High Performance	Avik Sengupta	Packt Publishing	2 nd Edition, 2019
3	The Julia Programming Language: High Performance Numerical Computing and Scientific Computation	Robert Smith and Leonard Berry	CRC Press	1 st Edition, 2021
4	Mastering Julia 1.0: Programming Julia for High-Performance Numerical Analysis and Computational Science	Malcolm Sherrington,	Packt Publishing	1 st Edition, 2019

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/111/106/111106164/

SEMESTER S8

PROGRAMMING AND DATA ANALYTICS WITH R

Course Code	PEIET862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to develop Proficiency in R Programming.
2. To equip students to apply statistical concepts and predictive modelling techniques in R to analyze real-world datasets.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to R: Setting up R and RStudio, overview of the R ecosystem, installing and loading R packages, getting help with R. Basic syntax and operations, introduction to R script, Basic data types, reading data into R, vectors, matrices and arrays, data frames, lists, factors and tables.	9
2	R programming: Functions, control statements and loops, math functions, simulation in R, object-oriented programming - S3 classes, S4 classes, R6 classes, input and output, reading and writing files, debugging, performance enhancement, parallel R. Rcpp: setup, basic syntax, writing functions, and interfacing. Optimize R performance with C++ for statistical computations and data processing. Develop, compile, and debug Rcpp modules, enhancing R code efficiency for real-world applications.	9

3	Data manipulation: Group manipulation - apply family, aggregate, plyr, data table, faster group manipulation with dplyr-pipes, tbl, select, filter, slice, mutate, summarize, group by, arrange, do; iterating with purr, data reshaping, tidyverse, string manipulation, manipulating time series data with xts and zoo packages.	9
4	Statistical analysis: Probability distributions, exploratory data analysis, creating and customizing graphs, inferential statistics, parametric and nonparametric inference, linear regression models - simple and multiple regression, generalized linear models, experimental designs, multivariate statistical analysis, time series analysis.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Make use of R data types and structures and prepare datasets for complex analysis and visualisation tasks.	K3
CO2	Develop functions, apply control structures, and automate tasks to solve practical data analysis and statistical computing challenges.	K3
CO3	Utilize various packages in R like dplyr and data.table for efficient data manipulation, preparing and transforming datasets for analysis and reporting.	K3
CO4	Perform robust statistical analyses, interpret data effectively, and produce actionable insights from various types of statistical models.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	2	2	-	-	-	-	-	-	2
CO2	2	2	2	2	2	-	-	-	-	-	-	2
CO3	2	2	2	2	2	-	-	-	-	-	-	2
CO4	2	2	2	2	2	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Art of R Programming: A Tour of Statistical Software Design	Norman Matloff	No Starch Press	1 st Edition, 2011
2	R for Everyone: Advanced Analytics and Graphics	Jared P Lander	Addison-Wesley	2 nd Edition, 2017
3	A Course in Statistics with R	Prabhanjan N. Tattar, Suresh Ramaiah and B.G. Manjunath	John Wiley & Sons, Ltd	1 st Edition, 2016
4	R for Data Science	Hadley Wickham & Garrett Grolemund	O'Reilly Media	1 st Edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data Analysis and Graphics Using R: An Example-Based Approach	John Maindonald and John Braun	Cambridge University Press	3 rd Edition, 2010
2	Practical Statistics for Data Scientists: 50 Essential Concepts Using R and Python	Peter Bruce, Andrew Bruce, and Peter Gedeck	O'Reilly Media	2 nd Edition, 2020
3	R Graphics Cookbook: Practical Recipes for Visualizing Data	Winston Chang	O'Reilly Media	2 nd Edition, 2019

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/111/104/111104146/
	https://archive.nptel.ac.in/courses/110/107/110107092/
	https://archive.nptel.ac.in/courses/106/106/106106179/

SEMESTER S8

QUANTUM COMPUTING WITH PYTHON

Course Code	PEIET863	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To develop a foundational understanding of quantum mechanics principles, quantum computing algorithms, and their applications using Python programming.
2. To equip students with skills to implement and simulate quantum algorithms and protocols using Python-based quantum computing frameworks like Qiskit.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to quantum mechanics: Basic concepts - dirac notations, qubits, bell state, superposition, entanglement, quantum states, quantum bits, and the Bloch sphere, introduction to quantum gates and circuits.	9
2	Quantum algorithms: Overview of quantum computation algorithms - Cirq, Qiskit, Deutsch-Jozsa, Bernstein-Vazirani, Simon's Algorithm, Grover's Algorithm, Shor's Algorithm.	9
3	Quantum protocols and quantum error correction: Quantum teleportation and superdense coding, quantum key distribution (BB84 and E91 protocols), basics of quantum error correction.	9
4	Practical quantum computing and quantum hardware: Review of current quantum computers and quantum supremacy, quantum hardware technologies (superconducting qubits, trapped ions), quantum machine learning and deep learning, future trends and challenges in quantum computing.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course, students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply quantum mechanics principles to construct and simulate quantum gates and circuits using Python.	K3
CO2	Apply quantum algorithms like Deutsch-Jozsa, Grover's, and Shor's to evaluate their efficiency in solving computational problems compared to classical approaches.	K3
CO3	Develop and simulate key quantum communication protocols and error correction schemes .	K3
CO4	Compare the performance of different quantum computing technologies and forecast potential developments and challenges in the field.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	2	3	-	-	-	-	-	-	2
CO2	2	2	3	2	3	-	-	-	-	-	-	2
CO3	2	2	3	2	3	-	-	-	-	-	-	2
CO4	2	2	3	2	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Quantum Computing for Everyone	Chris Bernhardt	MIT Press	1 st Edition, 2019
2	Programming Quantum Computers: Essential Algorithms and Code Samples	Eric R. Johnston, Nic Harrigan, Mercedes Gimeno-Segovia	O'Reilly Media	1 st Edition, 2019
3	Learn Quantum Computing with Python and Q#	Sarah C. Kaiser and Christopher E. Granade	Manning Publications	1 st Edition, 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hands-On Quantum Machine Learning With Python: Volume I: Get Started	Frank Zickert	Apress	1 st Edition, 2020
2	Quantum Computing: An Applied Approach	Jack D. Hidary	Springer	1 st Edition, 2019

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/106/106/106106232/

SEMESTER S8

INTRODUCTION TO NATURAL LANGUAGE PROCESSING

Course Code	PEIET864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to familiarize the fundamental principles of natural language processing (NLP) and implement NLP techniques in Python.
2. To equip students to construct and train machine learning models to address and evaluate common NLP problems using standard metrics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to NLP: The study of language, evaluating language understanding systems, different levels of language analysis, representations and understanding, organization of natural language understanding systems, linguistic background - An outline of english syntax, challenges in NLP. Machine learning, deep learning, and NLP: An overview, approaches to NLP, heuristics-based NLP, machine learning & deep learning for NLP, NLP pipeline, applications of NLP - machine translation, speech recognition, image captioning, spam detection, text prediction.	9
2	Regular expression and text processing: Importance, properties, working and python package, common regex function, meta characters- big brackets, cap, backslash, squared brackets, special sequences, asterisk, plus, and question mark, curly brackets Understanding pattern objects: Match method Vs search method,	9

	finditer method, logical Or, beginning and end patterns, parenthesis string modification, sentiment classification, language independent tokenization, string matching and spelling correction.	
3	Vector models and text preprocessing: Vectors, bag of words, countvectorizer, tokenization, stopwords, stemming, and lemmatization, Countvectorizer, vector similarity. TF-IDF, word-to-index mapping, building TF-IDF, neural word embeddings. Vector models, text preprocessing summary, steps of NLP analysis. Language modeling: Importance, types of language modeling, the curse of dimensionality, language model Markov assumption and N-Grams, language model implementation -setup, Ngrams function, update counts function, probability model function, reading corpus, language model implementation sampling text, Markov models, article spinning, cipher decryption with language modeling and genetic algorithm.	9
4	NLP using machine learning: Spam detection - problem, Naive Bayes theorem, intuition, spam detection using Naïve Bayes, class imbalance, ROC, AUC, AND F1 SCORE, implementing spam detection in python, sentiment analysis, text summarization, topic modeling, latent semantic modeling (Indexing). Deep learning: Word embeddings, nonlinear neural networks, Neuron - introduction, fitting a line, classification code preparation, text classification in tensorflow. Feed forward neural networks, convolution neural networks, recurrent neural networks.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course, students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Utilize regular expressions and Python libraries such as NLTK, spaCy, and Pandas for text processing tasks.	K3
CO2	Apply suitable machine learning models to solve various NLP tasks and compare them using appropriate performance metrics.	K3
CO3	Apply advanced NLP techniques to perform complex tasks like text summarization, topic modeling, and cipher decryption and compare their performance.	K3
CO4	Develop and implement deep learning architectures for advanced NLP tasks.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	2	3	-	-	-	-	-	-	2
CO2	2	2	2	2	3	-	-	-	-	-	-	2
CO3	2	2	3	2	3	-	-	-	-	-	-	2
CO4	2	2	3	2	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Natural Language Processing with Python	Steven Bird, Ewan Klein, and Edward Loper	O'Reilly Media	1 st Edition, 2009
2	Foundations of Statistical Natural Language Processing	Christopher D. Manning and Hinrich Schütze	The MIT Press	1 st Edition, 1999
3	Deep Learning for Natural Language Processing: Creating Neural Networks with Python	Palash Goyal, Sumit Pandey, and Karan Jain	Apress	1 st Edition, 2018
4	Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems	Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, and Harshit Surana	O'Reilly Media	1 st Edition, 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Natural Language Processing with Transformers: Building Language Applications with Hugging Face	Lewis Tunstall, Leandro von Werra, Thomas Wolf	O'Reilly Media	1 st Edition, 2022
2	Neural Network Methods for Natural Language Processing	Yoav Goldberg	Springer	1 st Edition, 2017
3	Taming Text: How to Find, Organize, and Manipulate It	Grant S. Ingersoll, Thomas S. Morton, Andrew L. Farris	Manning Publications	1 st Edition, 2013
4	Speech and Language Processing	Daniel Jurafsky, James H. Martin	Pearson	3 rd Edition, 2021
5	Natural Language Processing Recipes Unlocking Text Data with Machine Learning and Deep Learning using Python	Akshay Kulkarni, Adarsha Shivananda	Apress	2 nd Edition, 2021

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/106/101/106101007/

SEMESTER S8

DATA ANALYTICS AND MACHINE LEARNING WITH PYTHON

Course Code	PEIET865	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip students with the skills to perform advanced data preprocessing, feature engineering, and selection using Python libraries to optimize machine learning models.
2. Develop proficiency in building, evaluating, and refining machine learning models using Python and implementing deep learning techniques for real-world applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to data analytics with python: Introduction to Python; using pandas for data manipulation; cleaning and preparing data. Feature engineering with pandas and NumPy; techniques for creating and transforming features for analytical models. Introduction to data visualization with Matplotlib and Seaborn; exploring data distributions and relationships.	9
2	Statistical analysis and feature selection: Descriptive and inferential statistics with SciPy; understanding data through statistical metrics. Feature selection techniques; using scikit-learn for automated feature selection methods like RFE (Recursive Feature Elimination). Dimensionality reduction techniques; PCA and t-SNE with scikit-learn for feature extraction and data visualization.	9
3	Machine learning models and handling imbalanced data: Overview of machine learning algorithms (classification, regression)	9

	with scikit-learn. Handling imbalanced datasets; techniques like SMOTE, and balanced random forests with imbalanced-learn. Model evaluation and hyperparameter tuning using cross-validation and GridSearchCV in scikit-learn.	
4	Advanced machine learning techniques and applications: Introduction to deep learning with TensorFlow and Keras; basics of neural networks. Building and training deep learning models for classification and regression problems. Advanced topics in machine learning: ensemble methods, boosting, and stacking models to improve performance.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Examination</i>	<i>Analyse</i>	<i>Evaluate</i>	<i>Total</i>
5	15	10	10	40

Criteria for Assessment (Analyse and Evaluate): 20 marks

Phases	Assessment Criteria	Marks
Analyse	<i>Problem Definition</i> a. Clearly defines the real-world quality issue. b. Examine and identify relevant contextual factors.	5
	<i>Problem Analysis</i> a. Present a structured realistic solution methodology. b. Compare and justify the proposed solutions with evidence and logical reasoning	5
Evaluate	<i>Validation of Results</i> a. Thoroughly evaluate the proposed solutions. b. Compares trade-offs, advantages, and disadvantages. c. Considers feasibility, scalability, and practical implications.	5
	<i>Conclusion and Report Writing</i> a. Summarizes procedure, findings and insights, limitation, and scope for future work. b. Preparation of Report with all components of project report.	5

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply data manipulation and preprocessing techniques using pandas; construct and assess new features to enhance model accuracy.	K3
CO2	Identify statistical methods for insights; apply advanced feature selection strategies to optimize predictive models in scikit-learn.	K3
CO3	Apply various machine learning algorithms; address imbalanced data issues using specialized techniques to ensure robust model performance.	K3
CO4	Experiment deep learning with TensorFlow and Keras; enhance model predictions through ensemble methods and practical implementations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	2	3	-	-	-	-	-	-	2
CO2	2	2	3	2	3	-	-	-	-	-	-	2
CO3	2	2	2	2	3	-	-	-	-	-	-	2
CO4	2	2	3	2	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Python for Data Analysis	Wes McKinney	O'Reilly Media	2 nd Edition, 2017
2	Feature Engineering for Machine Learning: Principles and Techniques for Data Scientists	by Alice Zheng and Amanda Casari	O'Reilly Media	1 st Edition, 2018
3	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media	2 nd Edition, 2019
4	Deep Learning with Python	François Chollet	Manning Publications	1 st Edition, 2017
5	Imbalanced Learning: Foundations, Algorithms, and Applications	Haibo He and Yunqian Ma	Wiley-IEEE Press	1 st Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Applied Predictive Modelling	Max Kuhn and Kjell Johnson	Springer	1 st Edition, 2013
2	Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python	Peter Bruce, Andrew Bruce, and Peter Gedeck	O'Reilly Media	2 nd Edition, 2020

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/106/107/106107220/
	https://archive.nptel.ac.in/courses/106/105/106105152/

SEMESTER S8

FINANCIAL MANAGEMENT FOR ENGINEERS

Course Code	OEIET831	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to apply techniques of financial decisions and analysis.
2. To equip students to utilize tools of capital structure and working capital management.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Financial management: Finance, financial management and financial objectives. Scope of financial management, objectives of financial management, nature and scope of financial management, goal of financial management, financial management decisions. Time value of money - computations using compounding and discounting techniques. Securities and valuation of securities. Major financial decision areas – factors influencing financial decisions, internal and external factors. Financial management for engineers.	9
2	Financial statements and analysis: Financial statements, important concepts of financial accounting and financial statements. Major financial statements - income statement, balance sheet, fund flow statement and cash flow statement. Ratio analysis - balance sheet ratios, income statement ratios, cash flow statement ratios and composite ratios. Leverages - financial leverage, operating leverage and composite leverage.	9

3	Capital structure and capital structure theories: Capital structure - meaning, scope and patterns of capital structure, capital structure decisions, capital structure theories / approaches - Net Income Approach; Net Operating Income Approach; Traditional Approach and Modigliani-Miller (MM) Approach. Cost of capital. Planning for sources of finance (domestic and international).	9
4	Working capital management: Working capital - working capital planning, working capital management. management of cash (various theoretical models), inventories (including risk analysis), trade receivables, trade payables and short-term financing. Operating cycles. Capital budgeting – Conventional and discounted cash flow (DCF) Methods. Decision criteria in capital budgeting, net present value (NPV) – internal rate of return (IRR) comparisons, capital rationing and risk analysis. Introduction to international capital budgeting.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply the tools for financial management decisions.	K3
CO2	Identify techniques for the analysis of major financial reports.	K3
CO3	Make use of tools for capital structure decisions.	K3
CO4	Apply the techniques for working capital management.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	3	-	3	-	-	-	-	-	-	2
CO2	2	-	3	-	3	-	-	-	-	-	-	2
CO3	2	-	3	-	3	-	-	-	-	-	-	2
CO4	2	-	3	-	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Financial Management: Theory and Practice	Chandra P.	Tata McGraw Hill	9 th Edition, 2017
2	Management Accounting	Gupta, S. K., Sharma, R. K.	Kalyani Publishers	1 st Edition, 2017
3	Financial Management, Text, Problems & Cases	Khan, M. Y., Jain, P. K.	Tata McGraw-Hill	8 th Edition, 2018
4	Financial Management	Pandey I. M.	Vikas Publishing House	11 th Edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Financial Management and Policy: Text and Cases	Bhalla V. K.	Anmol Publications	9 th Edition, 2009
2	Fundamentals of Financial Management	Brigham E. F., Houston J. F.	Cengage Learning	10 th Edition, 2015
3	Principles of Managerial Finance,	Gitman L. J., Zutter C. J	Pearson Education	13 th Edition, 2017
4	Principles of Financial Management	Horne, J. C. V., Wachowicz J.	Pearson Education Limited	13 th Edition, 2010

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/107/110107144/
	https://archive.nptel.ac.in/noc/courses/noc21/SEM1/noc21-mg06/

SEMESTER S8

INTRODUCTION TO FINANCIAL MARKETS

Course Code	OEIET832	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to gain knowledge about various trading and investment activities.
2. To equip students to reap the benefits of financial markets for achieving financial freedom.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Primary and secondary equity market: Public issue, right issue, private placement. Secondary market - BSE, NSE, NASDAQ, trading and settlement, open outcry system, screen-based system, rolling settlement, transaction costs, buying and selling of shares, buying on margin and short sale. Stock market quotations and indices: Individual stock quotation, types of stock market indices, stock market indices in India - S&P CNX Nifty, Sensex.	9
2	Mutual funds: Definition of mutual fund, types and classification of mutual funds, advantages and disadvantages of passive investing over active investing, concepts about systematic investment plan and systematic withdrawal plan. Bonds: Bond characteristics, valuation of bonds, price of bond, yield, bond yield measures, yield to maturity, macaulay duration, intrinsic value of bond.	9

3	Investment management and algorithmic trading: Market impact and the order book, optimal execution- sell side perspective, impact models - buy side perspective, popular algorithmic trading strategies - volume weighted average price, time weighted average price, participation, market on close, arrival price.	9
4	Derivative Securities: Forward contracts, forward prices, swaps, futures contracts, futures prices, hedging, speculation. options theory- mechanics of options market, properties of stock options, nature of option values, put-call parity, European and American options.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the concepts of primary and secondary equity markets, and market indices.	K2
CO2	Identify suitable mutual funds and bonds for investments.	K3
CO3	Apply various strategies of algorithm trading in investments.	K3
CO4	Apply the concepts of option pricing and derivative trading.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	-	-	-	-	-	-	-	2	2
CO2	2	3	3	-	-	-	-	-	-	-	2	2
CO3	2	3	3	-	-	-	-	-	-	-	2	2
CO4	2	3	3	-	-	-	-	-	-	-	2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Financial Management, Text Problems and Cases	Khan M. Y., Jain P. K.	McGraw Hill Education	8 th Edition, 2014
2	Quantitative Trading: How to Build Your Own Algorithmic Trading Business	Chan E. P.	Wiley	1 st Edition, 2021
3	Applied Quantitative Methods for Trading and Investment	Dunis C.L., Laws J., Naim P.	Wiley	1 st Edition, 2003
4	An introduction to derivatives and Risk Management	Chance D.M., Robert Brooks	Cengage	9 th Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences	Cohen J., Cohen P., West S. G., Aiken L. S.	Routledge	3 rd Edition, 2003
2	Multivariate Data Analysis	Hair J. F., Black W. C., Babin B. J., Anderson R. E., Tatham R. L.	Pearson India	7 th Edition, 2015
3	Investment Analysis and Portfolio Management	Chandra P.	Tata McGraw-Hill	4 th Edition, 2013
4	Machine Learning for Algorithmic Trading	Jansen S.	Packt	2 nd Edition, 2020
5	Hedge Fund Investing: A Practical Approach to Understanding Investor Motivation, Manager Profits, and Fund Performance	Kevin R. Mirabile	John Wiley	1 st Edition, 2013
6	Python Algorithmic Trading Cookbook	Dagade P.	Packt	1 st Edition, 2020

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/111/103/111103126/

SEMESTER S8

COST AND MANAGEMENT ACCOUNTING FOR ENGINEERS

Course Code	OEIET833	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable the students to understand the main principles of cost and management accounting.
2. To apply various tools and techniques to ascertain the financial condition of organizations.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to accounting: Cost accounting, Management Accounting, Financial accounting. Scope, objectives and significance of cost accounting, elements of cost, cost accounting standards, cost accounting records, ledgers and cost statements.	9
2	Financial accounting: Salient features of balance sheet and profit and loss statement, cash flow and fund flow analysis(elementary), working capital management, ratio analysis - types, computation and interpretation of financial ratios. Depreciation-definition and methods.	9
3	Cost accounting systems: Cost concepts relating to income measurement and profit planning, cost concepts for control and decision making, job costing, process costing, allocation of overheads, activity based costing, standard costing and variance analysis, marginal costing, break even analysis.	9
4	Budget and budgeting systems: Introduction, planning process, budget - definition, meaning and purpose, requirements for a sound budget, fixed budgets - preparation of sales and production budgets,	9

	financial budgets, flexible budgets, zero based budgets and budgetary control, construction of various budgets.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the basic concepts of accounting in organisations.	K3
CO2	Apply techniques of accounting principles to prepare financial statements.	K3
CO3	Apply cost accounting concepts for managerial decision making.	K3
CO4	Construct various types of budgets for planning and decision making in organisations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	-	3	-	-	-	-	-	3	2
CO2	2	3	2	-	3	-	-	-	-	-	3	2
CO3	2	3	2	-	3	-	-	-	-	-	3	2
CO4	2	3	2	-	3	-	-	-	-	-	3	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cost and Management Accounting	M. P. Gupta, Ajai Gupta	Sultan Chand & Sons	1 st Edition, 2023
2	Management Accounting	M. Y. Khan, P. K. Jain	McGraw Hill	8 th Edition, 2021

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Management Accounting	Horngren, Surdem, Stratton, Burgstahler, Schatzberg	PHI Learning	16 th Edition, 2013
2	Cost Accounting: Text, Problems and Cases	Jawahar Lal, Seema Srivastav, Manisha Singh	McGraw Hill	6 th Edition, 2019
3	Financial and Managerial accounting - The basis for business decisions	Jan Williams	Tata McGraw Hill	16 th Edition, 2017
4	Fundamental of Financial Management	James C., Van Horne	Pearson Education	13 th edition, 2015

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/110/101/110101003/

SEMESTER S8

INTRODUCTION TO FINANCIAL ECONOMICS

Course Code	OEIET834	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable students to apply tools of financial economics, derivative markets, bonds and securities.
2. To equip students to utilize models and tools for financial risks and asset returns.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to financial economics: Investment theory and structure of interest rates. Time value of money - future value, present value, future value of an annuity, present value of annuity, present rate of perpetuity. investment criteria: net present value (NPV), benefit cost ratio (BCR), internal rate of return (IRR), modified internal rate of return. Apply time value of concepts to practical problems in financial economics.	9
2	Financial derivative markets: Introduction to financial derivatives: types and uses of derivatives; forward contracts: determination of forward prices, futures contract: theories of future prices - the cost of carry model, the expectation model, relation between spot and future prices. Options: types, value of an option and important option pricing models with application cases. Bonds and securities: Fundamentals of bonds and securities, valuation of securities, valuation of bonds and stocks, bond yield, yield to maturity, equity valuation and dividend discount models.	9

3	Financial risks and returns: Introduction to financial risks and returns, types of risks, historical returns, and risk, computing historical returns, average annual returns, variance of returns, estimation of risk and return of an asset, measurement of risk and return of a portfolio, determinants of beta, risk-return trade off. Effect of diversification, minimum variance portfolio, perfectly correlated assets, minimum variance opportunity set, optimal portfolio choice, mean- variance frontier of risky and risk-free assets.	9
4	Financial models for asset returns: Introduction to advanced asset return models, multi-index models, single index model, equilibrium models, capital asset pricing model (CAPM), arbitrage pricing theory (APT), capital market line, security market line and estimation models of beta for investors. Cost of Capital - debt and equity, cost of debt, cost of preference capital and equity capital, capital market lines, security market lines and use of models in investment analysis.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcomes		Bloom's Knowledge Level (KL)
CO1	Apply time value of money tools in financial economics	K3
CO2	Identify techniques on derivative markets, bonds, and securities.	K3
CO3	Make use of tools for estimation of financial risks and returns	K3
CO4	Apply advanced tools and models for asset returns	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	3	-	3	-	-	-	-	-	-	2
CO2	2	-	3	-	3	-	-	-	-	-	-	2
CO3	2	-	3	-	3	-	-	-	-	-	-	2
CO4	2	-	3	-	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Financial Institutions and Markets	L. M. Bhole, J. Mahukud	Tata McGraw Hill	5 th Edition, 2011
2	Options, Futures and Other Derivatives	John C. Hull	Pearson Education	6 th Edition, 2005
3	Financial Theory and Corporate Policy	Thomas E. Copeland, J. Fred Weston, Kuldeep Shastri	Prentice Hall	4 th Edition, 2003
4	Principles of Corporate Finance	Richard A. Brealey, Stewart C. Myers	McGraw-Hill	7 th Edition, 2002
5	Fundamentals of Corporate Finance	Stephen A. Ross, Randolph W. Westerfield, Bradford D. Jordan,	McGraw-Hill	7 th Edition, 2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Financial Theory and Corporate Policy	Copeland, T. E., J. F. Weston	Addison Wesley	1 st Edition, 1992
2	Modern Portfolio Theory & Investment Analysis	Elton, E. J., M. J. Gruber	John Wiley & Sons	4 th Edition, 1991
3	Economics of Financial Markets	Houthakker H. S., P. J. Williamson	Oxford University Press	1 st Edition, 1996

Video Links (NPTEL, SWAYAM...)	
Link ID	https://archive.nptel.ac.in/courses/130/104/130104116/